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General Relativistic Magnetohydrodynamic Simulations of Binary Neutron Star Mergers

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Introduction

Binary neutron star mergers have an incredible scientific potential. The complex interplay of several different phenomena and the extreme conditions in which such catastrophic events occur make them unique laboratories to explore a variety of interesting area of physics and astrophysics. Indeed, coalescing neutron star binaries serve as probes to test gravity in strong field regime and represent a perfect arena for relativistic magnetohydrodynamics. In addition, these systems offer the extraordinary opportunity of investigating the state of matter at supranuclear density - a task which is impossible to realize on Earth - hence providing powerful insights into nuclear and hadronic physics. Furthermore, they offer the possibility to deepen our understanding of relevant high-energy astrophysical phenomena such as gamma-ray bursts, but also to shed light on the origin of heavy elements in the universe. They also open up new interesting possibilities in cosmology, such as providing a new way of measuring the Hubble constant.

These mergers are accompanied by the emission of gravitational radiation and electromagnetic signals across the entire spectrum, from gamma-ray to radio waves. Their first combined detection in August 2017, using both gravitational wave interferometers and electromagnetic telescopes covering the full spectrum, has represented a significant breakthrough for multi-messenger astronomy. Within this context, the tools of computational astrophysics play a crucial role, allowing to make accurate predictions of gravitational and electromagnetic signatures and linking the near-future observational data with the physics underlying these processes. Specifically, this requires the combination of numerical relativity with computational methods in relativistic magnetohydrodynamics, radiation transport and nuclear astrophysics. In particular, in this work I performed General Relativistic Magnetohydrodynamic (GRMHD) simulations of binary neutron star mergers, using codes such as the GRMHD code **Spritz** ([1],[2]) and the GRHD code **WhiskyTHC** [3], in a computational framework known as Einstein Toolkit ([4],[5]).

Before providing a description of the simulations, I will first present the theoretical background required to understand these phenomena. Chapter 1 contains a description of our current understanding of neutron stars and binary neutron stars, with particular focus on their observational properties such as gravitational and electromagnetic emissions. Chapter 2 is devoted to present Numerical Relativity and its most important formulations, as well as Magnetohydrodynamics in the context of General Relativity. In Chapter 3, I describe the computational infrastructure employed for my simulations, known as Einstein Toolkit, which delivers powerful

numerical tools for solving problems in relativistic astrophysics and gravitational physics; then, I focus on describing the numerical methods implemented in the code **Spritz**, which I used for a part of my simulations.

In Chapter 4 and Chapter 5 I finally present the results of my simulations. In Chapter 4, I carry on the work discussed by Alice Gambaro in her Master’s thesis [6]; in particular, I describe a series of tests performed in order to investigate an issue, the “fast-inspiral problem”, encountered in evolving binary neutron stars systems with tabulated equations of state via **Spritz**. In Chapter 5, I outline the results extracted from two simulations performed through **WhiskyTHC**. After that, the work is concluded by a brief final summary.

Technical Note

All the simulations have been performed on the **Galileo100** cluster at **CINECA** thanks to a **CINECA-INFN** agreement (allocation INF22_teongrav_1).

Chapter 1

Binary Neutron Stars

1.1 Neutron Stars

In 1934, the astronomers Walter Baade and Fritz Zwicky, in the two remarkable papers *On Super-novae* [7] and *Remarks on Super-novae and Cosmic Rays* [8], postulated that the remnants of supernovae - a term coined by themselves - could be *neutron stars*, a new kind of object predominantly composed of neutrons, characterized by high density and small radius, more gravitationally bound than normal stars. In their words: “With all reserve we advance the view that a super-nova represents the transition of an ordinary star into a *neutron star*, consisting mainly of neutrons. Such a star may possess a very small radius and an extremely high density.”

Besides the important work by J. Robert Oppenheimer and George Volkoff in 1939 [9] - which performed the first accurate calculation of neutron star models, assuming an ideal gas of free neutrons at high density - these objects have been ignored for the next 30 years. The discovery of cosmic non-solar X-rays in 1962 and the identification of the first quasar in 1963 awakened interest in the physics of compact stars, thought to be related to those astrophysical phenomena; however, the majority of physicists and astronomers were still skeptical about the possibility that such exotic objects could exist.

The situation changed in 1967, when Jocelyn Bell discovered the first *pulsar*, observed as a pulsating radio signal. Shortly after the discovery, in 1968 Thomas Gold suggested that a rotating neutron star model could explain the pulsed radiation observed. The discoveries in the same year of *Crab Pulsar* and *Vela Pulsar*, both placed in supernova remnants, supported the original idea from Baade and Zwicky that neutron stars are formed in supernova explosions. Neutron stars have also been identified as sources of X-ray emission (*X-ray pulsars*), resulting from accretion processes in binary systems. As of today, more than 3000 pulsars have been discovered. The existence of neutron stars is now a well-established fact. However, we are still far from a complete understanding of their extreme nature. The rise and rapid advancement of gravitational-wave astronomy in recent years promises to offer new possibilities to explore their properties, in ways that can not be achieved

through electromagnetic astronomy.

1.1.1 Stellar Death and Origin of Neutron Stars

Neutron stars are *compact objects*. More in general, by this term one usually refers to three different types of object: white dwarfs, neutron stars and black holes. They share the feature of having small size relative to ordinary stars with comparable mass: this implies much stronger gravitational fields.

These species are pretty different from each other, but have in common an important fact regarding their origin: their life begins when normal stars die, i.e. nuclear fuel has been consumed. At this point, stars begin to collapse, since self-gravitational force is no longer balanced by thermal pressure from nuclear reactions. White dwarfs and neutron stars are instead supported by quantum degeneracy pressure exerted by fermions (electrons in white dwarfs, neutrons in neutron stars), while black holes are completely collapsed stars.

According to stellar evolution theory, the main factor determining the fate of a star - i.e. whether it becomes a white dwarf, a neutron star or a black hole - is the star's mass. In particular, neutron stars are thought to be originated from massive stars ($\gtrsim 10M_{\odot}$).

In these stars, helium burning phase in the core - occurring after the star exhausts the hydrogen - is succeeded by burning stages of progressively heavier elements (carbon, neon, oxygen and silicon), until an inner core made of iron group elements is eventually formed. The iron core is surrounded by concentric shells of different compositions - silicon, oxygen, neon, carbon, helium (from the innermost to the outermost) - and the envelope. Since iron is the most bound nucleus in terms of binding energy per nucleon, it will not undergo fusion: this reaction would require energy rather than release it. For this reason, no nuclear burning happens in the iron core, which starts collapsing and electrons become a degenerate gas. Due to the nuclear burning taking place in the outer layers, it keeps growing in mass. When the core's mass goes beyond the Chandrasekhar limit, the electron degeneracy pressure is no longer able to support it against gravity, causing it to rapidly collapse. At this point, the loss of support due to *electron capture* and *photodisintegration* contribute to the core collapse; in addition, these processes increase the neutron fraction. Finally, the neutron gas becomes degenerate, and exerts sufficient pressure to stop the collapse. The infalling matter rebounds and produces a shock wave propagating outwards and powering a *supernova explosion*, which expels the outer layers and leaves behind the neutron core, known as *neutron star*.

1.1.2 TOV Equations and Equilibrium Configurations

Modeling the structure of neutron stars requires a general relativistic framework, due to their nature of compact objects. A suitable formalism for this aim is provided by the *Tolman-Oppenheimer-Volkoff equations*, also known as *TOV equations*.

This set of equations describes the stellar structure in general relativistic regimes. In particular, they describe a spherically symmetric star in hydrostatic equilibrium within a general relativistic framework.

In natural units ($c = G = 1$), they take the following form:

$$\frac{dp}{dr} = -(e + p) \frac{d\phi}{dr}, \quad (1.1)$$

$$\frac{dm}{dr} = 4\pi r^2 e, \quad (1.2)$$

$$\frac{d\phi}{dr} = \frac{4\pi r^3 p + m}{r(r - 2m)}, \quad (1.3)$$

where p is the pressure, ρ is the rest-mass density, e is the energy density ($e = \rho + \rho\epsilon$, with ϵ specific internal energy), ϕ is a metric function which is the general relativistic equivalent to the Newtonian gravitational potential, r is the radial distance from the center of the star and m is the mass contained inside that radius.

In order to obtain TOV equations, the following assumptions have been made:

- spherical symmetry; hydrodynamic and spacetime quantities only depends on radial coordinate;
- static equilibrium, i.e., no time dependence of the solutions;
- no rotation;
- absence of magnetic fields;
- an *equation of state* in form $p = p(\rho)$, with no temperature dependence; this aspect will be discussed later.

The absence of magnetic fields is a good assumption from the point of view of stellar structure. Indeed, even in case of very powerful magnetic fields, magnetic energy density is lower by orders of magnitude than gas energy density (and magnetic pressure is significantly lower than gas pressure, as well). Magnetic fields of the order of 10^{17} G start having an effect on star deformation; for values of magnetic field around 10^{18} G, magnetic energy density becomes comparable to gas energy density and beyond this order of magnitude magnetic field would even destroy the star. Of course, the study of electromagnetic emission (for example, the *pulsar emission mechanism*) requires star magnetic field to be considered but - as said before - it can be neglected in studying the stellar structure. Spherical symmetry and absence of rotation are not always acceptable: certainly, the structure of rapidly rotating stars such as millisecond pulsars cannot be modeled by taking these assumptions into account, while these are reasonable (with a certain degree of error) for stars having longer rotation periods.

The physical meaning of TOV equations can be better understood in the *Newtonian limit*, i.e. $p \approx \rho\epsilon \rightarrow 0$ and $m \ll r$:

$$\frac{dp}{dr} = -\rho \frac{d\phi}{dr}, \quad (1.4)$$

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad (1.5)$$

$$\frac{d\phi}{dr} = \frac{m}{r^2}, \quad (1.6)$$

from equations (1.1), (1.2), (1.3) respectively.

Therefore, it is clear that - identifying ϕ with gravitational potential - (1.1) is the general relativistic version of hydrostatic equilibrium equation, which expresses the balance between gravity and a force due to the pressure gradient. Equation (1.2) is the relativistic version of the mass conservation equation; indeed, in a relativistic framework, it becomes a conservation equation including also a term due to internal energy. Equation (1.3) is a general relativistic relation for the gravitational potential; in the Newtonian limit, it is simply the relation satisfied by a spherically symmetric potential and as a matter of fact by taking the integral of that equation one gets the expression for the gravitational potential of a spherically symmetric mass distribution.

In the general case, in addition to (1.1), (1.2), (1.3), equations of state $p = p(\rho)$ and $e = e(\rho)$ have to be supplied in order to solve the system of equations; by imposing boundary conditions (such as, for example, fixing the value of central rest-mass density ρ_c) one finds star configurations with different masses and radii. Therefore, for a given equation of state, it is possible to build a sequence of TOV stellar solutions, and obtain the gravitational mass as a function of the central density or the radius. Then, one can identify the maximum mass supported by that equation of state, or eventually the radius corresponding to a typical-mass neutron star ($1.4M_\odot$) and compare the results extracted from the model with astrophysical observation of neutron stars masses and radii, placing constraints on the EoS.

In a sequence of TOV stellar solutions, it is possible to distinguish two branches:

- a *stable* branch, corresponding to stable-equilibrium solutions: they oscillate around the equilibrium solution, when perturbed;
- an *unstable* branch, corresponding to unstable-equilibrium solutions: they can either collapse to a black hole or migrate to the stable branch, when perturbed.

Even in the more realistic case of *rotating* neutron stars there exist analogous sequence of equilibrium configurations corresponding to a given *angular frequency* Ω . It is important to note that, for a given value of the central density ρ_c , larger rotation frequencies allow to obtain equilibrium configurations with larger masses: this is due to the additional support to the hydrostatic equilibrium given by the centrifugal force. There is a maximum value for the angular frequency at which

a neutron star can spin before losing mass due to the centrifugal force, known as *mass-shedding frequency* or *Keplerian frequency* Ω_K . The maximum mass for a uniformly rotating model with $\Omega = \Omega_K$ is larger than the maximum mass for a non-rotating model ($\Omega = 0$), but is also the largest mass for uniformly rotating model. In particular, for any uniformly rotating model:

$$M_{\max}(\Omega = 0) < M_{\text{supra}} \leq M_{\max}(\Omega = \Omega_K), \quad (1.7)$$

which is referred to as *Supra-Massive Neutron Star* (SMNS), to indicate that it has a larger mass than the one allowed for a non-rotating star.

There also exists the possibility of *differential rotation*, i.e. the spin frequency is not a constant, but depends on the distance from the rotation axis. While old and cold neutron stars are expected to uniformly rotating, differential rotation could be present in proto-neutron stars. The neutron star will now have some fixed degree of differential rotation. In this case, one can build equilibrium models with larger masses with respect to uniformly rotating ones. In particular, it is possible to define a *Hyper-Massive Neutron Star* (HMNS), a differentially rotating neutron star with larger mass than the maximum allowed for a uniformly rotating neutron star:

$$M_{\text{hyper}} \geq M_{\max}(\Omega = \Omega_k). \quad (1.8)$$

Once the differential rotation is lost due to dissipative processes, the star has a mass which is too high to be supported by uniform rotation and it will inevitably collapse to black hole. These differentially rotating configurations are expected to be found frequently as remnant of binary neutron star mergers.

1.1.3 Neutron Star Structure and Equation of State

The state of matter (and its composition as well) inside a neutron star is still uncertain: a naive model which describes a neutron star as a degenerate neutron gas doesn't match the astrophysical observations of neutron-star masses and radii, and becomes more and more inadequate as one proceeds inward: as stated before, in the core regions the density of matter is comparable or even higher than the nuclear density and strong interactions are not negligible anymore. Moreover, in these regions also the composition of matter is not very clear: exotic forms of matter may exist, including strange quark matter (free up, down and strange quarks) or matter containing hyperons, high energy pions and kaons in addition to neutrons, but also quark degenerate matter.

Since the temperature of neutron stars in most cases is well below the *Fermi temperature* of constituents particles, the picture of a *cold* neutron star is a reasonable approximation of reality, since temperature effects are negligible. Therefore, in these cases, neutron star matter can be described by a *cold* equation of state, i.e. a no temperature-dependent relation between the pressure p and the baryon mass density ρ usually written in the form $p = p(\rho)$.

However, this is not an acceptable approximation for neutron stars in some specific situation, such as the merger of a binary neutron star system. Here, temperature effects must be taken into account. More in general, the equation of state will also depend on the *electronic fraction* Y_e (i.e. the ratio of electron number density to baryon number density), so that $p = p(\rho, T, Y_e)$. In this case, we talk about *hot* equations of state. In order to obtain an equation of state depending on the density only - which is required for instance to solve TOV equations - it is possible to *slice* the three-dimensional equation of state by fixing two conditions: usually, one imposes a β -*equilibrium* condition, and select a specific temperature.

An equation of state which adequately describes the state of matter inside a neutron star must take into account the different density of its four main internal layers: outer crust, inner crust, outer core and inner core. Just to give some orders of magnitude:

- at the outer crust-inner crust boundary, where neutrons start dripping out of the nuclei, the density is $\rho_{drip} \sim 4 \times 10^{11} \text{ g cm}^{-3}$;
- at the crust-core boundary, the matter density is comparable to the nuclear density $\rho_{nuc} \sim 2.3 \times 10^{14} \text{ g cm}^{-3}$;
- at the outer core-inner core boundary, the matter density is even larger than the nuclear density, with a value $\sim 6 \times 10^{14} \text{ g cm}^{-3}$;
- at the center, density can reach values $\rho_c \sim 10^{15} \text{ g cm}^{-3}$, an order of magnitude higher than the nuclear density.

By making assumptions on the composition, and considering all the interactions at stake, one can obtain *tabulated* equations of state; these EoS are not analytic functions; instead, numerical values for the pressure are obtained for specific values of the density. These tabulated EoS can be used, for example, to solve TOV equations for a neutron star, or to evolve a binary neutron star system, by means of computational methods. At this stage, a problem shows up: while the system is evolving, one may need values of the pressure which are not present in the table; interpolation is needed to obtain those missing values. This procedure can be realized by specific tools but it can be very expensive from a computational point of view; in addition, one has to ensure that no violations in the laws of thermodynamics occur. For this reason, tabulated (cold) EoS are sometimes approximated by *Piecewise Polytropic* EoS, described by a specific polytropic EoS ($p = K\rho^\Gamma$) for each density interval: the analyticity of this equation allows to avoid the high computational cost of interpolation.

1.2 Binary Neutron Stars

The study of *compact binary systems* containing neutron stars offers a unique opportunity for investigating the properties of these objects. In fact, many neutron stars are found in binary systems with another compact object - i.e. a white dwarf, a black hole or another neutron star.

First of all, strong gravitational fields are involved. Therefore, these systems represent a singular testing arena for the predictions of General Relativity, also because general-relativistic effects in binary orbits can be measured with great precision. Moreover, due to the presence of strong magnetic fields, they provide an excellent occasion to deepen our understanding of *General Relativistic Magnetohydrodynamics* (GRMHD).

The binary components are expected to lose energy due to the emission of gravitational radiation, causing the orbits to shrink. The first phase of the *inspiral* can take a very long time as the two objects are distant and gravitational radiation emitted is weak. However, when the two companions are close, the orbit shrinks rapidly, eventually leading to the *merger*.

Gravitational emission can give interesting information about the properties of neutron stars, including their mass, spin and equation of state. Also, a wide variety of electromagnetic signals across the entire spectrum accompanies gravitational emission: these *electromagnetic counterparts* to gravitational radiation provide powerful insights into important high-energy astrophysical phenomena such as *Short Gamma Ray Bursts* and give the chance to explore the origin of heavy elements in the local universe, through the radioactively powered *kilonova* transients. Many others research fields in physics and astrophysics can be probed by studying binary neutron star mergers, witnessing their extraordinary scientific potential.

1.2.1 Gravitational Waves

The Birth of Gravitational-Wave Astronomy

Gravitational waves are perturbations of the fabric of spacetime, propagating at the speed of light. The strongest gravitational waves are produced mostly by catastrophic events such as colliding black holes, colliding neutron stars or supernovae. They have been detected directly for the first time in 2015, confirming a prediction made by Einstein in 1916 [10].

Even though the existence of gravitational waves as a mathematical entity has been proved so early, doubts about their physical reality remained until 1974, when two astronomers - using the Arecibo Radio Observatory in Puerto Rico - discovered the first *binary pulsar* (PSR 1913+16), whose companion is also a neutron star. According to General Relativity, this kind of system was expected to emit gravitational radiation, leading to an *orbital decay*, i.e. a gradual decrease of the distance between the two orbiting bodies. Since then, the orbits have shrunk in remarkable

agreement with the loss of energy due to gravitational emission predicted by General Relativity, providing the first indirect confirmation of the physical existence of gravitational radiation, before the aforementioned first direct detection in 2015.

In particular, on September 14, 2015, the Laser Interferometer Gravitational Wave Observatory (LIGO) observed a gravitational signal originating from a binary black-hole system. Since then, several other signals from merging binary black holes have been detected, as well as gravitational radiation emitted by merging binary neutron stars. Gravitational waves from binary systems containing a black hole and a neutron star could be also revealed in the next years of observations.

Gravitational Waves from Inspiring Binaries

By definition, gravitational waves are perturbations of spacetime - i.e. the metric - which propagate at speed of light. From a mathematical point of view, they must be wave-like solutions of some wave equation. In fact, this emerges naturally from Einstein's equations.

Consider a small perturbation of a background metric; in particular, assume that our background metric is the flat Minkowski spacetime. The perturbed metric can be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad (1.9)$$

where $\eta_{\mu\nu}$ is Minkowski metric and $h_{\mu\nu}$ is a small perturbation ($|h_{\mu\nu}| \ll 1$). This is all we need to compute the Einstein tensor. Dropping all the terms which of higher order than linear in $h_{\mu\nu}$, imposing certain gauge conditions and inserting the result for the Einstein tensor into Einstein's equations, we get the linearized form of field equations for weak gravitational field:

$$\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu}, \quad (1.10)$$

where $\square \equiv \eta^{\mu\nu} \partial_\mu \partial_\nu$ is the D'Alembertian, $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \eta_{\mu\nu} h/2$ is the *trace-reversed* metric perturbation with $h \equiv \eta^{\mu\nu} h_{\mu\nu}$ being the trace of $h_{\mu\nu}$ and $T_{\mu\nu}$ is the stress-energy tensor. It is clear that Einstein's equations reduce to a wave equation for the perturbation $h_{\mu\nu}$, whose solutions are waves propagating at speed of light, i.e. *gravitational waves*.

In particular, the gravitational emission of compact binary mergers can be modeled using a variety of techniques, depending on the regime we are exploring. For small orbital velocities and large separations between the binary companions, a natural tool is *Post-Newtonian* approximation of General Relativity, which incorporate general-relativistic corrections to the Newtonian solutions. These techniques are adequate for describing the *early* inspiral. It can be shown that, to leading order in expression for the gravitational wave frequency of an inspiralling binary as a function of time is determined primarily by a specific combination of the binary masses m_1 and m_2 , known as the *chirp mass*

$$\mathcal{M} = (m_1 m_2)^{\frac{3}{5}} (m_1 + m_2)^{-\frac{1}{5}}. \quad (1.11)$$

The frequency of gravitational waves as a function of the chirp mass is

$$\nu_{GW} = \frac{5^{3/8}}{8\pi} \frac{1}{\mathcal{M}^{5/8}(T-t)^{3/8}}, \quad (1.12)$$

where T is the *coalescence time*. By re-inserting the constants c and G , and expressing the chirp mass in terms of the solar mass $M_\odot \simeq 2 \times 10^{33}$ g, one obtains for (1.12)

$$\begin{aligned} \nu_{GW} &= \frac{5^{3/8}}{8\pi} \left(\frac{c^3}{GM_\odot} \right)^{5/8} \left(\frac{M_\odot}{\mathcal{M}} \right)^{5/8} \frac{1}{(T-t)^{3/8}} \\ &\simeq 155.7 \text{ Hz} \left(\frac{M_\odot}{\mathcal{M}} \right)^{5/8} \left(\frac{1 \text{ s}}{T-t} \right)^{3/8} \end{aligned} \quad (1.13)$$

For binary neutron stars having chirp masses of order of a solar mass, current detectors can observe the late inspiral but not the merger itself.

The chirp mass in terms of the frequency of the GW signal ν and its time derivative $\dot{\nu}$ reads

$$\mathcal{M} = \left(\frac{5}{96} \pi^{-8/3} \nu^{-11/3} \dot{\nu} \right)^{3/5}. \quad (1.14)$$

The magnitude of the metric perturbation is approximately

$$|h| \simeq \frac{4\mu R^2 \Omega^2}{d}, \quad (1.15)$$

where d is the distance from the source of gravitational waves, $\mu \equiv \frac{m_1 m_2}{m_1 + m_2}$ is the *reduced mass* and R is the binary separation.

The previous formulae show that both frequency and amplitude of gravitational waves emitted increase during the inspiral. However, to obtain these results one has to make the assumption that inspiral is *adiabatic*, i.e. a sequence of circular orbits in which the orbital radius change on much longer timescales with respect to the orbital period. This adiabatic approximation is valid below some *critical frequency*, which indicates the transition from inspiral to plunge and merger. Note that higher-order post-Newtonian approximations can be used to improve the previous estimates for smaller separations and higher frequencies. However, these completely break down when the two objects merge: here one has to rely on the tools of *Numerical Relativity*.

The characteristics of gravitational signals emitted are very sensitive to the nature of the binary system, but also to the properties of single components. For instance, consider the case of binary systems of neutron stars. The equation of state has an important impact on the gravitational signal: the binary companions exhibit tidal interactions during the late inspiral, whose effects leave an EoS signature on the gravitational waveform. In this context, it is important to introduce a specific parameter, called *tidal deformability*, which roughly measures the degree of a body deformation due to tidal forces, defined as

$$\Lambda \equiv \frac{2}{3} k_2 \left(\frac{R}{M} \right)^5, \quad (1.16)$$

where R and M are respectively the radius and the mass of the body, and k_2 is a parameter called *Love number*.

In binary neutron star systems of masses m_1 and m_2 , one defines a combination of tidal deformabilities of the two stars as

$$\tilde{\Lambda} = \frac{16}{13} \frac{(m_1 + 12 m_2) m_1^4 \Lambda_1 + (m_2 + 12 m_1) m_2^4 \Lambda_2}{(m_1 + m_2)^5}, \quad (1.17)$$

where Λ_1 and Λ_2 are tidal deformabilities of the components. This parameter is directly related to the gravitational wave frequency at merger. Therefore, by measuring this frequency, one obtains $\tilde{\Lambda}$ and places constraints on the equation of state.

Not only the EoS affects the gravitational signal of the inspiral. Other parameters such as *mass ratio*, *spin* and *eccentricity* can also leave a specific mark on the gravitational waveform.

In addition, precious information can be obtained by the gravitational wave signal in the *post-merger* phase. For example, it has been proved by a large number of numerical simulation that there exists a strong correlation between the *frequency peak* in the post-merger - i.e. the post-merger frequency with the largest amplitude - and properties of the neutron stars which can be related to the equation of state. Also, other factors which may have an impact on the post-merger gravitational signal are *magnetic fields* and neutrinos.

1.2.2 Electromagnetic Counterparts to Gravitational Waves

Two kinds of electromagnetic emissions are relevant in the phenomenology of compact binary mergers involving neutron stars:

- *Short Gamma Ray Bursts* (SGRB), extremely powerful and collimated electromagnetic emissions in the gamma-ray region of the electromagnetic spectrum, with a time duration of less than two seconds and powered by highly-relativistic *jets* launched by the merger remnant (in particular, by the accretion disk surrounding the black hole remnant);
- *kilonovae*, UV/optical/IR transient astronomical processes powered by the radioactive decay of r-process elements produced and ejected during the merger process.

In relation to gamma-ray bursts, there can be also electromagnetic signals called *precursors*, which are produced before the main GRB event, and the *afterglow* emission, from X-ray to radio, produced when the highly relativistic jets slow down due to the interaction with the external medium.

Short Gamma Ray Bursts

Before illustrating how SGRBs can be related to coalescences of binary systems involving neutron stars, it is convenient to recall some information about the nature of Gamma Ray Bursts.

Gamma Ray Bursts (GRB) are bursts of extremely energetic gamma rays which are thought to be the most powerful explosions in the Universe. They were serendipitously discovered in the late 1960s by U.S. Vela satellites, built to detect gamma radiation emitted by nuclear weapons; further investigations ruled out a terrestrial or solar origin. Originally, many theories placed the source of these powerful phenomena inside our galaxy; however, from 1991, Compton Gamma Ray Observatory (CGRO) and its Burst And Transient Source Explorer (BATSE) - an extremely sensitive gamma detector - found that the distribution of GRBs was isotropic, with no concentration near the galactic plane. Later, precise measurements of distances and energy involved confirmed that sources were not galactic but of cosmological origin.

GRBs come in two categories, depending on their duration. Indeed, the distribution of these events in terms of duration is bimodal. In this context, one introduces a variable known as T_{90} , defined as the time interval during which 90% of the total counts have been detected (it starts when 5% of the counts have been detected, and ends when 95% of the counts have been detected). In particular, GRBs can be classified into:

- Long GRB ($T_{90} > 2\text{ s}$);
- Short GRB ($T_{90} < 2\text{ s}$).

This classification is also supported by the hardness ratio - i.e., the ratio between photon counts in 100-300 keV channel and 50-100 keV channel - showing that SGRBs are harder than LGRBs.

The existence of two distinct types of Gamma Ray Bursts (LGRBs and SGRBs) suggests that they arise from different progenitors and have different observational properties, besides their duration. In particular:

- LGRBs are associated with the core-collapse of short-lived massive stars or *collapsar* (> 30 solar masses) in *hypernova* explosions (very similar to type Ic supernova but with broad spectral lines) and are typically found in galaxies with active star formation populated by young stars. Their observed rate is ~ 500 per year. Other important observational properties are isotropic energy $E_{iso} \sim 10^{53}$ erg, median redshift $z \sim 2$ and a cone opening angle $\theta_j \sim 3^\circ - 5^\circ$.
- SGRBs are associated with compact binary mergers of neutron stars: binary neutron star (BNS) or a stellar-mass black hole and a neutron star (BH-NS); they are found in galaxies with and without active star formation, with offsets between the location of the burst and host galaxy. This is coherent with the

fact that such binary systems receive a kick from the supernovae which originated the compact objects, causing them to be in the peripheral regions or even outside the host galaxy. Their observed rate is ~ 170 per year. SGRBs have an isotropic energy $E_{iso} \sim 10^{50}$ erg, median redshift $z \sim 0.5$ and a cone opening angle $\theta_j \sim 16^\circ \pm 10^\circ$.

Let's now focus on SGRBs and their association with compact binary mergers involving neutron stars. As previously mentioned, they are powered by highly-relativistic jets launched by the accretion disk of the black hole remnant: therefore, mainly the BNS and BH-NS channels which lead to the formation of a black hole surrounded by an accretion disk can potentially emit SGRBs. On top of that, the mass of the accretion disk has to be sufficiently large, so that the duration of the accretion process is long enough for a SGRB to be emitted.

In the NS-NS case, it's not difficult to produce a massive accretion disk. By keeping constant the mass ratio between the two NS, the mass of the accretion disk gets smaller as the total mass of the BNS increases; by keeping constant the total mass of the BNS and decreasing the mass ratio, the mass of the accretion disk increases because the less compact NS gets easily tidally disrupted, and it is more likely that an accretion disk will form. However, if the BNS system has very high mass, it could collapse directly to a black hole without forming an accretion disk.

In the BH-NS case, the formation of a sufficiently massive accretion disk strongly depends on the type of BH-NS event. In particular, a massive accretion disk can be formed only in the case of tidal disruption of the neutron star by the black hole, and this of course depends on the mass ratio between the BH and the NS, the spin of the black hole and the compactness of the neutron star. However, astrophysical models show that the most likely mass ratio is $M_{BH} \simeq 7 M_{NS}$; in this case an accretion disk will form if the black hole has very high spin, because in this case the ISCO - i.e. the innermost stable circular orbit - radius would be small and the neutron star would be tidally disrupted outside the ISCO (leading to the formation of the accretion disk).

Once this post-merger configuration (black hole surrounded by an accretion disk) has been formed, one can ask whether the relativistic jet will be launched or not. Numerical simulations have focused in particular on magnetically driven relativistic jets. Since neutron stars are magnetized, the accretion disk formed after a BNS or a BH-NS merger will be magnetized as well; these magnetized accretion disks can extract rotational energy from the magnetically dominated regions of the ergosphere of the rotating black hole and convert it to kinetic energy for particles, accelerating them and producing a relativistic jet (Blandford-Znajek mechanism). This is the most efficient process to produce relativistic jets, but potentially not the only one: another possible mechanism - way less efficient - is due to the annihilation process of neutrino-antineutrino pairs (emitted from the accretion disk) in electron-positron pairs: a jet produced in that way, however, would not be powerful enough to launch a SGRB.

Actually, there could be another interesting channel for SGRB production in the case of BNS mergers. There are observational evidences which suggest that the configuration where a black hole is surrounded by an accretion disk may not be the only source of SGRB power (the so-called *central engine*): in fact, recent analysis of X-ray afterglows in SGRB detections - especially by Swift mission - have shown evidence for energy injection; the X-ray afterglow emission deviates from the power-law decay predicted by the standard afterglow model and a flattening of X-ray light curves - known as *X-ray plateau* - is clearly observed, indicating that the SGRB energy source has to be active on a longer timescale with respect to the prompt emission. A plausible interpretation of this plateau emission is provided by the so-called *magnetar model* for SGRB: the remnant of the BNS merger could not collapse immediately to a black hole surrounded by a massive accretion disk (or even it couldn't collapse at all); instead, a long-lived *magnetar* - i.e., a neutron star with extremely powerful magnetic fields (up to 10^{16} G) - could form, which is expected to lose rotational energy emitting relativistic winds on timescales comparable to those observed, powering the X-ray plateau phase in light curves. Moreover, a statistical analysis based on the mass distribution of NS binaries in our galaxy (assuming that this population is representative for the entire population of NS binaries in the universe) predicts that for many reasonable equations of state, an important fraction of the remnants are neutron stars or SMNS [11]. This analysis supports the idea of a long-lived magnetar as central engine for SGRBs; however, this scenario has been investigated by recent numerical simulations with not very promising results: the properties of the collimated outflow (for example, the low Lorentz factor) are found to be inconsistent with the emission of a SGRB [12] and - although there is not a definitive answer - the most promising scenario is still the one with a black hole surrounded by an accretion disk as central engine.

Kilonova

Kilonova is the other kind of electromagnetic emission which occurs in BNS mergers or in BH-NS mergers. Unlike GRBs, this emission is not collimated and therefore easier to measure.

In the case of a BNS merger, the less compact neutron star can be tidally disrupted by its NS companion and a significant amount of matter can be ejected; the same happens in a BH-NS event where the neutron star is tidally disrupted by its black hole companion: part of the neutron star matter will form the massive accretion disk surrounding the black hole remnant and part will be ejected. In the case of a BNS merger, due to shock-waves generated during the collision, further matter will be expelled. Moreover, baryon-loaded winds can be launched by the metastable neutron star remnant or by the accretion disk surrounding the black hole remnant.

These matter ejecta are neutron-rich and undergo rapid neutron capture (r-process) nucleosynthesis, producing certain amounts of heavy elements (i.e., elements heavier than iron). On timescales of ~ 1 day, radioactive decay of unstable nuclei powers the approximately isotropic electromagnetic transient (UV, optical,

IR) known as *kilonova*. Later on, on timescales of years, interaction between matter ejected and interstellar medium produce a radio emission.

Depending on the location of the peak, one identifies two kilonova components: a blue (optical) and a red (infrared) component. In particular, blue kilonova is produced by matter ejecta which are not very neutron-rich (with electronic fractions $Y_e \gtrsim 0.25$) while more neutron-rich matter ejecta (with electronic fractions $Y_e \lesssim 0.25$) lead to the red kilonova; for a given ejecta component, the amount of neutrons is strongly determined by the ejection mechanism.

Consider, for example, the case of a BNS merger. As said before, a large amount of material can be ejected by dynamical mechanisms during the merger process (*shock-driven ejecta*). In the case of black hole formation, a short-living HMNS is formed after the merger: this object - before collapsing to a black hole - emits a large amount of neutrinos which radiate the dynamical ejecta, interacting with neutrons and producing protons and electrons (Y_e increases); the amount of neutrons decreases and elements up to atomic mass number $A \lesssim 140$ can be formed: this ejecta component will give rise to the blue kilonova. When the black hole is formed, matter can be ejected by the accretion disk (*disk winds*) but this component remains more neutron-rich and allows for the production of elements with $A \gtrsim 140$; this will give rise to the red kilonova; the so-called *tidal tail* produced before the merger contributes to the red component as well.

The analysis of BH-NS kilonovae emissions is similar but the intermediate phase of HMNS formation is absent and therefore one expects a red component only. Instead, if one consider the case of a long-lived central engine in a BNS merger, the same ejected components are present but the remnant neutron star keeps irradiating neutrinos, raising Y_e : the blue component will be predominant here.

Electromagnetic Precursors of SGRBs

A gamma-ray event in GRBs can be anticipated by a less powerful emission, called *precursor*. A systematic search of these signals have been carried out [13] on a sample of SGRBs observed by Swift/Burst Alert Telescope (BAT). Sometimes ($\sim 8 - 10\%$ of SGRBs), light-curves showed precursor activity, with a spike observed a few seconds before the main GRB. Different theoretical models have been proposed in the last years, and attempts have been made to figure out whether BNS and BH-NS systems can produce electromagnetic emissions before the merger.

In general, numerical simulations have proved that there can be mechanisms producing such emissions: for example, in a BNS, the interaction between stellar magnetospheres can power radiative Poynting flux [14]; also in BH-NS promising precursor emissions - due to the interaction between the stellar magnetosphere and the black hole - have been found in numerical simulations [15].

In the last years, however, the attention has been mainly focused on electromagnetic signals emitted during or after the merger.

1.2.3 Main Observations

GW170817 and Electromagnetic Counterparts

On 2017 August 17, the first gravitational wave signal consistent with a binary neutron star merger - known as GW170817 - has been observed by the Advanced LIGO and Advanced Virgo detectors. The combination of LIGO and Virgo allowed to localize the position of the source within a sky region of 28 deg^2 (with 90% probability). Moreover, a gamma ray burst - known as GRB170817A - was observed 1.7 s after the coalescence by Fermi Gamma-Ray Burst Monitor and INTEGRAL; they indicated an area of the sky which overlapped the one identified by gravitational-wave detectors. This enabled a campaign which identified an electromagnetic counterpart near the galaxy NGC 4993, consistent also with the distance obtained by gravitational-wave data (the source had a luminosity distance of $40_{-14}^{+8} \text{ Mpc}$) and compatible with the predicted kilonova model.

The joint observation of a gravitational-wave signal and a gamma ray burst, with subsequent identifications of electromagnetic counterparts across the EM spectrum corroborated the interpretation of this event as a binary neutron star merger. This first combined detection of gravitational waves and electromagnetic counterparts marked an important breakthrough for the so-called *Multimessenger Astronomy* and allowed to gain insight into astrophysics, nuclear physics, gravitation and also cosmology. For example, from the near simultaneous detection of GW170817 and GRB170817A, the speed of gravitational waves was found to be equal to the speed of light to one part in 10^{15} : this ruled out a class of modified theories of gravity and confirmed a fundamental prediction of General Relativity. A relevant result for cosmology was the measurement of the Hubble parameter H_0 : combining the redshift measurement using the association with galaxy NGC 4993 and the luminosity distance inferred by GW signal, it has been possible to obtain $H_0 = 70_{-8}^{+12} \text{ km s}^{-1} \text{ Mpc}^{-1}$; uncertainty in the result is quite large so it overlaps both the Cosmic Microwave Background (CMB) and supernovae values and that is not enough to solve the crisis in cosmology, i.e. divergent measurements of H_0 based on two independent methods (CMB and supernovae).

Inspections on LIGO data have revealed the presence of the characteristic chirp signal in the time-frequency representation of the GW signal: as neutron stars come closer, they produce higher frequency gravitational waves. From the GW signal, the chirp mass of the system has been precisely measured, giving the result $\mathcal{M} = 1.188_{-0.002}^{+0.004} M_\odot$. While the chirp mass is well established, the component masses depend on assumptions made on the values of spins. Since the fastest-spinning NS within a BNS system has dimensionless spin ≈ 0.02 , it is reasonable to restrict the analysis to $|\chi| \leq 0.05$ (low-spin prior scenario). In this case, one recovers the mass ratio $q \in (0.7, 1.0)$ with component masses $m_1 \in (1.36, 1.60) M_\odot$ and $m_2 \in (1.17, 1.36) M_\odot$ and a total mass $M = 2.74_{-0.01}^{+0.04} M_\odot$.

Recalling (1.17), an upper bound for the dimensionless tidal deformability parameter $\tilde{\Lambda}$ has been obtained, ruling out equations of state which produce less com-

compact neutron stars. When spins are considered to be within the range observed in galactic binaries, it is possible to establish a lower bound for this parameter, excluding the possibility of $\tilde{\Lambda} = 0$, as would occur in a (low-mass) binary black hole merger.

Therefore, the observation of GW170817 has provided the possibility to put further constraints on the NS EoS: in particular, the stiffest EoSs (which, for a fixed mass, produce larger radii and hence less compact stars) have been excluded, in favour of soft EoSs (which produce more compact stars).

Shifting the focus on electromagnetic counterparts, the observation of gamma ray burst GRB170817A 1.7 s after the merger has provided the first direct evidence of the link between BNS mergers and SGRBs.

The gamma ray pulse was much closer than any other previous SGRB and way less energetic, with an isotropic energy of the order of 10^{46} erg. Within a day of the GW trigger, early UV, optical and IR have been detected, with a quasi-thermal spectrum consistent with a kilonova transient. However, the source has been detected in X-rays 9 days later the GW event by Chandra, and the radio emission a few days later.

Different scenarios have been explored in order to explain the observational properties of GRB170817A.

- a) A jet seen on-axis, generating the gamma rays and the afterglow. It cannot explain the late X-ray emission, and within this scenario, the SGRB should be way more energetic with respect to the one observed. Therefore, it has been ruled out.
- b) A jet seen off-axis. It could explain the energetics associated to the event, but it's not able to provide a valid explanation for the afterglow, in particular for the radio emission and therefore has been ruled out as well.
- c) The jet may be choked inside the ejecta but a mildly relativistic cocoon ($\gamma \sim 2 - 3$) eventually breaks out, generating gamma rays and the afterglow. Although this was considered one of the most promising scenarios [16], the model doesn't fit the observations across the electromagnetic spectrum and has been discarded.
- d) A similar scenario to the previous one, where the jet is choked, and a slower outflow ($\gamma \sim 1.25 - 1.67$) arise from the fast tail of the merger ejecta. This scenario is not able to explain the emission in gamma rays and has been rejected.
- e) A successful jet - i.e. a jet breaking out the merger ejecta - driving a cocoon, observed off-axis ($\sim 15^\circ - 20^\circ$) and with a semi-opening angle of $\theta_{jet} \sim 5^\circ$. This scenario can also be visualized as a structured jet, with a relativistic narrow core surrounded by a mildly-relativistic outflow. Studies involving numerical simulations (see for example [17]) have shown that this scenario could

produce electromagnetic emissions which match the observational properties of GRB170817A and its afterglow: thus, it is considered the most likely one. Within the model of a structured jet, the delay between the GWs and gamma ray emission (1.7 s after the merger) is due to the time required by the jet to reach the photospheric radius [18].

The UV, optical and near-IR emissions were consistent with being powered by the radioactive decay of nuclei produced in the merger ejecta by the r-process. The UV-optical-NIR transient, first detected 11 hours after the merger, evolved from blue colors with a peak at optical wavelengths to redder colors, with a peak at $\sim 1.5 \mu\text{m}$ on timescales of several days to week. The emissions were in agreement with thermal emission expected in current models of kilonovae. In particular, the photometric and spectral evolutions are well described by a multi-component (red + blue) kilonova. The general picture is the one where a BH is surrounded by the accretion disk is formed after a HMNS intermediate phase: the blue and the red components of kilonova have different origins (dynamical ejecta for the blue kilonova, accretion disk outflows for the red kilonova). Light r-process nuclei (e.g. xenon and silver) are associated to the blue component while heavy r-process nuclei are associated to the red component (e.g. gold and uranium): it is estimated that ~ 200 Earth masses of pure gold have been produced from this neutron star merger.

GW190425

On 2019 April 25, the LIGO Livingston detector observed a compact binary merger; this was the second gravitational-wave signal consistent with a BNS inspiral, known under the name of GW190425.

The 90% credible intervals for the binary masses range from 1.12 to $2.52 M_{\odot}$ ($1.46 - 1.87 M_{\odot}$ with the low-spin prior $\chi \leq 0.05$). However, if the source of the GW signal is a BNS system, it is significantly larger than any other known, with total mass $M_{\text{tot}} = 3.4^{+0.003}_{-0.001} M_{\odot}$ and chirp mass $\mathcal{M} = 1.44^{+0.002}_{-0.002} M_{\odot}$.

The possibility that this signal was originated by a black hole-neutron star system could not be discarded in principle. In fact, in that case, the inferred masses are not in opposition to the possibility that the lighter component was a neutron star and the other one a low-mass black hole. Nevertheless, studies have shown that the kilonova associated to the BH-NS event would have been sufficiently bright to be detected compared to the one produced by a BNS system having those masses ([19] and [20]); this is coherent with absence of electromagnetic counterparts identified in association with this gravitational-wave event, even if this is also due to a bad sky localization (only one detector observed the GW event) and a larger distance compared to GW170817.

Chapter 2

Numerical Relativity

Describing the dynamics of binary neutron stars is a difficult task. First of all, since extremely strong gravitational fields are involved, we are in the regime of General Relativity, so that spacetime background evolves according to Einstein's equations

$$G_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (2.1)$$

Tensor equation (2.1) tells how the spacetime geometry described by the metric $g_{\mu\nu}$, which appears in the left-hand side of the equations contained in the Einstein tensor $G_{\mu\nu}$ evolves in the presence of matter and energy, whose distribution is encoded in the stress-energy tensor $T_{\mu\nu}$ in the right-hand side of the equation. We recall that, the Einstein tensor appearing on the left-hand side of the equation (2.1) is defined

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R, \quad (2.2)$$

where $R_{\mu\nu}$ is the Ricci curvature tensor and R is the Ricci scalar.

Einstein's field equations contain only four-dimensional quantities, with space and time on the same footing. Despite the elegance of the covariant formulation of General Relativity, it is not advantageous from a numerical point of view. Indeed, in numerical simulations, one has to solve initial value problems: from certain initial values, the physical quantities of interest has to be evolved as time advances. So, in practice, it is convenient to keep space and time separate, splitting the spacetime into three space coordinates and one time coordinate. Performing this procedure in the context of General Relativity gives rise to the *3+1 formulation of General Relativity*, which is the theoretical background of Numerical Relativity.

However, in the specific case of binary neutron stars, evolving the background metric is not the whole story. In fact, solving (2.1) requires the knowledge of $T_{\mu\nu}$ as well. Since the dynamics of neutron star matter can be well-described by a fluid in a general relativistic regime, Einstein's equations have to be coupled with the equations of relativistic hydrodynamics, known as *relativistic Euler equations*

$$\nabla_\mu(\rho u^\mu) = 0, \quad (2.3)$$

$$\nabla_\mu T^{\mu\nu} = 0, \quad (2.4)$$

where in equation (2.3) the variable ρ represents the rest-mass density and u^μ is the fluid four-velocity. While (2.3) expresses the conservation of the number of particles, or equivalently the conservation of baryonic mass, (2.4) is a continuity equation for the stress-energy tensor. In addition to that, an equation of state for the neutron star matter must be provided, establishing a relation between pressure and density, temperature and, in the more general case, the electron fraction, which can be expressed in the form

$$p = p(\rho, T, Y_e). \quad (2.5)$$

Another important aspect to be considered is that most neutron stars are endowed with strong magnetic fields. Moreover, their strength could undergo severe amplification after a binary merger. For these reason, we should include in our description Maxwell's equations for the evolution of electromagnetic fields. In covariant form, they read

$$\nabla_\nu F^{\mu\nu} = J^\mu, \quad (2.6)$$

$$\nabla_\nu {}^*F^{\mu\nu} = 0. \quad (2.7)$$

where $F_{\mu\nu}$ is the electromagnetic tensor, ${}^*F_{\mu\nu}$ its dual, and J^μ the electromagnetic four-current.

Also, emission and absorption of neutrinos have an impact, especially if we are interested in the composition properties of the merger ejecta, affecting its electron fraction Y_e and hence determining the properties of the kilonova emission.

More details about the topics presented in this chapter can be found in the books [21], [22], [23]. For Valencia formulation of General Relativistic Hydrodynamics and Magnetohydrodynamics see [24] and [25].

2.1 The 3 + 1 Decomposition of Spacetime

The starting point of 3 + 1 formulation is the so-called *foliation* of spacetime: one considers constant time hypersurfaces of spacetime - i.e., three dimensional spacelike level surfaces of constant time t . These hypersurfaces are also known as *time slices*. At this point, we introduce the *normal* n_μ as proportional to the gradient of t

$$n_\mu = -\alpha \nabla_\mu t \quad (2.8)$$

and this gives

$$n_\mu = (-\alpha, 0, 0, 0). \quad (2.9)$$

The quantity α - known as *lapse* - is in general a function of spacetime coordinates and will be an elementary unit of the 3 + 1 metric.

The *normal vector* n^μ is timelike by the assumption that our hypersurfaces are spacelike. This four-vector can be associated to the four-velocity of a special class

of observers, called *normal observers* or *Eulerian observers*. Being n^μ a timelike four-vector, if we require that it represents a four-velocity, the condition

$$n_\mu n^\mu = -1 \quad (2.10)$$

must hold¹. This is automatically satisfied by

$$n^\mu = \frac{1}{\alpha}(1, -\beta^i) \quad (2.11)$$

for any spatial vector β^i . The vector β^i - in general a function of spacetime coordinates - is called *shift* and will be a building block of the 3 + 1 metric along with the lapse.

Note that a normal observer moves normal to the spacelike hypersurfaces. We also say that the normal observer's *worldline* is generated by the normal vector n^μ . It is easy to show that the increment of the proper time $d\tau$ measured by the normal observer corresponds to αdt as the time coordinate moves forward by dt , i.e. moving from the hypersurface Σ_t to Σ_{t+dt} . In particular, this reads

$$d\tau = \alpha dt, \quad (2.12)$$

so that the lapse measures the proper time elapsed according to a normal observer moving from Σ_t to Σ_{t+dt} . Using a similar argument, it can be shown that a normal observer measures a change in the spatial coordinates

$$dx^i = -\beta^i dt \quad (2.13)$$

as a proper time $d\tau$ has elapsed, i.e. moving from Σ_t to Σ_{t+dt} . In this context, one defines a *coordinate observer* as an observer whose spatial coordinates remain unchanged, moving on a worldline whose tangent four-vector is given by

$$t^\mu = \alpha n^\mu + \beta^\mu = (1, 0, 0, 0), \quad (2.14)$$

where the time component of β^μ is zero, and the spatial components are β^i .

Having introduced n_μ and n^μ , it is now possible to introduce the metric and its inverse on each hypersurface, which respectively take the form

$$\gamma_{\mu\nu} \equiv g_{\mu\nu} + n_\mu n_\nu, \quad \gamma^{\mu\nu} \equiv g^{\mu\nu} + n^\mu n^\nu, \quad (2.15)$$

where $\gamma^{0\mu} = 0$, $\gamma_{ij} = g_{ij}$ but $\gamma^{ij} \neq g^{ij}$ in general. Note also that $\gamma^{ik}\gamma_{kj} = \delta^i_j$ so that the metric can be used for raising and lowering indices of purely spatial vectors or tensors. It is also convenient to define a *spatial projection operator* as

$$\gamma^\mu_\nu \equiv \delta^\mu_\nu + n^\mu n_\nu \quad (2.16)$$

and a *time projection operator*

$$N^\mu_\nu \equiv -n^\mu n_\nu. \quad (2.17)$$

¹Note also that the choice $\alpha > 0$ makes the normal four-vector n^μ future-oriented.

Therefore, a generic four-vector U^μ can be decomposed in a spatial part and in a time part as

$$U^\mu = \gamma^\mu_\nu U^\nu + N^\mu_\nu U^\nu, \quad (2.18)$$

but the same could be done for tensors of any rank. This procedure is necessary to build a suitable formalism for numerical purposes.

The four-dimensional line element can be written as

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt). \quad (2.19)$$

The component of the metric and its inverse can be written explicitly as

$$g_{\mu\nu} = \begin{pmatrix} -\alpha^2 + \beta_i \beta^i & \beta_i \\ \beta_i & \gamma_{ij} \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} -1/\alpha^2 & \beta^i/\alpha^2 \\ \beta^i/\alpha^2 & \gamma^{ij} - \beta^i \beta^j/\alpha^2 \end{pmatrix} \quad (2.20)$$

From these explicit expressions, it is easy to get the relation

$$\sqrt{-g} = \alpha \sqrt{\gamma}, \quad (2.21)$$

where $g \equiv \det(g_{\mu\nu})$ and $\gamma \equiv \det(\gamma_{ij})$.

It is now time to introduce some useful relations which are ubiquitous in the 3+1 formulations of relativistic hydrodynamics. For this purpose, consider a fluid with four-velocity components u^μ . The spatial four-velocity measured by an Eulerian observer has contravariant components

$$v^t = 0, \quad v^i = \frac{\gamma^i_\mu u^\mu}{-n_\mu u^\mu} = \frac{1}{\alpha} \left(\frac{u^i}{u^t} + \beta^i \right), \quad (2.22)$$

and covariant components

$$v_t = \beta_i v^i, \quad v_i = \frac{\gamma_{i\mu} u^\mu}{-n_\mu u^\mu} = \frac{u^i}{\alpha u^t} = \frac{\gamma_{ij}}{\alpha} \left(\frac{u^j}{u^t} + \beta^j \right), \quad (2.23)$$

where the relation $-n_\mu u^\mu = \alpha u^t$ has been used. Being u^μ a four-velocity, the condition $u^\mu u_\mu = -1$ holds. Hence, by indicating the *Lorentz factor* as W one has

$$\alpha u^t = -n_\mu u^\mu = \frac{1}{\sqrt{1 - v_i v^i}} = W, \quad u_t = W(-\alpha + \beta_i v^i), \quad (2.24)$$

so that (2.22)-(2.23) finally become

$$v^i = \frac{u^i}{W} + \frac{\beta^i}{\alpha}, \quad v_i = \frac{u_i}{W}. \quad (2.25)$$

These expressions reduce to usual special-relativity relations as $\alpha = 1$ and $\beta = 0$. It is also possible to write the fluid four-velocity as sum of a spatial and temporal part:

$$u^\mu = W(n^\mu + v^\mu). \quad (2.26)$$

At this point, we are ready to present the main formulations of Numerical Relativity.

2.2 Spacetime Evolution

2.2.1 ADM formalism

An approach to General Relativity based on 3 + 1 formulation has been first introduced by Arnowitt, Deser & Misner in 1962 (*ADM formulation*). Having previously introduced the basic quantities of a generic 3+1 spacetime splitting, the next step is the introduction of three-dimensional counterparts of all four-dimensional geometrical quantities involved in Einstein's field equations.

Indeed, one defines - through the spatial projection operator - the three-dimensional covariant derivative D_α , connection $^{(3)}\Gamma_{\beta\gamma}^\alpha$, Riemann tensor $^{(3)}R_{\alpha\beta\sigma}^\tau$ (with Ricci tensor $^{(3)}R_\beta^\alpha$ and scalar $^{(3)}R$ as well) which are the spatial analogues of the same four-dimensional quantities. Moreover, a geometrical quantity called *extrinsic curvature*, which is defined as

$$K_{\alpha\beta} \equiv -\gamma_\alpha^\sigma \gamma_\beta^\tau \nabla_\sigma n_\tau, \quad (2.27)$$

measures how much a hypersurface is warped and can be related to the Lie derivative along n^μ through the relation

$$K_{\alpha\beta} = -\frac{1}{2} \mathcal{L}_n \gamma_{\alpha\beta}. \quad (2.28)$$

Using properties of the Lie derivative and recalling the relation (2.14) with $t^\mu = (1, 0, 0, 0)$ and $\beta^\mu = (0, \beta^i)$, one can also write

$$\mathcal{L}_n = \frac{1}{\alpha} (\partial_t - \mathcal{L}_\beta) \quad (2.29)$$

where $\mathcal{L}_\beta$ is the Lie derivative along β^μ . At this point, it is easy to show that equation (2.28) becomes

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + D_i \beta_j + D_j \beta_i. \quad (2.30)$$

An evolution equation for the extrinsic curvature is then obtained combining the Ricci equation, the usual relation (2.29) and Einstein's equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$:

$$\begin{aligned} \partial_t K_{ij} = & -D_i D_j \alpha + (\beta^k D_k K_{ij} + K_{ik} D_j \beta^k + K_{kj} D_i \beta^k) + \\ & \alpha ({}^{(3)}R_{ij} + K K_{ij} - 2K_{ik} K_j^k) + 4\pi \alpha [\gamma_{ij} (S - E) - 2S_{ij}] \end{aligned} \quad (2.31)$$

where $S_{\mu\nu} \equiv \gamma_\mu^\alpha \gamma_\nu^\beta T_{\alpha\beta}$, $S_\mu \equiv -\gamma_\mu^\alpha n^\beta T_{\alpha\beta}$, $S \equiv S_\mu^\mu$ and $E \equiv n^\alpha n^\beta T_{\alpha\beta}$.

Then, from *Gauss-Codazzi equations*, one obtains two constraint equations:

$${}^{(3)}R + K^2 - K_{ij} K^{ij} = 16\pi E, \quad (2.32)$$

$$D_j (K^{ij} - \gamma^{ij} K) = 8\pi S^i, \quad (2.33)$$

known respectively as *Hamiltonian constraint* and *momentum constraint*. Time-evolution equations for γ_{ij} and K_{ij} along with constraint equations (plus a free

choice for lapse and shift) form the ADM formulation.

Another comment should be made about the lapse and the shift. They represent the four degrees of freedom of General Relativity and therefore are not specified by the equations given above. As said before, one could chose them arbitrarily, but in practice it is important to adopt appropriate prescriptions. In fact, there are different possible *slicing conditions* but these should be chosen to avoid singularities and counteract coordinate distortions; in addition, these gauge conditions should not be computationally expensive. More details about slicing conditions I used in my simulations can be found in Chapter 3.

The main problem with this approach for numerical purposes is that the system of partial differential equations is *weakly hyperbolic*. The importance of determining the degree of hyperbolicity of a set of equations is related to the concept of *well-posedness*. In well-posed problems, the solutions of equations grow at most exponentially. This is a fundamental requirement for those solutions to be computed numerically. Otherwise, they could grow faster than exponential and this is a problem from a computational point of view; numerical instabilities arise causing simulations to crash. Therefore, it is difficult to obtain long-term stable numerical simulations within ADM formulation. This reason has motivated the search for alternative approaches which remove from the ADM equations those parts responsible for their weak hyperbolicity. The most commonly used one in numerical simulations is the so-called *BSSN formulation*.

2.2.2 BSSN formalism

This formulation of General Relativity modifies ADM equations in order to build a strongly hyperbolic PDE system: BSSN formulation permits long-term stable simulations and thus can be used to study numerically a large variety of strong-field gravitational phenomena. In particular, ADM equations can be put in hyperbolic form by means of a *conformal transformation* and the introduction of new variables.

First, one imposes a conformal transformation of the spatial metric γ_{ij} :

$$\tilde{\gamma}_{ij} = \phi^2 \gamma_{ij}, \quad \tilde{\gamma}^{ij} = \phi^{-2} \gamma^{ij} \quad (2.34)$$

where ϕ is a conformal factor, which in principle could be chosen in different ways². In BSSNOK, it is chosen such that the conformal metric $\tilde{\gamma}_{ij}$ has unit determinant:

$$\phi = \gamma^{-\frac{1}{6}}, \quad (2.35)$$

and this relation has to remain satisfied during the evolution.

²For example, a common choice of ϕ is such that $\tilde{\gamma}_{ij} = e^{-4\phi} \gamma_{ij}$. See Alcubierre (2008) and Baumgarte and Shapiro (2010). The resulting equations are modified accordingly.

Within this formulation, the extrinsic curvature is separated into its trace K and the traceless part

$$A_{ij} \equiv K_{ij} - \frac{1}{3}\gamma_{ij}K \quad (2.36)$$

and through the same conformal rescaling of the traceless part of the extrinsic curvature one defines

$$\tilde{A}_{ij} \equiv \phi^2 A_{ij}. \quad (2.37)$$

Now, the important point is that this formulation introduces three auxiliary variables, called *conformal connection functions*, defined as

$$\tilde{\Gamma}^i \equiv \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i, \quad (2.38)$$

where $\tilde{\Gamma}_{jk}^i$ are the Christoffel symbols of the conformal metric and the equality follows from the definition of Christoffel symbols when $\tilde{\gamma} = 1$.

This formulation uses the variables ϕ , K , $\tilde{\gamma}_{ij}$, $\tilde{A}_{ij}$ and $\tilde{\Gamma}_i$ instead of the variables γ_{ij} and K_{ij} used in the ADM formulation. The evolution equations for the other variables are obtained directly from the ADM equations. The complete system takes the following form:

$$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + 2\tilde{\gamma}_{k(i} \partial_{j)} \beta^k - \frac{2}{3} \tilde{\gamma}_{ij} \partial_k \beta^k + \beta^k \partial_k \tilde{\gamma}_{ij} \quad (2.39)$$

$$\partial_t \phi = \frac{1}{3} \phi \alpha K - \frac{1}{3} \phi \partial_i \beta^i + \beta^k \partial_k \phi \quad (2.40)$$

$$\begin{aligned} \partial_t \tilde{A}_{ij} = & \phi^2 \left[-D_i D_j \alpha + \alpha \left({}^{(3)}R_{ij} - 8\pi S_{ij} \right) \right]^{\text{TF}} + \beta^k \partial_k \tilde{A}_{ij} \\ & + 2\tilde{A}_{k(i} \partial_{j)} \beta^k + \alpha \left(\tilde{A}_{ij} K - 2\tilde{A}_{ik} \tilde{A}_j^k \right) - \frac{2}{3} \tilde{A}_{ij} \partial_k \beta^k \end{aligned} \quad (2.41)$$

$$\partial_t K = -D_i D^i \alpha + \alpha \left(\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} K^2 \right) + 4\pi \alpha (E + S) + \beta^i \partial_i K \quad (2.42)$$

$$\begin{aligned} \partial_t \tilde{\Gamma}^i = & \tilde{\gamma}^{jk} \partial_j \partial_k \beta^i + \frac{1}{3} \tilde{\gamma}^{ik} \partial_k \partial_j \beta^j + \frac{2}{3} \tilde{\Gamma}^j \partial_j \beta^i - \tilde{\Gamma}^j \partial_j \beta^j - 2\tilde{A}^{ij} \partial_j \alpha \\ & + \beta^j \partial_j \tilde{\Gamma}^i + 2\alpha \left(\tilde{\Gamma}_{jk}^i \tilde{A}^{jk} - 3\tilde{A}^{ij} \partial_j \ln(\phi) - \frac{2}{3} \tilde{\gamma}^{ij} \partial_j K \right) \\ & - 16\pi \alpha \tilde{\gamma}^{ij} S_j \end{aligned} \quad (2.43)$$

where D_i is the covariant derivative with respect to the spatial metric γ_{ij} , the superscript ^{TF} denotes the traceless part of the expression inside the brackets. Moreover, in the evolution equation for K , the Hamiltonian constraint has been used to eliminate the Ricci scalar ${}^{(3)}R$ in (2.42), i.e.

$${}^{(3)}R = K_{ij} K^{ij} - K^2 + 16\pi E = \tilde{A}_{ij} \tilde{A}^{ij} - \frac{2}{3} K^2 + 16\pi E, \quad (2.44)$$

and momentum constraints have been used to eliminate the divergence of the extrinsic curvature on the right-hand side of (2.43).

Among the several motivations behind the introduction of these new variables, the conformal transformation and the separation of the traceless component of extrinsic curvature allow to control slicing conditions, which generally involve K . The most important reason, however, after the introduction of the $\tilde{\Gamma}^i$ - considered as independent variables - the second derivatives of the conformal metric which appear on the right hand side of equation (2.41) - contained in the Ricci tensor - reduce to a simple *Laplace operator*, i.e. $\gamma^{lm}\partial_l\partial_m\tilde{\gamma}_{ij}$. All the problematic second-order mixed derivatives are absorbed by the first derivative of the variables $\tilde{\Gamma}^i$. The system is now strongly hyperbolic and thus well-posed. Therefore, it is well-suited for numerical implementation. Indeed, most of the modern codes in Numerical Relativity adopt BSSN as the standard background formalism for the evolution of Einstein's equations.

Actually, the BSSN formulation has a feature which is not desirable. In fact, constraint violations do *not* propagate and if in a simulation they increase due to strong and rapid modifications of spacetime, these violations will not decrease in the following evolution, spoiling the accuracy of the solution. Therefore, throughout the years, new formulations of Numerical Relativity including *constraint damping* techniques have been introduced, acting on Einstein's equations. For instance, the *Z4 formulation* was introduced as a covariant extension of the Einstein's equations (Bona *et al.*, 2003). Within this formulation, Hamiltonian and momentum constraints become evolution equations for a new four vector Z_μ , modifying the principal part of ADM equations converting it from *weakly hyperbolic* to *strongly hyperbolic*. This four-vector Z_μ measures the deviations from Einstein's field equations, with condition $Z_\mu = 0$ corresponding to realize the standard Hamiltonian and momentum constraints. For the purpose of controlling these constraints, damping terms are added to the original system of equations. There also exists a variant of the Z4 formulation, known as *conformally and covariant Z4 formulation*, or *CCZ4*, built in order to apply the constraint damping techniques within a conformal and traceless formulation such as BSSN. I used the CCZ4 formulation for evolving spacetime in some of my simulations.

2.2.3 CCZ4 formalism

Without giving details on derivation, which can be found in the sources at the beginning of this chapter, we report the system of CCZ4 equations.

$$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + 2\tilde{\gamma}_{k(i} \partial_{j)} \beta^k - \frac{2}{3} \tilde{\gamma}_{ij} \partial_k \beta^k + \beta^k \partial_k \tilde{\gamma}_{ij} \quad (2.45)$$

$$\partial_t \phi = \frac{1}{3} \phi \alpha K - \frac{1}{3} \phi \partial_i \beta^i + \beta^k \partial_k \phi \quad (2.46)$$

$$\begin{aligned} \partial_t \tilde{A}_{ij} = & \phi^2 \left[-D_i D_j \alpha + \alpha \left({}^{(3)}R_{ij} + D_i Z_j + D_j Z_i - 8\pi S_{ij} \right) \right]^{\text{TF}} + \beta^k \partial_k \tilde{A}_{ij} \\ & + 2\tilde{A}_{k(i} \partial_{j)} \beta^k - 2\alpha \tilde{A}_{ik} \tilde{A}_j^k + \alpha \tilde{A}_{ij} (K - 2\Theta) - \frac{2}{3} \tilde{A}_{ij} \partial_k \beta^k \end{aligned} \quad (2.47)$$

$$\begin{aligned} \partial_t K = & -D_i D^i \alpha + \alpha \left({}^{(3)}R + 2D_i Z^i + K^2 - 2\Theta K \right) + \beta^i \partial_i K \\ & - 3\alpha \kappa_1 (1 + \kappa_2) \Theta + 4\pi \alpha (S - 3E) \end{aligned} \quad (2.48)$$

$$\begin{aligned} \partial_t \hat{\Gamma}^i = & 2\alpha \left(\tilde{\Gamma}_{jk}^i \tilde{A}^{jk} - 3\tilde{A}^{ij} \partial_j \ln(\phi) - \frac{2}{3} \tilde{\gamma}^{ij} \partial_j K \right) - 2\tilde{A}^{ij} \partial_j \alpha + \tilde{\gamma}^{jk} \partial_j \partial_k \beta^i \\ & 2\tilde{\gamma}^{ij} \left(\alpha \partial_j \Theta - \Theta \partial_j \alpha - \frac{2}{3} \alpha K Z_j \right) + \frac{1}{3} \tilde{\gamma}^{ik} \partial_k \partial_j \beta^j + \frac{2}{3} \tilde{\Gamma}^j \partial_j \beta^i - \tilde{\Gamma}^j \partial_j \beta^j \end{aligned} \quad (2.49)$$

$$\begin{aligned} & 2\kappa_3 \left(\frac{2}{3} \tilde{\gamma}^{ij} Z_j \partial_k \beta^k - \tilde{\gamma}^{jk} Z_j \partial_k \beta^i \right) + \beta^j \partial_j \hat{\Gamma}^i - 2\alpha \kappa_1 \tilde{\gamma}^{ij} Z_j - 16\pi \alpha \tilde{\gamma}^{ij} S_j \\ \partial_t \Theta = & \frac{1}{2} \alpha \left({}^{(3)}R + 2D_j Z^j - \tilde{A}_{ij} \tilde{A}^{ij} + \frac{2}{3} K^2 - 2\Theta K \right) - Z^j \partial_j \alpha \\ & + \beta^j \partial_j \Theta - \alpha \kappa_1 + (2 + \kappa_2) \Theta - 8\pi \alpha E, \end{aligned} \quad (2.50)$$

where Θ is the projection along the normal of the Z4 four-vector, i.e.

$$\Theta \equiv -n_u Z^\mu \quad (2.51)$$

and new *gamma variables* have been introduced as

$$\hat{\Gamma} \equiv \tilde{\Gamma}^i + 2\tilde{\gamma}^{ij} Z_j. \quad (2.52)$$

The coefficients κ_1 and κ_2 are free parameters related to the characteristic time of exponential damping of constraint violations, while κ_3 has been introduced for numerical stability. More details can be found in Alic *et al.*, (2012a).

Although the potential of this formulation has not been fully investigated, numerical simulations (see for example Alic *et al.*, 2012a; Kastaun *et al.*, 2013) have shown that CCZ4 leads to stable evolutions and decrease violation of constraints of at least one order of magnitude with respect to the BSSN. I found a confirmation of this fact in my simulations performed with CCZ4 (see Chapter 4).

2.3 Magnetohydrodynamics

The different formulations of General Relativity introduced so far provide methods to solve numerically Einstein's equations. In the previous treatment, we made the assumption that the right-hand side of Einstein's equations is known, i.e. the properties of the matter described by the energy-momentum tensor $T_{\mu\nu}$ are known. However, this requires the solution of the equations of relativistic hydrodynamics, which express the time-evolution of rest-mass density, energy and momentum:

$$\nabla_\mu(\rho u^\mu) = 0, \quad (2.53)$$

$$\nabla_\mu T^{\mu\nu} = 0. \quad (2.54)$$

Different methods have been developed throughout the years in order to solve numerically the equations of relativistic hydrodynamics. As in the case of formulations of General Relativity, finding optimal numerical schemes which allow to achieve stable solutions is far from being trivial. Some methods could be more suitable than others in order to avoid undesired behaviour of the solutions. In particular, there exists a particular class of methods, known as *conservative methods*, which are particularly robust from a numerical point of view.

2.3.1 Conservative Form of Relativistic Hydrodynamics

Before describing the properties of conservative formulations, we introduce some basic dynamical variables - which will be useful later - as

$$D = \rho W, \quad (2.55)$$

$$S_j = \rho h W^2 v_j, \quad (2.56)$$

$$E = \rho h W^2 - p, \quad (2.57)$$

where ρ is the rest-mass density, p the pressure, h the specific enthalpy, given by $h = 1 + \epsilon + p/\rho$ with ϵ being the specific internal energy. The quantities (2.55)-(2.57) are respectively the conserved rest-mass density, the covariant momentum and the generalized energy density.

Moreover, we make use of the following expression for the energy momentum-tensor, i.e. the expression for an ideal fluid:

$$T^{\mu\nu} = \rho h u^\mu u^\nu + p g^{\mu\nu}. \quad (2.58)$$

Using the definitions (2.55)-(2.58) inside (2.53) and (2.54), one obtains a set of equations known as *Wilson formulation* of relativistic hydrodynamics, due to the pioneering work by Wilson. However, this formulation is *not* in conservative form.

In order to understand what that means, let's first take a step back. It can be shown that the equations of relativistic hydrodynamics can be written in the following form:

$$\partial_t \mathbf{U} + \mathbf{A} \cdot \nabla \mathbf{U} = \mathbf{S}, \quad (2.59)$$

where $\mathbf{U}$ is referred to as *state vector* and $\mathbf{S}$ is a source term. This system is said to be *hyperbolic* if the matrix of coefficients $\mathbf{A}$ is diagonalizable with a set of real eigenvalues and a set of N linearly independent right eigenvectors. As already said in the context of Numerical Relativity, the crucial point here is that hyperbolic equations are *well-posed*, making them suitable for numerical solutions. On top of that, if the matrix $\mathbf{A}(\mathbf{U})$ is the Jacobian matrix of a *flux vector* $\mathbf{F}(\mathbf{U})$ with respect to the *state vector* $\mathbf{U}$ - i.e. $\mathbf{A}(\mathbf{U}) \equiv \partial \mathbf{F} / \partial \mathbf{U}$ - then equation (2.59) can be written in the so-called *conservative form*:

$$\partial_t \mathbf{U} + \nabla \mathbf{F}(\mathbf{U}) = \mathbf{S}, \quad (2.60)$$

and $\mathbf{U}$ is called the vector of *conserved variables* in this case.

The conservative form of the equation reveals its importance when dealing with problems admitting shocks or discontinuities as solutions. In particular, there exist two complementary theorems, one due to Lax and Wendroff [26] and the other one due to Hou and LeFloch [27], which - in simple terms - state that if a conservative formulation is *used*, the numerical solution will converge to the correct one if a conservative formulation is *not used*, the numerical solution will not converge to the correct solution if a shock wave is present in the flow.

2.3.2 Valencia Formulation for GRMHD

The system of equations (2.53)-(2.54) can be cast in conservative form, after the introduction of a set of *conserved variables*. Consider the *continuity equation* (2.3), which is in conservative form but conservative variables have not been introduced yet. Making use of the following relation for the divergence of a generic four vector U^μ

$$\nabla_\mu U^\mu = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} U^\mu), \quad (2.61)$$

equation (2.3) becomes

$$\begin{aligned} \nabla_\mu (\rho u^\mu) &= \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} \rho u^\mu) \\ &= \frac{1}{\sqrt{-g}} [\partial_t (\sqrt{-g} \rho u^t) + \partial_i (\sqrt{-g} \rho u^i)]. \end{aligned} \quad (2.62)$$

Then, equation (2.62) reads

$$\partial_t (\sqrt{\gamma} D) + \partial_i [\sqrt{\gamma} D (\alpha v^i - \beta^i)] = 0. \quad (2.63)$$

We would like to write down an analogous expression for the conservation equation of the energy-momentum tensor (2.54). We can use the following expression for the divergence of a generic rank-2 tensor:

$$\nabla_\mu T^{\mu\nu} = g^{\nu\lambda} \left[\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} T^\mu_\lambda) - \frac{1}{2} T^{\alpha\beta} \partial_\lambda g_{\alpha\beta} \right], \quad (2.64)$$

which applied to the equation (2.4) leads to

$$\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}T^\mu_\lambda) = \frac{1}{2}T^{\alpha\beta}\partial_\lambda g_{\alpha\beta}, \quad (2.65)$$

which, separating time and space components, becomes

$$\frac{1}{\sqrt{-g}}[\partial_t(\sqrt{\gamma}\alpha T^t_\lambda) + \partial_i(\sqrt{\gamma}\alpha T^i_\lambda)] = \frac{1}{2}T^{\mu\nu}\partial_\lambda g_{\mu\nu}. \quad (2.66)$$

Finally, collecting (2.63) and (2.66) and recalling the definitions (2.55)-(2.57), and defining the new variable

$$\tau \equiv E - D \quad (2.67)$$

it is possible to express the conservative relativistic hydrodynamics equations in the conservative form:

$$\frac{1}{\sqrt{-g}}[\partial_t(\sqrt{\gamma}\mathbf{U}) + \partial_i(\sqrt{-g}\mathbf{F}^i)] = \mathbf{S}, \quad (2.68)$$

where the vector of conserved variables is given by

$$\mathbf{U} \equiv \begin{pmatrix} D \\ S_j \\ \tau \end{pmatrix} = \begin{pmatrix} \rho W \\ \rho h W^2 v_j \\ \rho W(hW - 1) - p \end{pmatrix}, \quad (2.69)$$

and the corresponding flux-vector in the i-direction is

$$\mathbf{F}^i \equiv \begin{pmatrix} D \left(v^i - \frac{\beta^i}{\alpha} \right) \\ S_j \left(v^i - \frac{\beta^i}{\alpha} \right) + p \delta_j^i \\ \tau \left(v^i - \frac{\beta^i}{\alpha} \right) + p v^i \end{pmatrix}, \quad (2.70)$$

while the source term reads

$$\mathbf{S} \equiv \begin{pmatrix} 0 \\ T^{\mu\nu} \left(\frac{\partial g_{\nu j}}{\partial x^\mu} - \Gamma_{\nu\mu}^\delta g_{\delta j} \right) \\ \alpha \left(T^{\mu 0} \frac{\partial \ln \alpha}{\partial x^\mu} - T^{\mu\nu} \Gamma_{\nu\mu}^0 \right) \end{pmatrix}, \quad (2.71)$$

where the source term has been expressed using the Christoffel symbols.

So far, electric and magnetic fields have been neglected, presenting a conservative formulation of relativistic hydrodynamics in curved spacetime (GRHD). However, for a large variety of astrophysical phenomena, including binary neutron star mergers, one is interested in studying *electrically-conducting* fluids in presence of *magnetic fields*, in a general relativistic framework, a field known as *General Relativistic Magnetohydrodynamics* (GRMHD).

Adding Electromagnetic Fields: GRMHD

For the purpose of writing the equations of General Relativistic Magnetohydrodynamics, let's first recall *Maxwell's equations* written in covariant form:

$$\nabla_\nu F^{\mu\nu} = J^\mu, \quad (2.72)$$

$$\nabla_\nu {}^*F^{\mu\nu} = 0, \quad (2.73)$$

where $F_{\mu\nu} = u^\mu E^\nu - u^\nu E^\mu - \eta^{\mu\nu\rho\sigma} u_\rho B_\sigma$ is the electromagnetic tensor and ${}^*F_{\mu\nu} = \frac{1}{2}\eta^{\mu\nu\rho\sigma} F_{\rho\sigma}$ its dual. In these expressions, $\eta^{\mu\nu\rho\sigma} \equiv \frac{1}{\sqrt{-g}}\epsilon^{\mu\nu\rho\sigma}$ with $\epsilon^{\mu\nu\rho\sigma}$ being the Levi-Civita antisymmetric tensor, E^μ and B^μ are respectively the electric and magnetic fields measured by an observer with four-velocity u^μ (Eulerian), while the four-current reads $J^\mu \equiv \rho_q u^\mu + \sigma F^{\mu\nu} u_\nu$, where ρ_q is the proper charge density and σ the electric conductivity.

These equations take a simpler form in the specific case of $\sigma = \infty$, i.e. the fluid is a *perfect conductor*. This is the case in most astrophysical situations of interest, including BNSs, where the gas is highly ionized and hence an excellent conductor. This is referred to as *ideal* MHD condition. Being σ infinite, the term $F^{\mu\nu} u_\nu$ must be zero, which means that the electric field four-vector vanishes in the fluid rest frame. In this circumstance, the component of the electric field measured by the Eulerian observer are

$$E^0 = 0, \quad (2.74)$$

$$E^i = -\alpha\eta^{0ijk}v_j B_k. \quad (2.75)$$

Equations (2.7) reduce to

$$\frac{\partial}{\partial x^i}(\sqrt{\gamma}B^i) = 0, \quad (2.76)$$

which is nothing but the *divergence-free* condition and the evolution equation for the magnetic field, i.e. the *induction* equation:

$$\frac{1}{\sqrt{\gamma}}\frac{\partial}{\partial x^0}(\sqrt{\gamma}B^i) = \frac{1}{\sqrt{\gamma}}\frac{\partial}{\partial x^j}\{\sqrt{\gamma}[(\alpha v^i - \beta^i)B^j - (\alpha v^j - \beta^j)B^i]\}. \quad (2.77)$$

Equations (2.76) and (2.77) can respectively be written in terms of the vector $\mathbf{B}$ as³

$$\nabla \cdot (\sqrt{\gamma}\mathbf{B}) = 0, \quad (2.78)$$

$$\partial_t(\sqrt{\gamma}\mathbf{B}) = \nabla \times [(\alpha\mathbf{v} - \mathbf{B}) \times \sqrt{\gamma}\mathbf{B}]. \quad (2.79)$$

For a magnetized fluid, the energy-momentum tensor can be written as the sum of a *fluid* part $T_{\text{Fluid}}^{\mu\nu}$, given by the expression (2.58) for a perfect fluid, and an *electromagnetic* contribution $T_{\text{EM}}^{\mu\nu}$, which can be defined in terms of $F^{\mu\nu}$ as

$$T_{\text{EM}}^{\mu\nu} \equiv F^{\mu\rho}F^\nu_\rho - \frac{1}{4}g^{\mu\nu}F^{\rho\sigma}F_{\rho\sigma}. \quad (2.80)$$

³It is well known that electromagnetism can be reformulated in terms of a four-vector potential, which however is not unique. More on gauge conditions can be found in Chapter 3.

In the particular case of ideal MHD, the previous expression takes the form

$$T_{\text{EM}}^{\mu\nu} = \left(u^\mu u^\nu + \frac{1}{2} g^{\mu\nu} \right) b^\mu b_\mu - b^\mu b^\nu, \quad (2.81)$$

where b^μ is the magnetic field measured by the fluid-comoving observer. The total energy-momentum tensor $T^{\mu\nu}$ can be finally written as

$$\begin{aligned} T^{\mu\nu} &= T_{\text{Fluid}}^{\mu\nu} + T_{\text{EM}}^{\mu\nu} \\ &= \rho \left(h + \frac{b^2}{\rho} \right) u^\mu u^\nu + \left(p + \frac{b^2}{2} \right) g^{\mu\nu} - b^\mu b^\nu. \end{aligned} \quad (2.82)$$

Following the same exact approach presented for the non-magnetized ideal fluid, i.e. starting from the conservation equations for the energy-momentum tensor (2.54) and the continuity equation (2.53), together with the evolution equation of the magnetic field, it is possible to obtain a first-order, conservative and hyperbolic system of the form (2.60).

The state vector $\mathbf{U}$ of conserved variables can be written as

$$\mathbf{U} \equiv \begin{pmatrix} D \\ S_j \\ \tau \\ B^k \end{pmatrix}, \quad (2.83)$$

where now:

$$D = \rho W, \quad (2.84)$$

$$S_j = \rho \left(h + \frac{b^2}{\rho} \right) W^2 v_j - \alpha b^0 b_j, \quad (2.85)$$

$$\tau = \rho \left(h + \frac{b^2}{\rho} \right) W^2 - \left(p + \frac{b^2}{2} \right) - \alpha^2 (b^0)^2 - D, \quad (2.86)$$

while the four components of b^μ in the comoving frame and the three-vector components B^i measured by the Eulerian observer are related by:

$$b^0 = \frac{W B^i v_i}{\alpha}, \quad (2.87)$$

$$b^i = \frac{B^i + \alpha b^0 u^i}{W}. \quad (2.88)$$

The flux vector in the i -direction now reads:

$$\mathbf{F}^i \equiv \begin{pmatrix} D \left(v^i - \frac{\beta^i}{\alpha} \right) \\ S_j \left(v^i - \frac{\beta^i}{\alpha} \right) + \left(p + \frac{b^2}{2} \right) \delta_j^i - \frac{b_j B^i}{W} \\ \tau \left(v^i - \frac{\beta^i}{\alpha} \right) + \left(p + \frac{b^2}{2} \right) v^i - \alpha \frac{b^0 B^i}{W} \\ \left(v^i - \frac{\beta^i}{\alpha} \right) B^k - \left(v^k - \frac{\beta^k}{\alpha} \right) B^i \end{pmatrix}. \quad (2.89)$$

The source term is exactly like (2.71), but with one more null component, i.e.

$$\mathbf{S} \equiv \begin{pmatrix} 0 \\ T^{\mu\nu} \left(\frac{\partial g_{\nu j}}{\partial x^\mu} - \Gamma_{\nu\mu}^\delta g_{\delta j} \right) \\ \alpha \left(T^{\mu 0} \frac{\partial \ln \alpha}{\partial x^\mu} - T^{\mu\nu} \Gamma_{\nu\mu}^0 \right) \\ 0^k \end{pmatrix}, \quad (2.90)$$

where 0^k means a spatial vector with null components.

The Valencia formulation of General Relativistic Magnetohydrodynamics has been originally introduced since the hyperbolic and conservative form of the equations allowed to import several numerical methods already existent for classical hydrodynamics. More properties of these schemes, known as *high-resolution shock-capturing (HRSC) methods*, will be shown in the next chapter.

Including Neutrinos

During the merger phase of binary systems containing neutron stars one reaches extremely high temperatures. In this scenario, neutrinos play a crucial role in transporting energy and determining the electron fraction Y_e . This aspect is considerably important if we are interested in studying the r-process nucleosynthesis taking place in the merger ejecta, which determines the properties of the observed kilonova emission. In particular, one should estimate the rates of all the different reactions involving neutrinos in order to determine nucleosynthesis yields. This is far from being an easy task. Indeed, solving the full neutrino transport problem requires an exceptionally high computational cost. Therefore, one usually introduces approximate schemes based on simplifying assumptions, for instance the *neutrino leakage scheme*, implemented in **Spritz** via the **ZelmaniLeak** code.

Chapter 3

Numerical Methods and Codes

The formulations of Numerical Relativity and of General Relativistic Magnetohydrodynamics presented in the previous chapter have been developed throughout the years aiming to write down equations in a well-suited form for numerical purposes. In this chapter, I present the numerical methods used to solve these equations and describe some relevant features of the codes I used for simulating the dynamics of binary neutron stars. Specifically, I performed simulations using the General Relativistic Magnetohydrodynamic (GRMHD) code **Spritz** and the General Relativistic Hydrodynamic (GRHD) code **WhiskyTHC**, in the framework of a computational infrastructure known as *Einstein Toolkit*. In addition, in the last part of the chapter I introduce **Kadath**, the code I used to generate binary neutron stars initial data for all my simulations.

Main references for Einstein Toolkit are the papers [4] and [5]. For more details on **Spritz**, see [1],[2],[28].

3.1 The Einstein Toolkit

The Einstein Toolkit is a community-led and publicly available numerical infrastructure providing computational tools for relativistic astrophysics and gravitational physics. This apparatus delivers a set of powerful instruments intended for simulating astrophysical objects such as black holes, neutron stars, compact binaries and also for analyzing and processing the results obtained from the simulations.

In particular, it is built upon some core components which allow to manage numerical simulations, from the generation of code to the storage of the results. An important feature of the Toolkit is in fact the separation of tasks for managing simulations - such as memory management, parallel I/O, data distribution and checkpointing - performed by specific drivers, from the physics ingredients. This is possible through a modular structure, with an underlying infrastructure which allows to facilitate communications between different modules. Let's first take a look at the infrastructural part, and then move to the actual physics components of the Toolkit.

3.1.1 Infrastructure and Core Technologies

The Computational Framework: Cactus

As just mentioned, the Einstein Toolkit relies on a basic infrastructure allowing to handle several aspects of numerical simulations. This is possible due to the **Cactus** computational framework, an open source programming environment which is based on a completely modular structure, with a central core known as *flesh* providing the communication interface between distinct modules, called *thorns*. Thorns having similar origin, or performing common tasks, are grouped into *arrangements*.

The existence of a common interface is a fundamental feature of this framework, having several advantages. For instance, this allows to use numerical codes written by others without worrying about the specific details of the implementation, and to use own modules, in supported programming language of one's choice, without changing the source code of other parts.

Adaptive Mesh Refinement: Carpet

In **Cactus**, the infrastructural tasks are carried out by *driver* components. The main driver used in astrophysical simulations within **Cactus** framework is **Carpet**. This component has the capability to handle simulation grids providing *Adaptive Mesh Refinement*.

A single uniform grid is not the most convenient way to deal with many astrophysical problems: some regions of the simulation domain could require higher resolutions than others. For instance, in the case of a binary neutron star system, we would like to have higher resolutions in the regions where the stars are located; at the same time, the grid should be sufficiently extended to properly model relevant phenomena in the weak-field region, such as gravitational radiation. A suitable technique to realize that is through *adaptive mesh refinement (AMR)*, meaning that the simulation region is dynamically adapted to the state of the simulation. Hence, regions requiring higher resolutions, such as the central region of the domain in a binary neutron star merger, are covered with blocks having finer grids; these levels of different grid spacing, adapted to the required resolution, are known as *refinement levels*. Typically, each level has a grid which is finer by a factor of two with respect to the previous one. Moreover, a procedure named *regridding* revisits resolution requirements at regular intervals, constantly updating the grid hierarchy.

Note that finer grids require smaller time steps, so that they make multiple time steps for each time step performed by a coarser grid, conducting to a recursive time evolution pattern. If n refinement levels are active, the finest and the coarsest grids have resolutions differing by a factor of 2^{n-1} both in space and time.

The Simulation Factory

Simulating the dynamics of complex systems such as binary neutron stars requires the use of supercomputers. These machines are generally very different from one

another in both hardware and software configurations. Managing this kind of simulations can be difficult, especially due to tasks which do not require physical insight, like low-level management ones. The *Simulation Factory*, or *SimFactory*, is a tool offering a set of abstractions which simplify the management of a simulation in **Cactus** framework, making it a uniform operation on a wide variety of machines, from laptops to supercomputers. For instance, it provides a convenient manner of building new configurations of the Einstein Toolkit to be used in simulations, and facilitate the operation of job submission and restart from checkpoints. I widely used *SimFactory* for simulation management throughout my thesis.

Generating Code from Equations: Kranc

Kranc is a Mathematica package which converts tensorial systems of partial differential equations into a module for **Cactus**, suitable for running on several infrastructures, from laptops to supercomputers. For instance, the thorn for spacetime evolution **McLachlan** and the wave extraction thorn **WeylScal4** are generated using **Kranc**.

3.1.2 Base Modules for Physics

ADMBase

In **Cactus**, there exist different methods for the evolution of spacetime. However, all are essentially based on the ADM formulation. For this reason, it serves as a foundation for exchanging data between modules evolving spacetime with different formalisms. In particular, in the **ADMBase** module the following variables are defined for the main spacetime quantities of the ADM formulation, in particular:

- Spatial metric γ_{ij} : **gxx**, **gxy**, **gxz**, **gyy**, **gyz**, **gzz**
- Extrinsic curvature K_{ij} : **Kxx**, **Kxy**, **Kxz**, **Kyy**, **Kyz**, **Kzz**
- Lapse α : **alp**
- Shift vector β^i : **betax**, **betay**, **betaz**

This module defines parameters which allow the communications between different modules, such as the type of requested initial data or the type of evolution method. However, **ADMBase** is not used for actual evolution: instead, this task is performed by evolution modules according to a specific evolution method; after each step, the internal variables of these modules are converted in **ADMBase** variables, so that modules performing analysis can use directly **ADMBase** variables without dependencies on other evolution modules.

HydroBase

In a similar manner, the **HydroBase** module provides a connection between modules performing the evolution of relativistic hydrodynamics, including an interface for magnetohydrodynamics. The aim of this module is the definition of a set of variables common to all hydrodynamics code, i.e. *primitive variables*:

- Rest-mass density ρ : `rho`
- Pressure p : `press`
- Internal energy density ϵ : `eps`
- Contravariant components of fluid three-velocity v^i , given by the formula (2.25): `vel[3]`
- Electron fraction Y_e : `Y_e`
- Temperature T : `temperature`
- Specific entropy per particle s : `entropy`
- Contravariant magnetic field vector B^i : `Bvec[3]`

Another task performed by `HydroBase` is scheduling blocks for organizing functions commonly present in hydrodynamics codes.

`TmunuBase`

The `TmunuBase` provides grid functions for the stress-energy tensor $T_{\mu\nu}$ and schedule when these functions are computed. Since different thorns can contribute to the stress-energy tensor, this module allows to do so without depending on each other. In particular, the grid functions computed by this module are:

- Time component T_{00} : `eTtt`
- Mixed components T_{0i} : `eTtx`, `eTty`, `eTtz`
- Spatial components T_{ij} : `eTxx`, `eTxy`, `eTxz`, `eTyy`, `eTyz`, `eTzz`

3.1.3 Spacetime Evolution

`McLachlan`

The spacetime evolution thorn `McLachlan` (actually, it is an arrangement) is auto-generated from tensor equations using `Kranc`. In particular, it generates BSSN equations, but also the CCZ4 ones. I used both the modules `ML_BSSN` and `ML_CCZ4` contained in `McLachlan` arrangement for my simulations. `McLachlan` uses finite difference schemes up to eight order with *Kreiss-Oliger dissipation* terms - i.e. a method to suppress high-frequency modes for numerical stability [29] - for the space-time variables. For instance, in the case of BSSN formulation, it solves the BSSN equations for the conformal factor ϕ , the conformal 3-metric $\tilde{\gamma}_{ij}$, the trace K of the extrinsic curvature, the conformal traceless extrinsic curvature $\tilde{A}_{ij}$ and the conformal connection functions $\tilde{\Gamma}_i$. It is designed to communicate with `ADMBase` and `TmunuBase`.

As mentioned in Chapter 2, evolving spacetime evolution equations requires a choice of the lapse and the shift. In particular, `McLachlan` uses for the following

slicing conditions (see [30] for a more detailed discussion on slicing conditions). For the lapse, it adopts an hyperbolic *K-driver* condition, which reads

$$(\partial_t - \beta^j \partial_j) \alpha = -\alpha^2 f(\alpha) (K - K_0), \quad (3.1)$$

where $f(\alpha) = \frac{2}{\alpha}$, known as *1+log* slicing, and $K_0 \equiv K(t=0) = 0$. For the shift, a hyperbolic Γ -*driver* condition is used:

$$(\partial_t - \beta^j \partial_j) \beta^i = \frac{3}{4} B^i \quad (3.2)$$

$$(\partial_t - \beta^j \partial_j) B^i = (\partial_t - \beta^j \partial_j) \tilde{\Gamma}^i - \eta B^i, \quad (3.3)$$

where B^i is a quantity related to the derivative of the shift. The factor $\frac{3}{4}$ is arbitrary but leads to good numerical results, while η is typically of order of $\frac{1}{2M}$, with M the total mass of the system. These gauge condition have been extremely successful in several simulations.

3.1.4 Other Thorns for Analysis

Among other useful thorns for analysis, I mention some of them particularly relevant for my simulations.

AHFinderDirect

This thorn finds a black-hole apparent horizon given the 3-metric and the extrinsic curvature. For details on how to find apparent horizons in Numerical Relativity, see for example [31] and [32].

WeylSca14

One of the main purposes of Numerical Relativity is to properly model gravitational waves, in order to obtain templates which can help analyzing signals from gravitational waves interferometers. The Toolkit can compute gravitational waves either with *Moncrief formalism* (see [33] for more details) or calculating the *Weyl scalar* Ψ_4 .

In particular, Ψ_4 is a complex scalar - defined within the *Newman-Penrose formalism* of General Relativity [34] - which encodes, in an appropriate frame, the outgoing gravitational radiation of an asymptotically flat spacetime, being very important for gravitational wave analysis. Specifically, in the radiation zone, the strain of the gravitational waves is related to Ψ_4 through the relation

$$h = h_+ - ih_\times = - \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' \Psi_4. \quad (3.4)$$

In the Einstein Toolkit, the Weyl scalar is computed by the thorn `WeylSca14`, generated via `Kranc`.

Multipole

The Weyl scalar Ψ_4 is provided as a grid function. For this, but actually for any other grid function, there exists a module called **Multipole**, which decomposes these functions into spin-weighted spherical harmonics.

3.2 The Spritz Code

In the first part of my thesis, I used the code **Spritz** in order to solve the equations of General Relativistic Magnetohydrodynamics (GRMHD) in 3D on a dynamical background spacetime. The aim of this code was to merge the main capabilities of three existent codes: **WhiskyMHD** [35] and the publicly available **GRHydro** [36] and **IllinoisGRMHD** [37]. In particular, **Spritz** can handle in principle any kind of equation of state, being able to use modern tabulated equations of state, adopting an approach similar to the one used in **GRHydro**. Moreover, it evolves the vector potential instead of the magnetic field in a staggered formalism which is presented within this section. Finally, as already mentioned, it is able to include the treatment of neutrinos via the **ZelmaniLeak** code.

In order to describe how **Spritz** deals with GRMHD equations, it is preliminarily convenient to introduce the vector potential formulation, along with the choice of appropriate gauge conditions.

3.2.1 Gauge Conditions for Vector Potential

We have already described how to include the magnetic field into GRMHD equations. In practice, however, it can be convenient to formulate the problem in terms of a *vector potential*. Considering the operator $\nabla \equiv (\partial_x, \partial_y, \partial_z)$, it is well known that¹

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (3.5)$$

where $\mathbf{A}$ is the vector potential. The divergenceless condition of magnetic field is automatically satisfied since

$$\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0. \quad (3.6)$$

Then, we introduce the four-vector potential

$$\mathcal{A}_\nu = n_\nu \Phi + A_\nu, \quad (3.7)$$

where A_ν is the purely spatial vector potential while Φ is the scalar potential. The divergence-free condition (2.76) and the induction equations (2.77) become respectively

$$B^i = \epsilon^{ijk} \partial_j A_k \quad (3.8)$$

and

$$\partial_t A_i = -E_i - \partial_i(\alpha \Phi - \beta^j A_j), \quad (3.9)$$

¹3.5 and 3.6 are for the non-relativistic case and here they serve as an example.

where ϵ^{ijk} is the three-dimensional Levi-Civita tensor. Clearly, the choice of the four-vector potential is not unique. The first simulations in magnetohydrodynamics employing the vector potential used the *algebraic gauge* (see e.g. [38] and [39]), where the scalar potential Φ and the purely spatial vector potential A_i are related by

$$\Phi = \frac{1}{\alpha}(\beta^j A_j) \quad (3.10)$$

so that equation (3.9) reduces to

$$\partial_t A_i = -E_i, \quad (3.11)$$

being extremely simplified and not requiring an evolution equation for the scalar potential.

In more recent simulations [38], in order to avoid spurious magnetic effects which appeared using the *algebraic gauge* with AMR, the gauge condition

$$\nabla_\nu \mathcal{A}^\nu = 0. \quad (3.12)$$

has been used. This is known as *Lorenz gauge* and introduces an evolution equation for Φ to be added to (3.9). Written explicitly, the previous equation becomes

$$\partial_t(\sqrt{\gamma}\Phi) + \partial_i(\alpha\sqrt{\gamma}A^i - \sqrt{\gamma}\beta^i\Phi) = 0 \quad (3.13)$$

Another robust gauge, named *generalized Lorenz gauge*, has been introduced in [40] through the condition

$$\nabla_\nu \mathcal{A}^\nu = \xi n_\nu \mathcal{A}^\nu, \quad (3.14)$$

with ξ usually set to $1.5/\Delta t_{max}$, where Δt_{max} is the time step of the coarsest grid. Hence, the evolution equation for the scalar potential is

$$\partial_t(\sqrt{\gamma}\Phi) + \partial_i(\alpha\sqrt{\gamma}A^i - \sqrt{\gamma}\beta^i\Phi) = -\xi\alpha\sqrt{\gamma}\Phi. \quad (3.15)$$

The generalized Lorenz gauge is the one adopted in **Spritz**. By the addition of the dumping term ξ , has been demonstrated to further reduce spurious magnetic field values at the boundary between refinement levels.

3.2.2 HRSC Methods

In the previous chapter we have commented on the importance of the conservative formulation of equations, in order to implement methods which lead to the correct solutions, especially when *shocks* are present in the solutions. In particular, there exists a particular class of conservative methods, known as *High-Resolution Shock-Capturing (HRSC)* methods, which are well-suited to deal with problems with discontinuities in the solution. Let us see in more detail what we mean by conservative methods.

Consider the conservative form of a hyperbolic system in one dimension:

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}(\mathbf{U}) = 0. \quad (3.16)$$

At this point, let us discretize the spatial domain into J cells $I_j = [x_{j-1/2}, x_{j+1/2}]$, with spacing $\Delta x = x_{j+1/2} - x_{j-1/2}$, with $j = 0, \dots, J-1$. In particular:

$$x_j \equiv j\Delta x, \quad j = 0, 1, \dots, J-1. \quad (3.17)$$

Also, discretize the time domain with a time step Δt , such that

$$t^n \equiv n\Delta t, \quad t = 0, 1, \dots, N-1. \quad (3.18)$$

Note that the time step can not be chosen arbitrarily. In fact, the spatial resolution and the time resolution must be related by the *Courant-Friedrichs-Lewy* condition [41], which is implemented via a dimensionless parameter known as *Courant factor*, as

$$C_F \equiv \frac{|a|\Delta t}{\Delta x} < 1, \quad (3.19)$$

where $|a|$ is the maximum speed at which information can travel in the system, e.g. c in General Relativity. This condition on the propagation of information is essential to produce correct numerical results.

Integrating in space equation (3.16), we obtain

$$\frac{d}{dt} \int_{x_{j-1/2}}^{x_{j+1/2}} \mathbf{U}(x, t) dx = \mathbf{F}(\mathbf{U}(x_{j-1/2}, t)) - \mathbf{F}(\mathbf{U}(x_{j+1/2}, t)) \quad (3.20)$$

and the in time between t^n and t^{n+1}

$$\begin{aligned} \int_{x_{j-1/2}}^{x_{j+1/2}} \mathbf{U}(x, t^{n+1}) dx &= \int_{x_{j-1/2}}^{x_{j+1/2}} \mathbf{U}(x, t^n) dx \\ &+ \int_{t^n}^{t^{n+1}} [\mathbf{F}(\mathbf{U}(x_{j-1/2}, t)) - \mathbf{F}(\mathbf{U}(x_{j+1/2}, t))] dt \end{aligned} \quad (3.21)$$

which is the integral form of (3.16). Now we can define the *cell (volume) averages* as

$$\mathbf{U}_j^n \equiv \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} \mathbf{U}(x, t^n) dx \quad (3.22)$$

and the *numerical fluxes* as

$$\mathbf{F}_{j\pm 1/2} \equiv \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} \mathbf{F}(\mathbf{U}(x_{j\pm 1/2}, t)) dt, \quad (3.23)$$

so that equation (3.21) becomes

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n + \frac{\Delta t}{\Delta x} (\mathbf{F}_{j-1/2} - \mathbf{F}_{j+1/2}). \quad (3.24)$$

These methods of solving (3.16) are known as *finite volume* methods. Up to this point, this expression does not represent a numerical scheme yet, because (3.24) is still exact, without mathematical approximation performed. It will become a *numerical scheme* when cell averages $\mathbf{U}_j$ and numerical fluxes $\mathbf{F}_{j\pm 1/2}$ are computed with approximations. However, we see that in order to compute $\mathbf{U}_j^{n+1}$ by (3.24), we do not have the numerical fluxes $\mathbf{F}_{j\pm 1/2}$. In order to compute them, an ingenious idea is due to Godunov [42].

Godunov Methods

The observation behind Godunov's idea was that at each interface between cells, the quantity $\mathbf{U}_j^n$ - the average of $\mathbf{U}(x, t)$ over the cell $[x_{j-1/2}, x_{j+1/2}]$ at $t = t^n$ - has a *jump*, so that its values on the left and on the right of the interfaces can be regarded respectively as left and right states of a local *Riemann problem*, i.e. an initial value problem composed of a conservation equation and piecewise constant initial data having a discontinuity in the domain of interest. In the original form, the initial data for the local Riemann problem read

$$\mathbf{U}(x, 0) = \begin{cases} \mathbf{U}_j^n & x < x_{j-1/2}, \\ \mathbf{U}_{j+1}^n & x > x_{j+1/2}. \end{cases} \quad (3.25)$$

The great advantage here is that there exist methods for solving Riemann problems exactly, known as *Riemann solvers*. However, exact Riemann solvers are expensive from a computational point of view, so that in practice *approximate* Riemann solvers are adopted, as we will see. Their solution will give $\mathbf{U}(x_{j\pm 1/2}, t)$ which can finally be used to compute the fluxes $\mathbf{F}_{j\pm 1/2}$.

To sum up, *Godunov methods* allow to compute the fluxes $\mathbf{F}_{j\pm 1/2}$ by solving Riemann problems at the interfaces $x_{j\pm 1/2}$. Since this method can be applied to any cell, more in general a Godunov method provides an efficient way to compute numerical fluxes as *sequence* of Riemann problems.

Riemann Solvers

In **Spritz**, the default Riemann solver is *Harten-Lax-van Leer-Einfeldt (HLL)* [43], where numerical fluxes at interfaces are computed as

$$\mathbf{F}^i = \frac{c_{\min} \mathbf{F}_r^i + c_{\max} \mathbf{F}_l^i - c_{\max} c_{\min} (\mathbf{U}_r - \mathbf{U}_l)}{c_{\max} + c_{\min}}, \quad (3.26)$$

where $\mathbf{U}$ are as usual the vectors of conserved variables. The subscript $r(l)$ means that the functions are computed at the right(left) side of the cell interface, $c_{\max} = \max(0, c_{+,l}, c_{+,r})$, $c_{\min} = -\min(0, c_{-,l}, c_{-,r})$ and $c_{\pm,r}(c_{\pm,l})$ are the right-going (+) and left-going (−) maximum waves speeds, computed from the primitive variables at the right and left side.

Another Riemann solver implemented in **Spritz** is the *Lax-Friedrichs (LxF)* scheme [44], which is useful when strong discontinuities in pressure have to be considered since it is more dissipative than the HLL solver. The fluxes are computed as

$$\mathbf{F}^i = \frac{\mathbf{F}_r^i + \mathbf{F}_l^i - c_{\text{wave}} (\mathbf{U}_r - \mathbf{U}_l)}{2}, \quad (3.27)$$

where $c_{\text{wave}} = \max(c_{\max}, c_{\min})$.

There exist some techniques which aim to give a polynomial representation of $\mathbf{U}$ inside each cell, providing better estimates for the right and left states of the Riemann problem. These are called *reconstruction algorithms*. In particular, **Spritz** adopts the third-order *piecewise parabolic method (PPM)* [45], but the second-order *total variation diminishing (TVD) minmod method* [44] is also implemented in case of strong shocks (minmod is indeed more dissipative than PPM and therefore it may smooth out strong discontinuities).

Recovering Primitive Variables

Actually, the values of the conserved variables does not allow to directly compute the fluxes. In fact, recalling equation (2.89), we see that they depend on primitive variables, although we evolve the conserved ones $\mathbf{U}$. Therefore, at each time of the evolution, we have to invert the equation (2.83) in order to get the values of primitive variables from the conservative ones. This is a very delicate operation, with no analytic solution, because of the presence of the Lorentz factor. Therefore, one has to apply numerical methods, usually Newton-Raphson schemes.

In **Spritz**, both the methods present in **WhiskyMHD** [35] and **GRHydro** [46] have been implemented. Recently, new tests have been performed in order to implement the new robust and accurate scheme **RePrimAnd** (for more details, see [47] and [48]) into **Spritz** [28].

3.2.3 Electromagnetic Fields

For convenience, let us first define the variable

$$\Psi_{\text{mhd}} \equiv \sqrt{\gamma}\Phi. \quad (3.28)$$

Recall that the spatial domain is a grid divided into cells, whose resolution is defined by the user. While each fluid's state variable is stored on grid-cell center, electric and magnetic fields are stored respectively on cell edges and faces. The locations of the different quantities over the grid are reported in the table on next page. As previously mentioned, **Spritz** evolves the magnetic field as the curl of a vector potential, whose component are staggered (stored on cell edges). The electric and magnetic fields are not evolved. Following the same approach implemented in **WhiskyMHD** [35], using the *Flux-CT* method [49] the electric field at the edges is computed from magnetic field HLLE fluxes. Instead, the magnetic field is computed at cell faces using the vector potential components stored at the edges and then interpolated at the center of the cell. Note also that, since the magnetic field is obtained by the curl of a vector potential, the divergence-free condition is automatically satisfied. Specifically, **Spritz** evolves the vector potential $\mathbf{A}$ and the scalar potential Ψ_{mhd} through the generalized Lorenz gauge. The evolution equations for the vector and scalar potentials can be written, recalling equations (3.9) and (3.15):

$$\partial_t A_m = -E_m - \partial_m(G_A), \quad G_A \equiv \alpha \frac{\Psi_{\text{mhd}}}{\sqrt{\gamma}} - \beta^j A_j \quad (3.29)$$

$$\partial_t(\Phi_{\text{mhd}}) = -\partial_j(F_{\Psi}^j) - \xi\alpha\Psi_{\text{mhd}}, \quad F_{\Psi}^j \equiv \alpha\sqrt{\gamma}A^j - \beta^j\Psi_{\text{mhd}}. \quad (3.30)$$

Symbol	Definition	Location
α	lapse	(i, j, k)
β^m	m -component of the shift	(i, j, k)
γ^{mn}	mn -component of the spatial metric	(i, j, k)
γ	determinant of the spatial metric	(i, j, k)
ρ	rest-mass density	(i, j, k)
p_{gas}	pressure	(i, j, k)
ϵ	energy-density	(i, j, k)
v_m	m -component of the fluid velocity	(i, j, k)
B^1	x -component of the magnetic field	$(i + \frac{1}{2}, j, k)$
B^2	y -component of the magnetic field	$(i, j + \frac{1}{2}, k)$
B^3	z -component of the magnetic field	$(i, j, k + \frac{1}{2})$
A_1	x -component of the vector potential	$(i, j + \frac{1}{2}, k + \frac{1}{2})$
A_2	y -component of the vector potential	$(i + \frac{1}{2}, j, k + \frac{1}{2})$
A_3	z -component of the vector potential	$(i + \frac{1}{2}, j + \frac{1}{2}, k)$
Ψ_{mhd}	scalar potential	$(i + \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2})$

It is important to note that interpolations are needed in order to relate quantities which are stored in different locations, as we can see in the previous equations.

- If a variable f is stored on cell vertices, its value at the center is given by

$$\begin{aligned}
 f(i, j, k) = \frac{1}{8} & \left[f\left(i - \frac{1}{2}, j - \frac{1}{2}, k - \frac{1}{2}\right) + f\left(i - \frac{1}{2}, j + \frac{1}{2}, k - \frac{1}{2}\right) \right. \\
 & + f\left(i + \frac{1}{2}, j + \frac{1}{2}, k - \frac{1}{2}\right) + f\left(i + \frac{1}{2}, j - \frac{1}{2}, k - \frac{1}{2}\right) \\
 & + f\left(i - \frac{1}{2}, j - \frac{1}{2}, k + \frac{1}{2}\right) + f\left(i - \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2}\right) \\
 & \left. + f\left(i + \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2}\right) + f\left(i + \frac{1}{2}, j - \frac{1}{2}, k + \frac{1}{2}\right) \right].
 \end{aligned} \tag{3.31}$$

- If f is located on the edge, for instance along x , its value at the center is interpolated using the formula

$$\begin{aligned}
 f(i, j, k) = \frac{1}{4} & \left[f\left(i, j - \frac{1}{2}, k - \frac{1}{2}\right) + f\left(i, j + \frac{1}{2}, k - \frac{1}{2}\right) \right. \\
 & \left. + f\left(i, j - \frac{1}{2}, k + \frac{1}{2}\right) + f\left(i, j + \frac{1}{2}, k + \frac{1}{2}\right) \right].
 \end{aligned} \tag{3.32}$$

- To interpolate from the center to the edge, the expression used is

$$\begin{aligned}
 f(i, j + \frac{1}{2}, k + \frac{1}{2}) = \frac{1}{4} & [f(i, j, k) + f(i, j + 1, k) \\
 & + f(i, j, k + 1) + f(i, j + 1, k + 1)].
 \end{aligned} \tag{3.33}$$

- Finally, to interpolate from the center to a vertex:

$$\begin{aligned}
 f(i, j, k) = & \frac{1}{8} [f(i, j, k) + f(i, j + 1, k) \\
 & + f(i, j, k + 1) + f(i, j + 1, k + 1) \\
 & + f(i + 1, j, k) + f(i + 1, j + 1, k) \\
 & + f(i + 1, j, k + 1) + f(i + 1, j + 1, k + 1)] .
 \end{aligned} \tag{3.34}$$

In particular, **Spritz** uses these interpolation within the following scheme:

1. the functions Ψ_{mhd} and A_j are interpolated at the cell centers, in order to compute the functions F_{Ψ}^j and G_A ;
2. then, they are interpolated back to the original locations;
3. after the interpolation, the values are subjected to a numerical differentiation by means of a *centered difference scheme*, which means that values of the variables in neighboring points are used. For example, the derivative of G_A along x direction on the edge $(i, j + \frac{1}{2}, k + \frac{1}{2})$ is

$$\frac{d}{dx} G_A \equiv \frac{G_A(i + \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2}) - G_A(i - \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2})}{\Delta x} . \tag{3.35}$$

In this section, I focused on the numerical methods implemented in the **Spritz** code, which I employed in the first part of my work. As I said before, the final part of my simulations has been performed using **WhiskyTHC**, which is a GRHD code and does not deal with magnetic fields. For a detailed description of the algorithms employed in **WhiskyTHC**, see [3].

3.3 Initial Data: Kadath

In a numerical simulation, before evolving the system by means of evolution equations, one has to generate an initial configuration which satisfies the constraint equations and accurately describes the physics of interest. For all the simulations I performed, initial data have been generated using **Kadath**. **Kadath** is a library written in C++ implementing methods for solving a variety of problems in theoretical physics, using different types of coordinates and geometries. Partial differential equations are discretized through spectral methods, and the resulting system solved using *Newton-Rapson* algorithms. Hence, the library is able to deal with strongly non-linear situations, such as problems in General Relativity. See [50] for a detailed description of the library. Among the tools provided by this library, a set of *initial data solvers* have been developed. In particular, the Frankfurt/Kadath (FUKA) initial data codes provide initial data for asymmetric and highly-spinning compact binaries such as binary black holes, binary neutron stars and black hole-neutron star binaries.

Chapter 4

Numerical Simulations: Spritz

In this chapter, I present the results obtained performing numerical simulations using **Spritz**. This part is a continuation of the work carried out by Alice Gambaro in her Master’s thesis [6]. The original intent of her thesis was to simulate - by means of **Spritz** - a binary neutron star system using the hot version of the tabulated *Bombaci-Logoteta* (BLh) EoS, which would have produced a short-lived neutron star remnant after the coalescence. However, the system reached the merger after an unexpectedly short time, in contrast with results present in the previous literature. Despite performing a series of tests to find out the origin of the issue, they did not prove effective.

Therefore, I tried to furtherly examine this “fast-inspiral problem”. Here, after an introductory section summarizing the results obtained by Alice Gambaro, I describe the strategies adopted to tackle the issue. Unfortunately, even in my case, all the tests performed in order to identify its origin have been unsuccessful.

4.1 Fast-Inspiral Problem

As already mentioned, this part of my work begins carrying on the investigations performed by A. Gambaro in her Master’s thesis [6]. These aimed to shed light on the origin of an issue arising in numerical simulations of binary neutron stars with a hot tabulated equation of state, the *Bombaci-Logoteta* (BLh) EoS, performed via the **Spritz** code. From now on, I will refer to this issue as “fast-inspiral problem”, since it is related to the fact that the neutron stars forming the binary system manifest an unexpectedly short inspiral phase.

4.1.1 Initial Data and Computational Setup

The original aim of the work [6] was to produce a short-lived BNS remnant, a Hyper-massive Neutron Star (HMNS), thought be the central engine of Short Gamma-Ray Bursts (SGRBs).

The equation of state of neutron star matter employed in the simulations - the BLh EoS - is a new finite-temperature EoS derived in the context of the non-relativistic many-body Brueckner-Hartree-Fock (BHF) approximation [51]. It is an extension of the zero-temperature and β -equilibrated version presented in [52] and used for binary neutron star mergers in [53]; also the BLh EoS has already been employed in BNS (e.g. [54] and [55]). In this EoS, the interactions between nucleons are described by means of a potential derived perturbatively in the chiral effective field theory.

In particular, A. Gambaro generated the initial dataset for the simulations using a 0.1 MeV *slice* of the BLh EoS. Then, on the basis of physical considerations on the relation between the total baryonic mass and the maximum mass for a TOV inspired by the paper [56] - in which the system had produced a HMNS which launched a relativistic jet - she deduced that a total baryonic mass $\sim 1.31M_{\text{TOV}}$ might have produced a hypermassive neutron star. As a result, a stability configuration (along a TOV sequence of neutron star models) satisfying that condition has been selected. This neutron star model was characterized by the following parameters:

- baryonic mass $M_b = 1.625738M_{\odot}$;
- radius $R = 12.363066$ km;
- logarithm of central enthalpy $\log_{10} h_c = 0.241414$.

This model has been used to generate initial conditions for an equal mass *irrotational* BNS system, by means of LORENE, the main code for delivering general relativistic initial data of binary systems containing neutron stars. The two neutron stars were set at an initial coordinate separation between the centers of 45 km^1 , in order to follow the last few orbits and the merger.

The initial data has been evolved on a square grid, having a size of $L \sim 5670$ km per side ($L = 3840$ in **Cactus** units), with a resolution of $dx \sim 295$ m ($dx = 0.2$ in **Cactus** units) on the finest refinement level.

For a more detailed discussion see [6]. Here I will focus on the main issue encountered throughout the evolution, i.e. the fast-inspiral problem.

4.1.2 Investigating the Issue

The evolution of the BNS system was characterized by a short inspiral phase: the neutron stars reached the merger after a few ms (only ~ 1.5 orbits). This was unexpected, especially considering the previous literature. For instance, in [53] - where the cold version of BL EoS has been employed - the BNS system merged after ~ 6 orbits; in [55] the original separation of 45 km corresponds to $\sim 3 - 4$ orbits before merger. Therefore, in order to investigate the origin of this unexpected behaviour, A. Gambaro performed a series of tests, which I briefly summarize here.

¹This value was inspired by [55].

Different Resolutions

Three simulations have been performed with same initial data and parameters, changing the *resolution* only. The following values for the resolution have been used:

- $dx \sim 295$ m ($dx = 0.20$ in **Cactus** units)
- $dx \sim 221$ m ($dx = 0.15$ in **Cactus** units)
- $dx \sim 162$ m ($dx = 0.11$ in **Cactus** units)

Result: changing the resolution does not have an influence on the rapidity of the inspiral².

Different Courant Factors

Two simulations have been performed with same initial data and parameters, changing the *Courant factor* only. The following values for the Courant factor have been used:

- $C_F = 0.35$
- $C_F = 0.15$

Result: changing the Courant factor does not have an influence on the rapidity of the inspiral.

OpenMP vs MPI

Two simulations have been performed with same initial data and parameters, changing the *parallelization scheme*, i.e. the handling of the operations needed to run the simulation on multiple processors. Specifically, the following parallelization schemes have been adopted:

- Pure MPI
- Hybrid MPI + OpenMP

Result: changing the parallelization scheme does not have an influence on the rapidity of the inspiral.

²I should also note that in the two latter simulations the magnetic field was turned off, but this did not impact the evolution of the system.

Different Datasets

Two simulations have been performed employing different *initial datasets*³:

- SMNS initial data (ID for producing a Supermassive NS remnant)
- HMNS initial data (ID for producing a Hypermassive NS remnant)

Result: although small differences in the evolution are visible, which could be easily explained by the different initial masses, changing initial datasets does not have an influence on the rapidity of the inspiral.

Ideal Fluid Test

At this point, A. Gambaro performed another test with different initial conditions, in order to furtherly explore a possible relation between ID generated via **LORENE** and the fast-inspiral problem. In particular, this test aimed to reproduce the ID employed in the Missing Link paper [56] (referred to as ML ID) and evolve them using an Ideal Fluid EoS, which is an analytic EoS. In this case, the BNS system reached the merger after ~ 3 orbits, showing an evolution consistent with [56]. This suggested to exclude a problem related to the initial data, given that **Spritz** is able to properly simulate the evolution of ID generated via **LORENE** in the Ideal Fluid EoS case. Since one could argue that tabulated EoS ID might be less accurate, she also compared the violation of the Hamiltonian constraint of the simulations with ML ID and HMNS ID at $t = 0$, finding no substantial differences.

Result: the problem does not seem to lie in the initial data, since in the case of the Ideal Fluid EoS the ID generated via **LORENE** are evolved as expected; moreover, the Hamiltonian constraint violation test seems to suggest that also the BLhot EoS ID generated via **LORENE** are delivered correctly: the problem could instead reside in the evolution of these tabulated EoS ID with **Spritz**.

Since the resolution of this problem is crucial to realistically describe the evolution of BNS system, it is convenient to explore all the possibilities. Therefore, even the last Ideal Fluid test suggested that the issue did not seem to lie in initial conditions provided by **LORENE**, there was still room for inspecting whether the issue could arise from initial data. Here I report the words used by A. Gambaro in the final part of her investigation: “a possible path for future tests would be to generate an already known set of ID with a different code and check whether the issue persists in that case as well”. Indeed, this has been the starting point of my examination.

³While in the SMNS case the temperature was kept fixed, in the HMNS simulation it was let free to evolve, as this variable seemed to have no impact on the “rapidity” of the inspiral.

4.2 Trial 1: *Kadath* Initial Data

My work begins carrying on these investigations. Specifically, I first explored the possibility that the fast inspiral problem could be related to the fact that initial data generated with **LORENE** were not formulated or read correctly in the case of tabulated EoS, given that **Spritz** did not show any troubles in dealing with analytic EoS, such as the Ideal Fluid EoS. Following the path suggested by A. Gambaro, I generated an already known set of initial data by means of a different code, so as to check whether the problem persisted or not. As already mentioned previously, I used the code **Kadath** instead of **LORENE** for generating the same set of HMNS initial data used by A. Gambaro.

For all the tests I present in this chapter, I adopted the same computational setup described before, i.e. a square grid of size $L \sim 5670$ km per side ($L = 3840$ in **Cactus** units), with a resolution of $dx \sim 295$ m ($dx = 0.2$ in **Cactus** units) on the finest refinement level.

In this first simulation, I did not find substantial differences in the evolution of the system. In fact, the BNS system with **Kadath** initial data evolves in a similar manner with respect to the one with **LORENE** initial data: as it is shown in Figure 4.1 and in Figure 4.2, the two neutron stars merge after slightly more than 1.5 orbits (~ 4 ms), as in the previous case. Moreover, from the snapshots in Figure 4.3 and in Figure 4.4, comparing the rest-mass density on xy and xz planes at equal times, the binary neutron stars seem to follow an almost identical evolution, with modest differences probably due to the differences between **LORENE** initial data and **Kadath** initial data.

In this chapter I will not focus on the behaviour of other physical quantities since the main purpose of these tests were to examine the duration of the inspiral. In this particular case, as the simulation evolution appeared to be essentially identical to the one performed by A. Gambaro, for a detailed description of the physics extracted by this simulation see [6].

To sum up, this first test excluded that the fast inspiral problem could be related to the initial data generated using **LORENE**. In fact, the BNS system manifested the same behaviour when the initial data are generated with a different code, in this case **Kadath**.

Since the method for generating initial data (**LORENE** or **Kadath**) did not affect the evolution of the system, I continued to use **Kadath** for generating initial data for the other simulations as well (see the next sections).

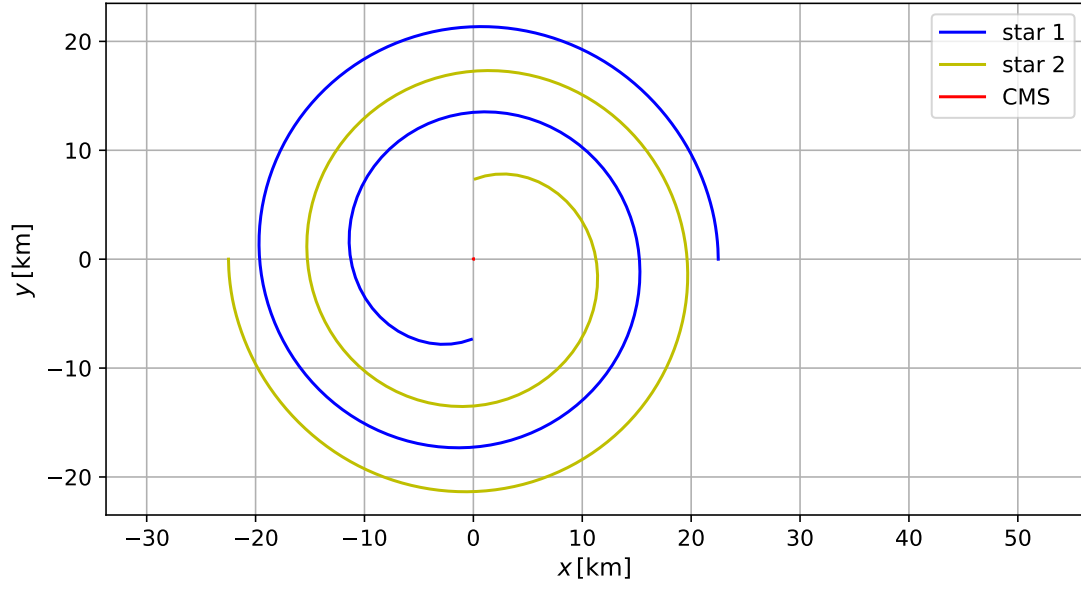


Figure 4.1: Orbits followed by the neutron stars up to the merger.

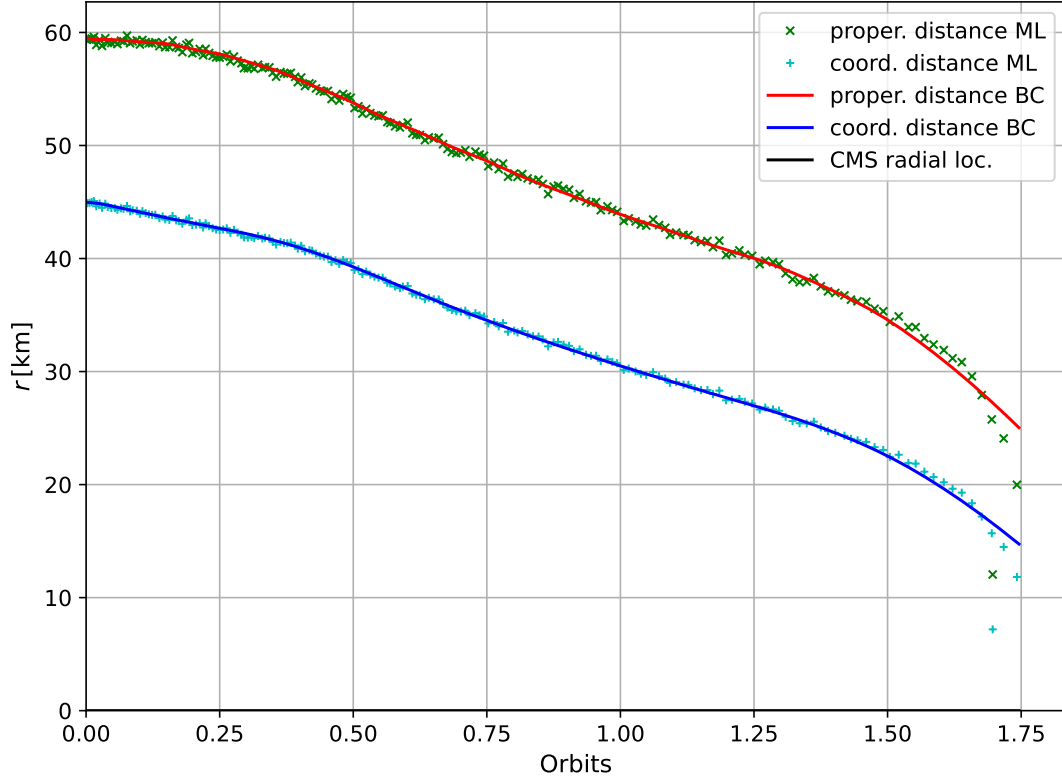


Figure 4.2: Coordinate and proper separation as a function of the number of orbits. ML is referred to density maxima, BC means barycenters while CMS is the center of mass.

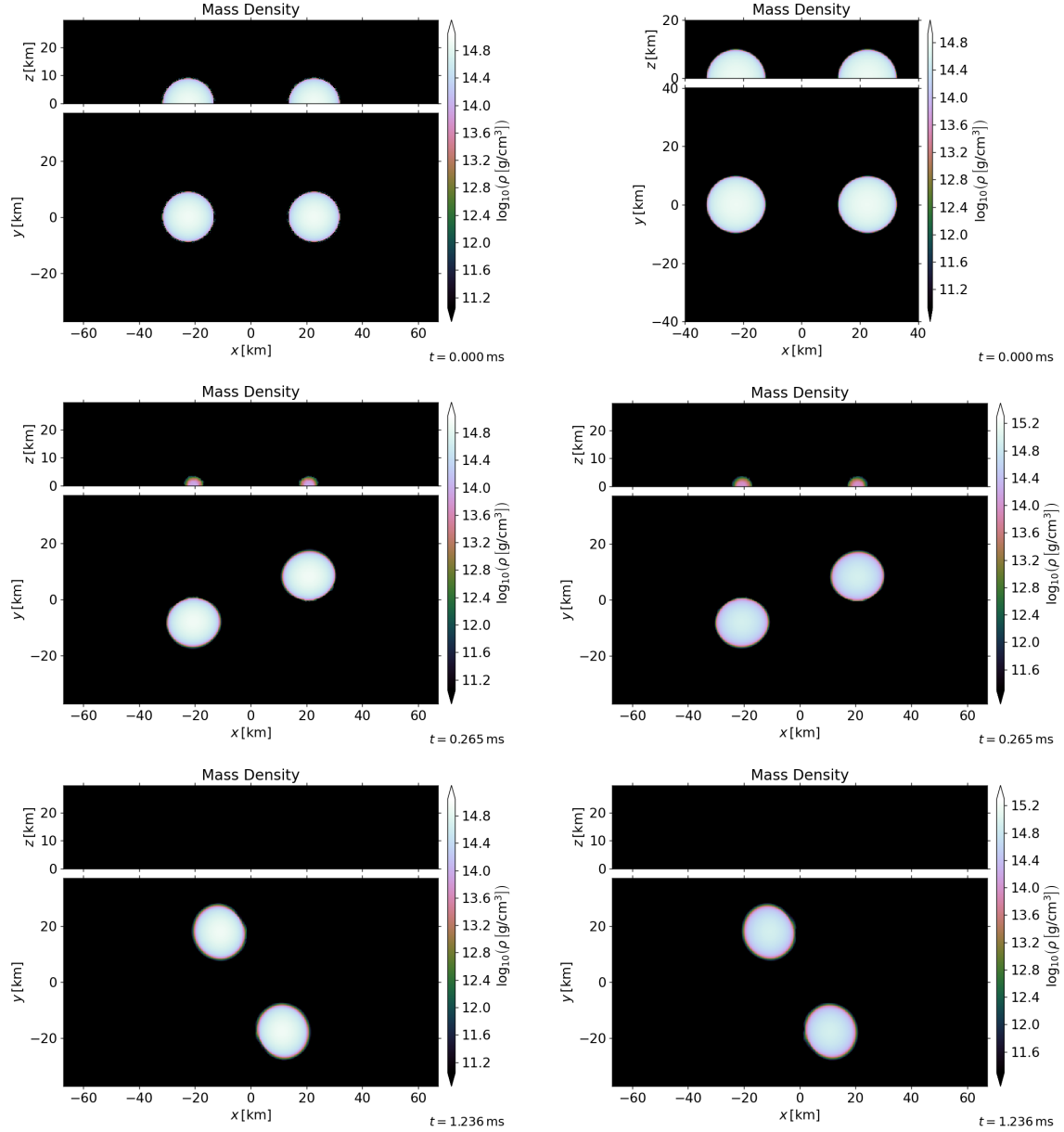


Figure 4.3: Evolution of the rest-mass density (in logarithmic scale) on the xy and xz planes, in the first part of the inspiral. The snapshots on the left are relative to LORENE initial data, those on the right come from the simulation with Kadath initial data.

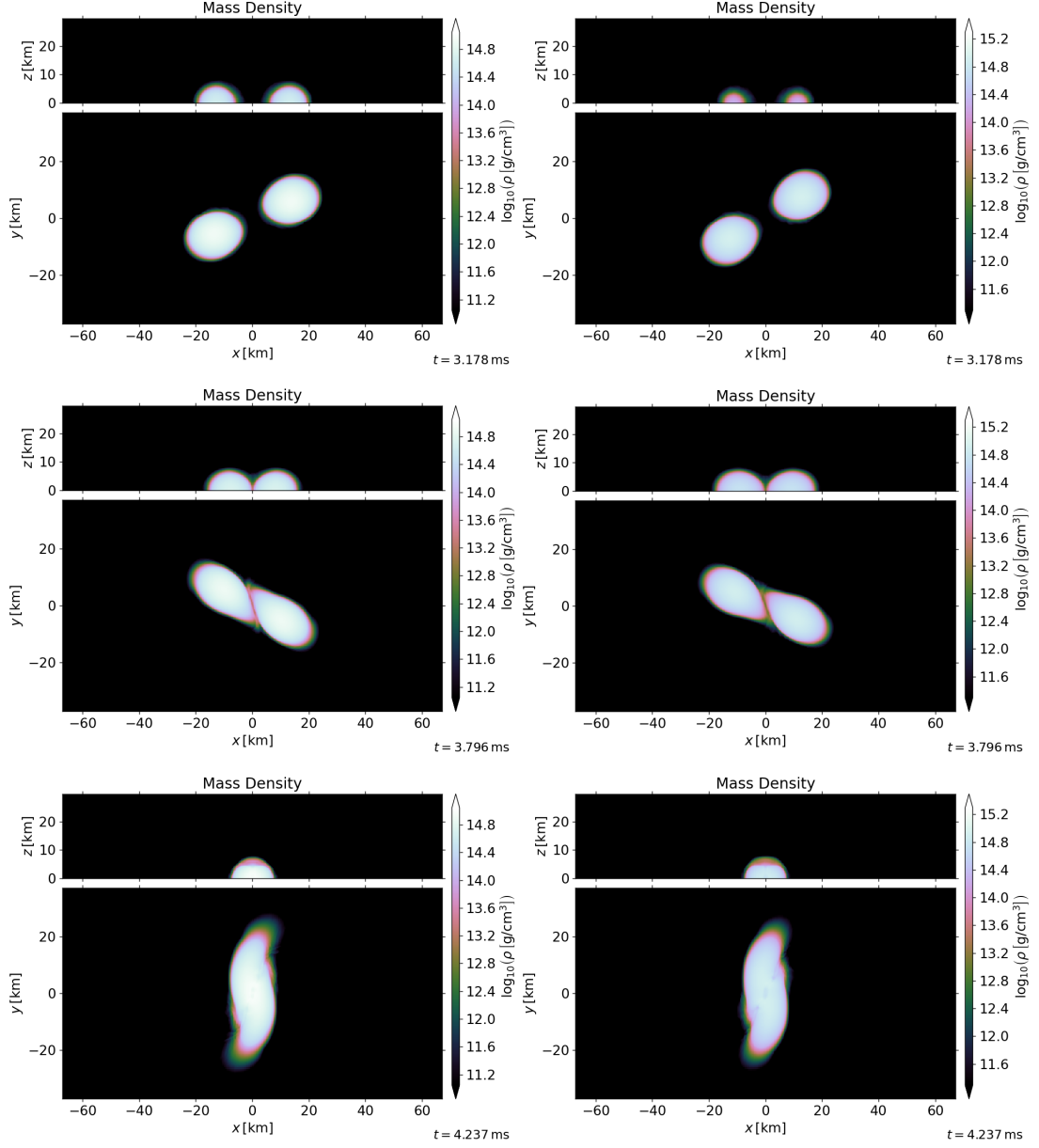


Figure 4.4: Evolution of the rest-mass density (in logarithmic scale) on the xy and xz planes, in the last part of the inspiral up to the merger. As before, the snapshots on the left are relative to LORENE initial data, those on the right come from the simulation with Kadath initial data.

4.3 Trial 2: Changing Tabulated EoS

Having excluded that the origin of the issue lied in the initial data and considering all the other tests, it seemed likely that the trouble had to do with **Spritz** treatment of tabulated EoS, or at least of *this* tabulated EoS. At this point, a natural check was to verify whether the BNS system manifested the same behaviour when employing other tabulated equations of state to describe the neutron stars' matter. For this test, I picked up another tabulated EoS, known as DD2, which was among the equations of state used in [55].

The DD2 is an EoS based on the relativistic mean-field (RMF) theory of high-density nuclear matter (see [55] and references therein). It contains neutrons, protons, light nuclei such as deuterons, helions, tritons, and alpha particles, and heavy nuclei in nuclear statistical equilibrium. In particular, it is a *stiff* EoS, which means that the pressure increases a lot for a given increase in density, denoting also that the star is harder to compress.

For this test I employed new “DD2 initial data” generated by means of **Kadath** for a system with same gravitational mass and coordinate separation used in the BLhot case, changing the equation of state only.

I evolved the simulation up to the merger. The results are shown in Figure 4.5 and Figure 4.6. It is clear that, again in this case, the binary neutron star system evolution is characterized by an extremely rapid inspiral phase. As said before, the DD2 was among the equations of state employed in [55]: in the paper we find that the same initial separation between the two stars should allow the completion of the last 3 – 4 orbits before merger. However, in this instance as well, I have found that the two stars merge after ~ 1.5 orbits (slightly less than the BLh case), corresponding to ~ 4 ms.

This check seems to suggest that the problems arising when **Spritz** deals with the BLh EoS are not attributable to that equation of state. Since the evolution of the system exhibited the same behaviour also when utilizing the DD2 EoS, it appears to be a more general problem, not related to a specific tabulated equation of state.

In the next section, I present the last trial I performed in order to investigate the issue: changing the method for evolving the spacetime.

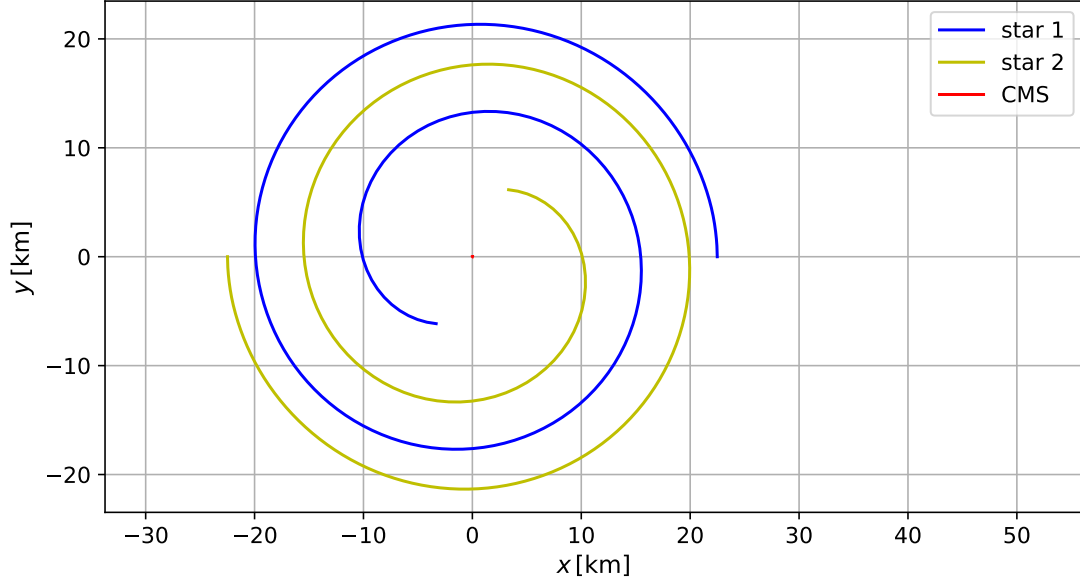


Figure 4.5: Orbits followed by the neutron stars up to the merger.

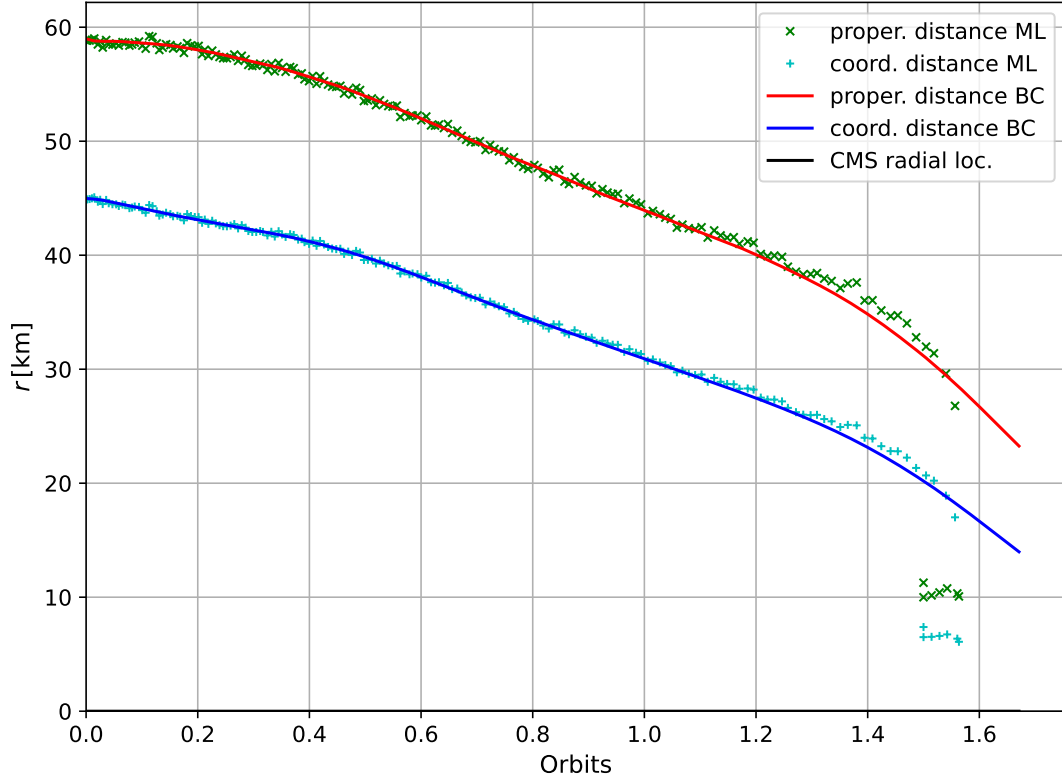


Figure 4.6: Coordinate and proper separation as a function of the number of orbits.

4.4 Trial 3: Changing Spacetime Evolution

Although the previous tests seemed to suggest that the issue was related to how *Spritz* manages tabulated equations of state, I performed another test in order to explore all the possibilities. In particular, this is related to the evolution of the spacetime.

As already explained in the previous chapter, the Einstein Toolkit provides methods for the evolution of the spacetime. In particular, this is provided by the thorn *McLachlan*, which is able to evolve BSSN equations as well as CCZ4 equations for the evolution of spacetime quantities (see Chapter 2 for BSSN and CCZ4 formulations of General Relativity). In the simulations I presented so far, I used the module *ML_BSSN*. For this test, where I used the DD2 EoS (keeping fixed all the rest), I switched to *ML_CCZ4*, by setting the following parameters:

```
ML_CCZ4::dampk1 = 0.05
ML_CCZ4::dampk2 = 0
ML_CCZ4::GammaShift = 0.5
```

which correspond respectively to the parameters $\kappa_1, \kappa_2, \kappa_3$ in CCZ4 formulation (see the section dedicated to CCZ4 in Chapter 2). The choices of κ_1 and κ_2 are “standard” values already used in previous BNS simulations (see e.g. [57]), while the value for κ_3 was inspired by [58]⁴.

The results are shown in Figure 4.7-4.8-4.9. From Figure 4.7 it is clear that there is no significative difference with respect to the BSSN evolution.

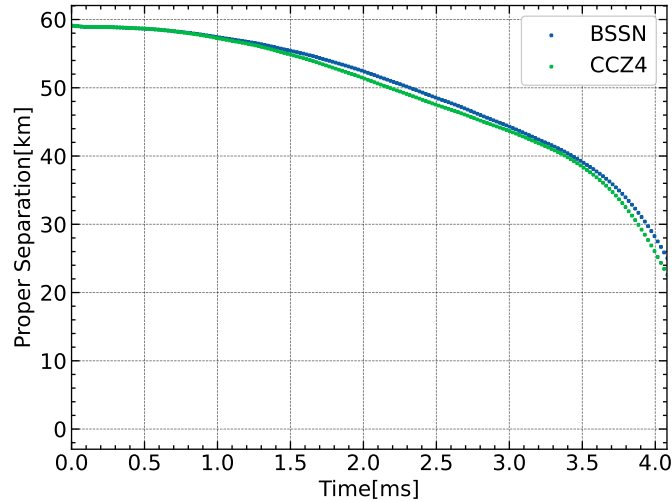


Figure 4.7: Proper separation comparison between BSSN and CCZ4.

⁴In particular, this choice led to accurate and stable evolutions for binary black hole spacetimes.

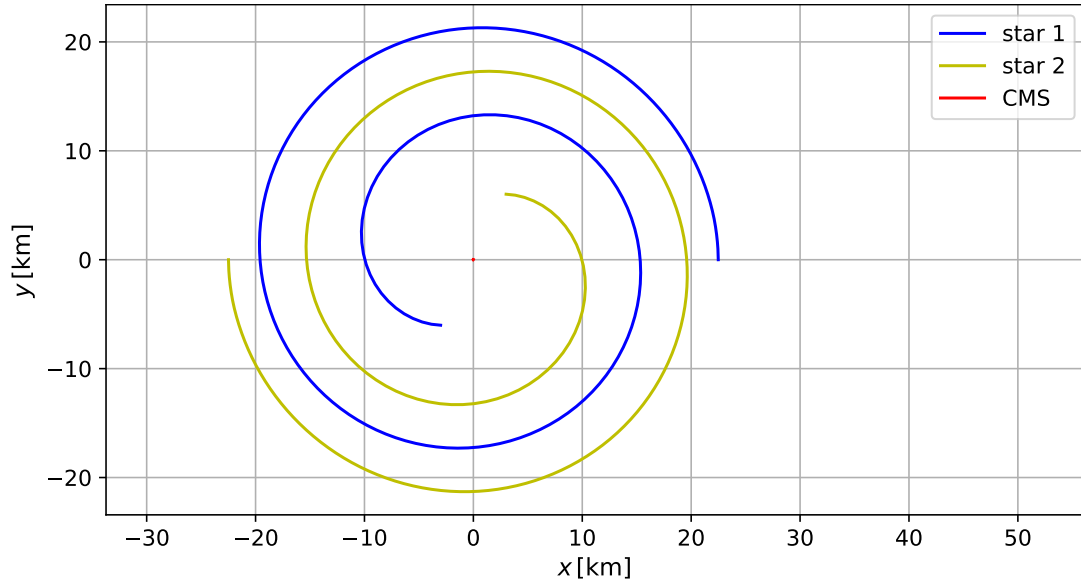


Figure 4.8: Orbits followed by the neutron stars up to the merger.

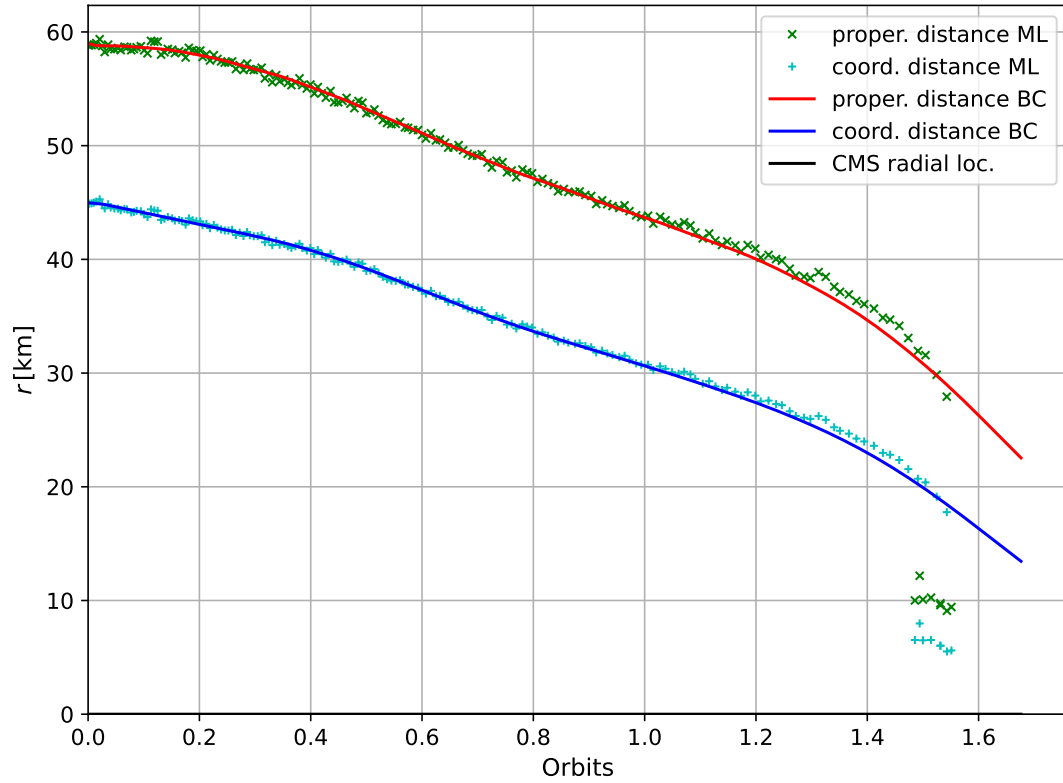


Figure 4.9: Coordinate and proper separation as a function of the number of orbits.

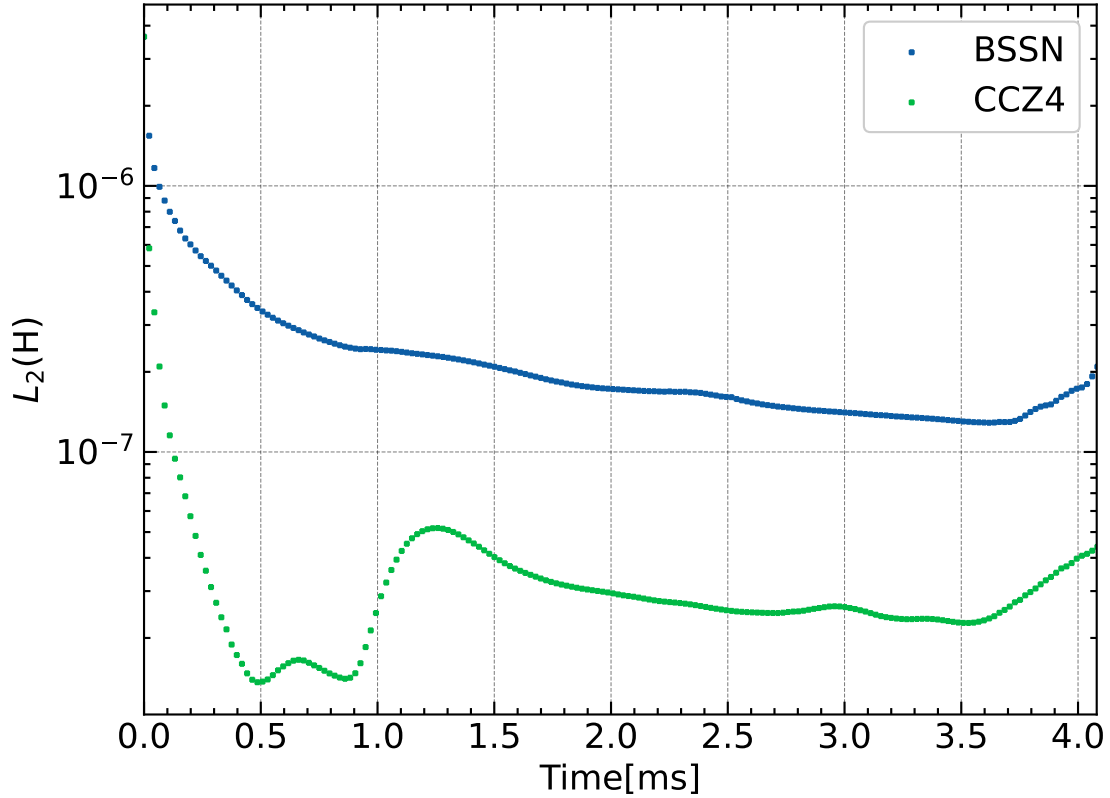


Figure 4.10: Evolution of L2-norm of the Hamiltonian constraint (in logarithmic scale) for BSSN and CCZ4.

Although the evolution of the system is pretty similar in the two cases, it can be interesting to look at the constraint violations, which are supposed to be different for BSSN and CCZ4 formulations (see Chapter 2). In order to do that, we can monitor the L2-norm of the Hamiltonian constraint, which is a global spacetime property. From Figure 4.10 it is clear that the CCZ4 shows a violation of the Hamiltonian constraint which is significantly smaller than for the BSSN case (the L2-norm of the Hamiltonian constraint being at least 1 order of magnitude smaller), leading to a more accurate solution of Einstein's equations. Thus, even if it was not the main purpose of the test, this simulation has confirmed that the CCZ4 formulation represents a considerable improvement over the standard BSSN formulation. I therefore have adopted the CCZ4 formulation for the simulations I will present in the next chapter.

This last test has proven that the fast-inspiral problem has nothing to do with the method employed for spacetime evolution. Instead, taking also into account all the tests previously carried out, I can conclude that it is very likely that the origin of the issue resides in **Spritz** treatment of tabulated equations of state. This is also confirmed by the results I will present in the next chapter, where I have evolved the same system (using the same initial data) by means of a different code, **WhiskyTHC**: in this case, the system does not manifest the fast-inspiral problem, and the evolution is coherent with results in literature.

Chapter 5

Numerical Simulations: WhiskyTHC

In this chapter I present the results obtained from the **WhiskyTHC** simulations. Specifically, I have performed two different simulations of the same BNS system described in Chapter 4, one employing the DD2 EoS and the other using the BLh EoS. As already stated at the end of the previous chapter, in this case the evolution of the system appears to be coherent with the expectations. In fact, in both cases, there is no fast inspiral and the neutron stars merge after $\sim 4 - 5$ orbits. This provides a confirmation that **Spritz** has an issue when dealing with tabulated EoS, given that adopting **WhiskyTHC** produces the expected results, very different from the **Spritz** case.

In addition, these two simulations provide an illustration of how the equation of state of neutron star matter can impact the evolution of a BNS system. In particular, while the BLh system collapsed to a black hole a few milliseconds after the merger, the DD2 system survived for longer times. However, due to lack of time and resources, I did not follow the complete evolution of the DD2 up to the collapse.

I then show a series of results, including gravitational wave signals and other physical quantities of interest, in order to make some discussion over the aftermath of the merger, in particular about properties of the merger ejecta, since they are crucial for determining the nature of electromagnetic emissions such as kilonovae.

5.1 DD2 EoS

Before describing the results, I remark that the computational setup of these two simulations is the one already described in Chapter 4. Moreover, I have used initial data generated by **Kadath**, as in the previous simulations.

I will start by describing the inspiral. Figure 5.1 and Figure 5.2 show that the neutron stars merge after ~ 4.5 orbits (almost 12 ms), to be compared with the ~ 1.5 orbits (~ 4 ms) of **Spritz**. In fact, in Figure 5.3 we see a remarkable difference in the evolution between **Spritz** and **WhiskyTHC**.

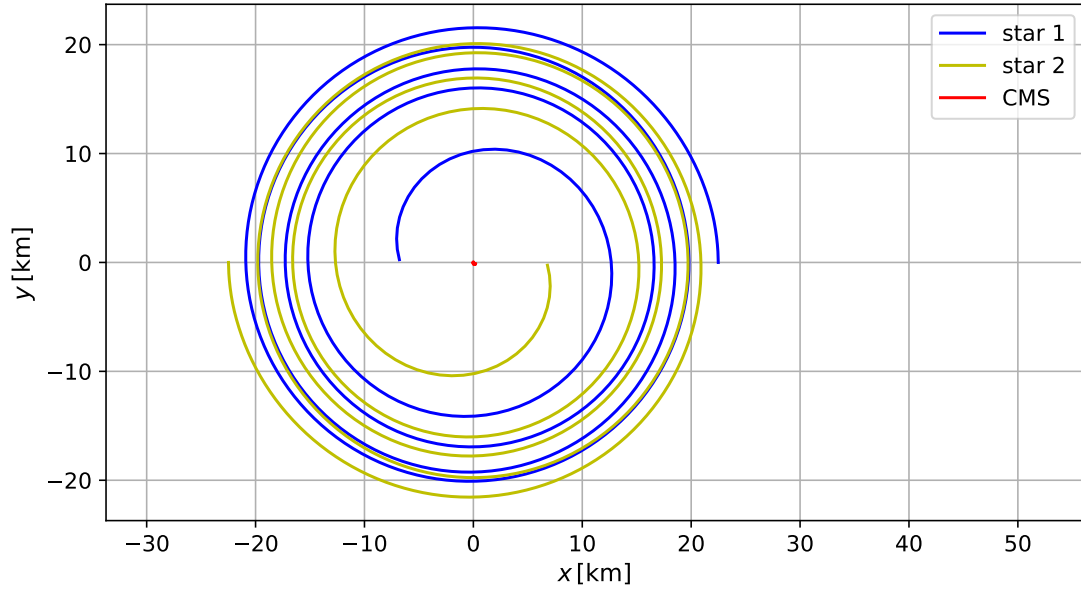


Figure 5.1: Orbits followed by the neutron stars up to the merger.

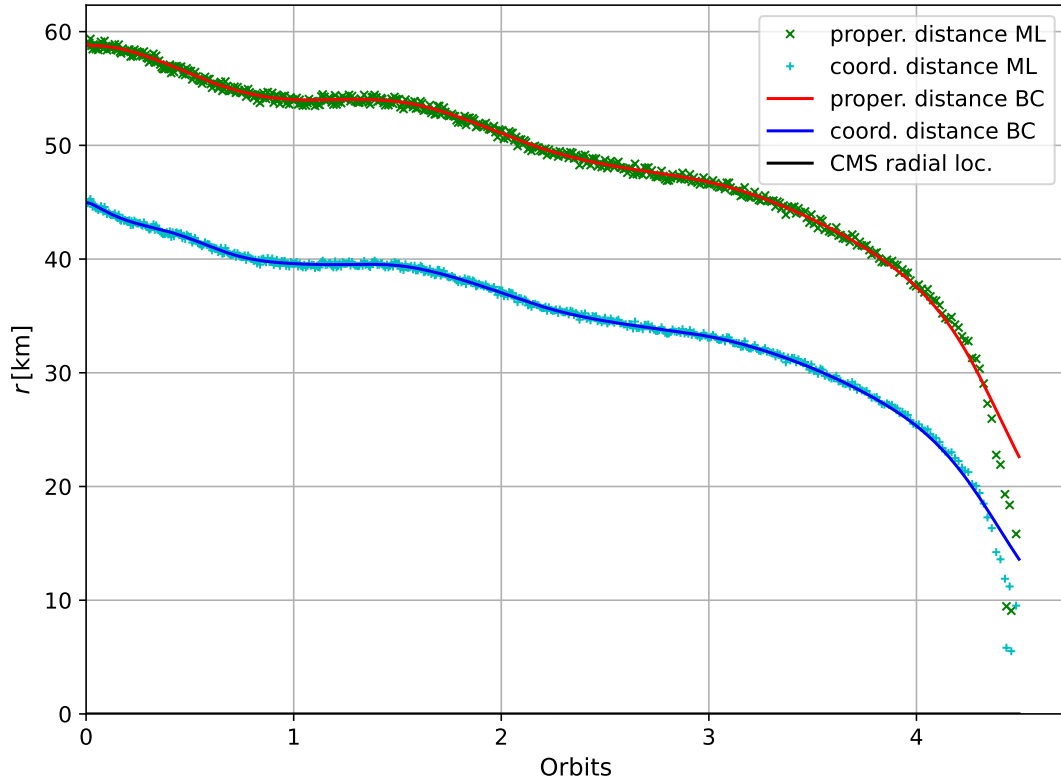


Figure 5.2: Coordinate and proper separation as a function of the number of orbits.

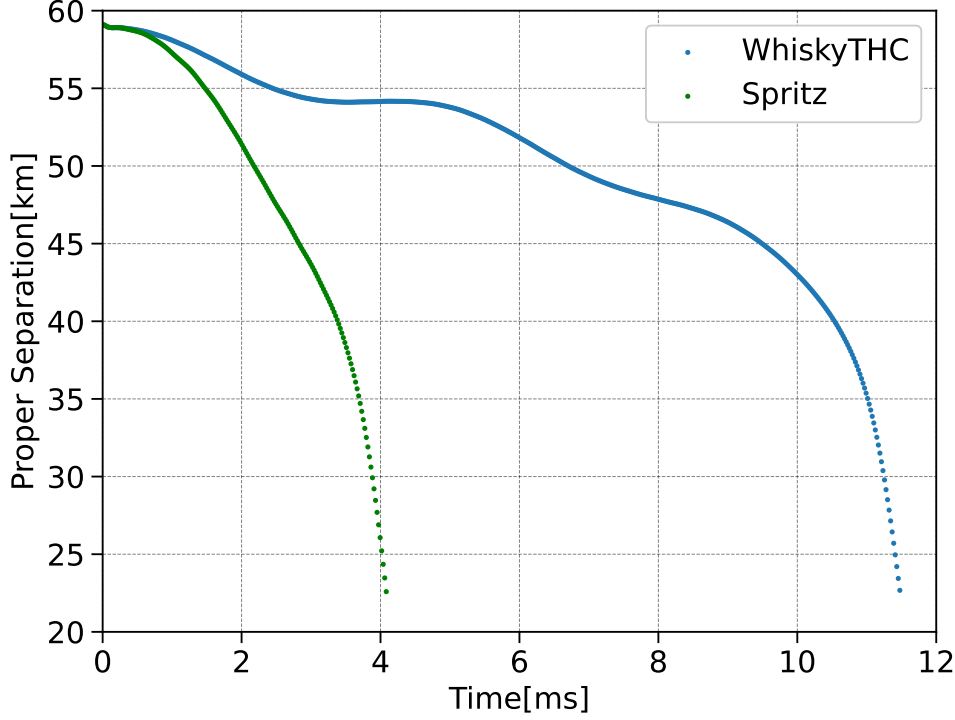


Figure 5.3: Proper separation comparison between **Spritz** and **WhiskyTHC**.

In Figure 5.4, six snapshots show the rest-mass density on xy and xz planes throughout the evolution of the system. At the beginning, the stars are at the initial coordinate separation of 45 km. Throughout the inspiral, it can be seen that - as the evolution advances - tidal forces begin to deform the stars (see the snapshot at $t = 9.710$ ms), which lose their spherical symmetry to assume a more “stretched” shape, clearly visible from the snapshot at 10.857 ms. Just before the merger, from the snapshot at $t = 11.828$ ms it is well discernible the formation of spiral arms due to tidal interactions (*tidal tails*). These zones are characterized by a lower density ($\rho \sim 10^{13} - 10^{14}$ g/cm³) and expand outwards in the equatorial plane due to transport of angular momentum by hydrodynamic processes after the merger (see the snapshot at $t = 15.889$ ms). This matter forms the disk surrounding the central remnant (see the last snapshot). In this case, the system has not promptly collapsed to a black hole, even if the increasing of the maximum of the rest-mass density seems to suggest that it will happen at some time (see Figure 5.5).

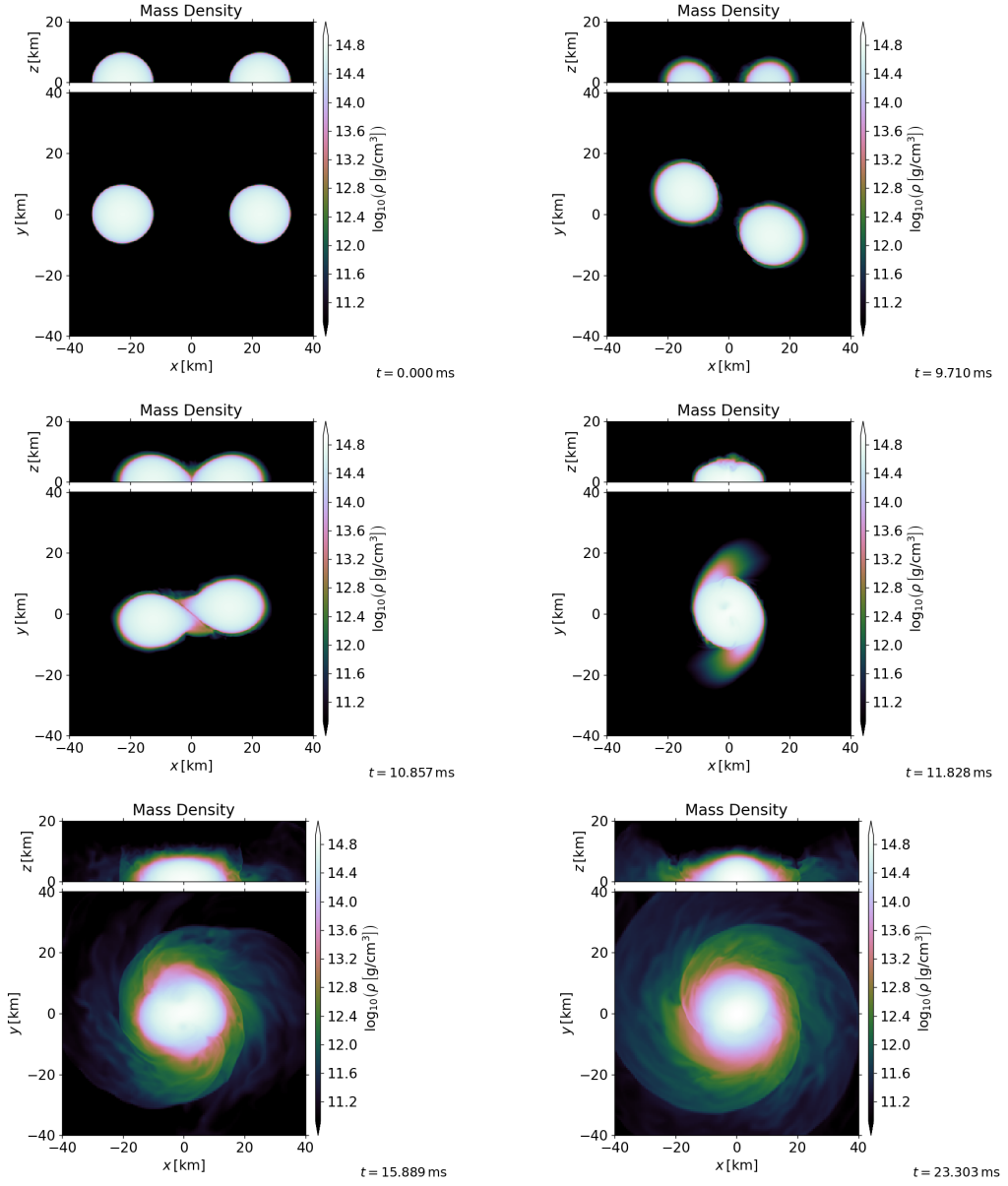


Figure 5.4: Evolution of the rest-mass density (in logarithmic scale) on the xy and xz planes.

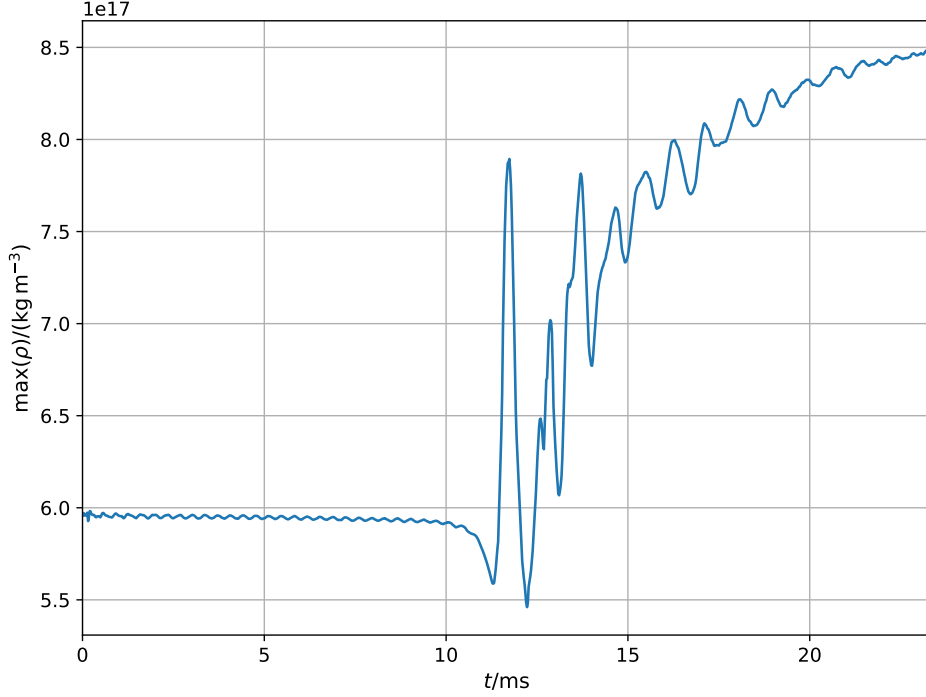


Figure 5.5: Evolution of the maximum rest-mass density.

It is also interesting to examine the evolution of the temperature (see Figure 5.6). At the beginning of the evolution, the stars are colder and have the temperature of the EoS slice used to generate the initial data. Throughout the evolution, the temperature is free to evolve and it is visible from Figure 5.6 that the surfaces of the stars heat up as they become closer. Just before the merger (see the snapshot at $t = 11.034$ ms), it can be seen a region with higher temperature between the two stars. The highest temperatures are reached during and after the merger; from there, the central remnant sheds mass and angular momentum outward through spiral density waves streaming from the shock interfaces [55]. This shock-heated matter is ejected in a different range of angular directions (see the snapshot at $t = 12.887$ s) and experiences the highest temperatures ($T > 10$ MeV).

Another important property of these dynamical ejecta is the electron fraction Y_e . Since this simulation does not take into account weak interactions, the matter ejected is characterized by $Y_e \lesssim 0.1$, sufficiently low to produce a robust abundance pattern for heavy nuclei with $A \gtrsim 130$ (see e.g. [59]). In Figure 5.7 it is visible that the ejected matter (*shock-driven* ejecta) has in fact $Y_e \lesssim 0.1$, as expected. These ejecta would contribute to the red kilonova component, having lower Y_e . However, including the effect of neutrino irradiation and $e^\pm$ capture, dynamical ejecta would have a wider electron fraction distribution ($Y_e \sim 0.1 - 0.4$), producing lighter r-process elements, contributing to the blue kilonova component.

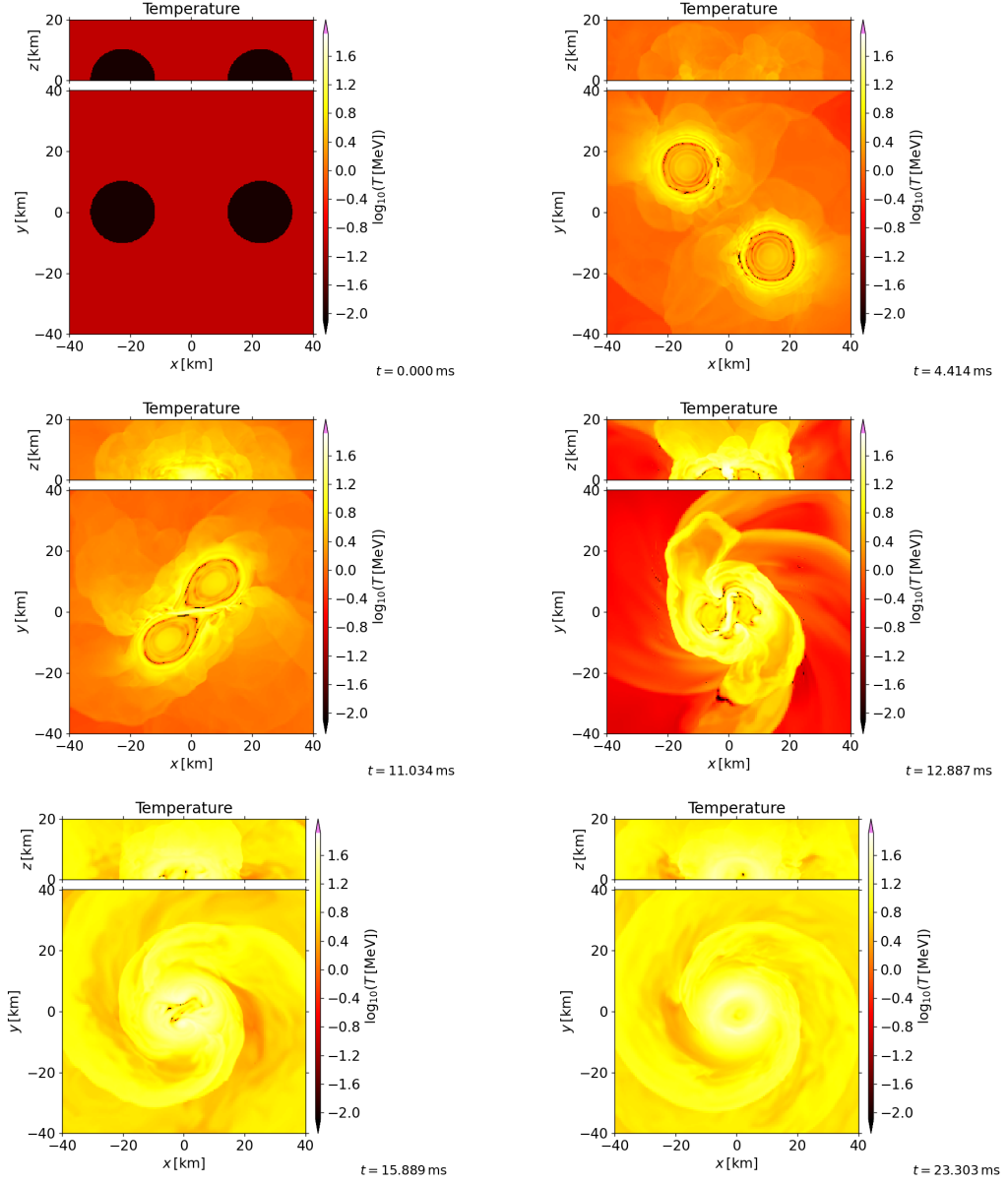


Figure 5.6: Evolution of the temperature (in logarithmic scale) on the xy and xz planes.

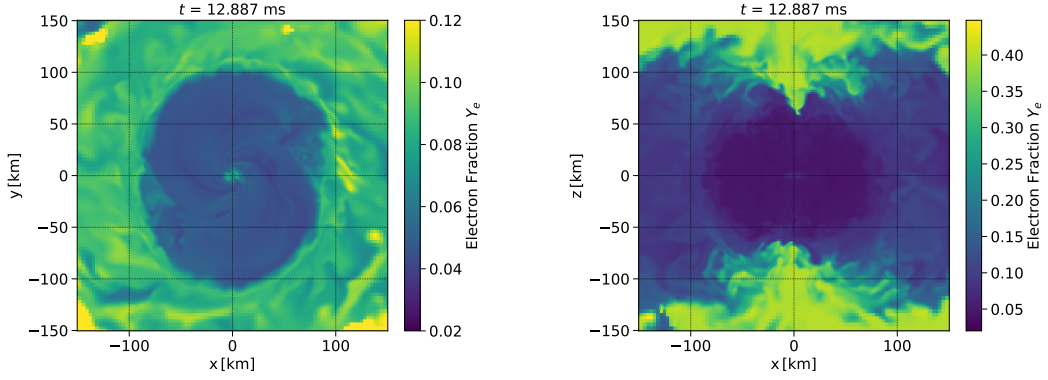


Figure 5.7: Evolution of the electron fraction on the xy and xz planes. In the right panel, the color scale has been changed due to the presence of regions with $Y_e \gtrsim 0.1$.

Gravitational Waves

One of the main purposes of Numerical Relativity simulations is the extraction of gravitational wave signals, providing templates which can be used to analyze and interpret signals revealed by gravitational waves interferometers.

I recall that, in the radiation zone, the strain of the gravitational waves is related to the Weyl scalar Ψ_4 through the relation

$$h = h_+ - ih_\times = - \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' \Psi_4. \quad (5.1)$$

Once extracted Ψ_4 - which is computed in the Einstein Toolkit by the thorn `WeylScal4` - I obtained the strain h through post-processing tools such as `PostCactus` [60] and `Kuibit` [61]. The gravitational wave strain can be decomposed in spin-weighted spherical harmonics. In particular, from Figure 5.8 it is clear that the dominant multipole component is $l = m = 2$, which - among all the components - carries away the greatest amount of energy. Therefore, we can focus on this particular component. For instance, in Figure 5.9, the top panel shows the “plus” and “cross” polarizations of the gravitational wave strain (estimated from Ψ_4 at the outermost detector) as functions of the retarded time $t - r$; in the bottom panel, it is represented the instantaneous frequency of the gravitational wave as a function of the retarded time, in which is clearly visible the distinctive *chirp* signal of an inspiraling binary. This signal, being of order of hundreds of Hertz, could be detected by LIGO and Virgo gravitational wave detectors. Note also that both the frequency and the amplitude of the gravitational wave increases during the inspiral. It is also evident that the merger, which occurred after 11 – 12 ms, is characterized by the maximum in the gravitational wave amplitude.

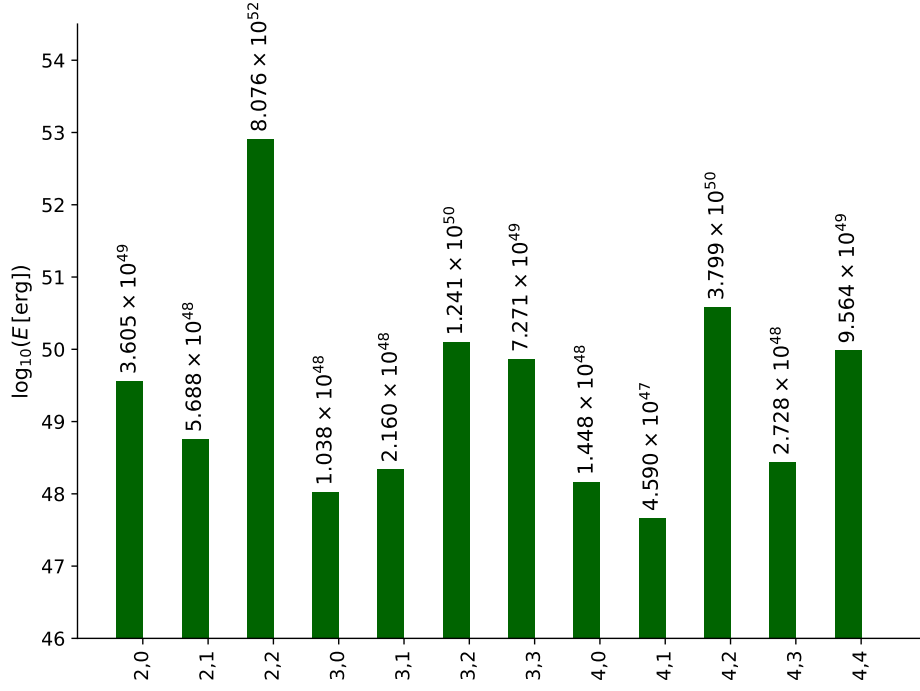
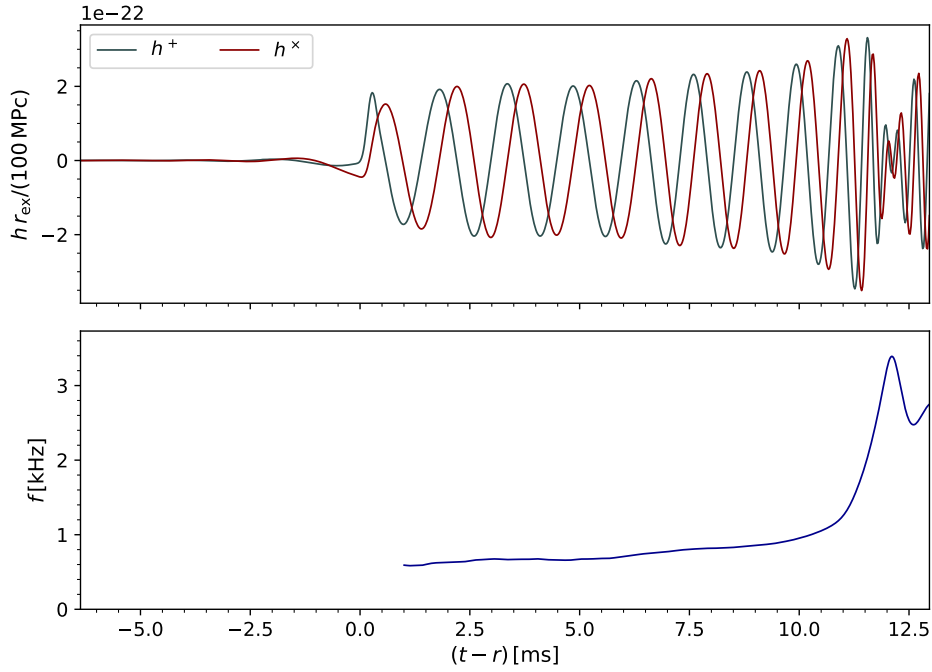


Figure 5.8: Total energy radiated for each multipole component.

Figure 5.9: Gravitational wave strain for $l = m = 2$ and evolution of instantaneous frequency.

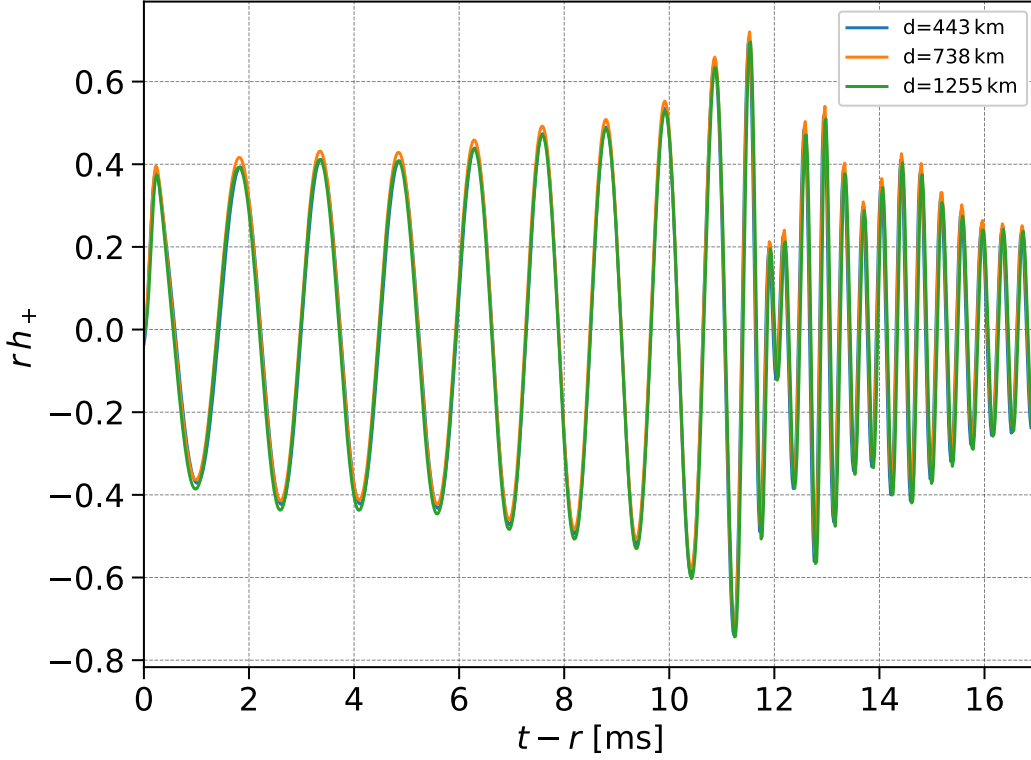


Figure 5.10: Gravitational wave strain for three different detectors.

In order to check whether it is actual gravitational radiation, one can extract the signal using detectors placed at different distances from the source. It is important that these detectors are sufficiently far from the source since the spacetime is highly perturbed near the location of the merger. If the signals overlap at different detectors, we are actually observing gravitational radiation. In particular, I have extracted the “plus” polarization of the gravitational wave strain ($l = m = 2$) from detectors at 443 km, 738 km and 1255 km. Figure 5.10 shows that the signals overlap very well, confirming that it is actual gravitational radiation.

5.2 BLh EoS

I am now going to present the results of the BLh simulation.

First of all, Figure 5.11 and Figure 5.12 show that the BNS system reaches the merger after ~ 5 orbits (~ 13 ms), to be compared with (slightly more than) 1.5 orbits (~ 4 ms) of *Spritz*. Again in this case, from Figure 5.13 is evident a remarkable difference between *Spritz* and *WhiskyTHC*.

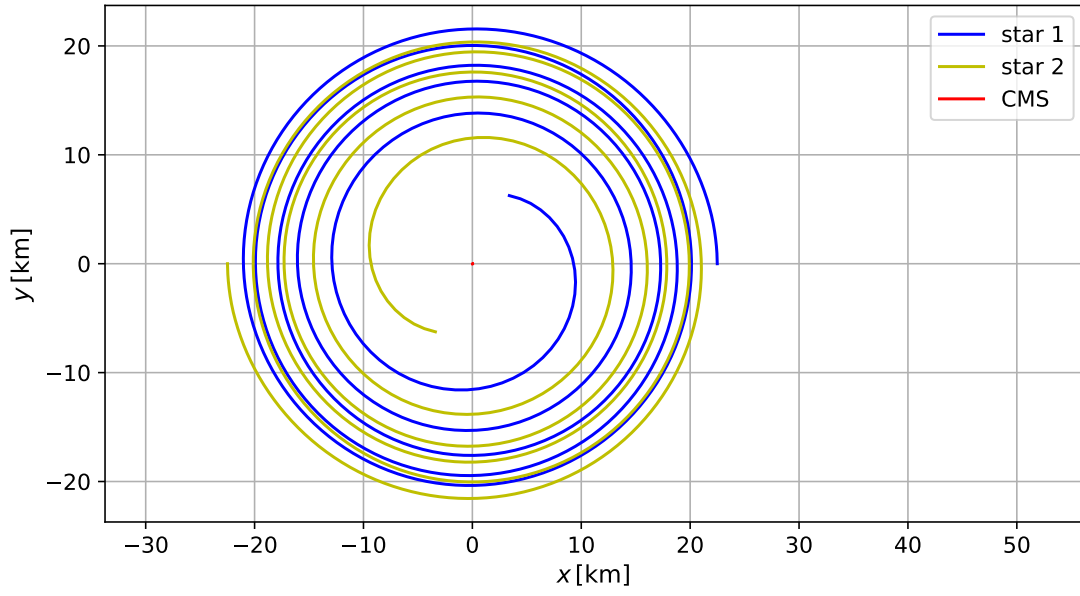


Figure 5.11: Orbits followed by the neutron stars up to the merger.

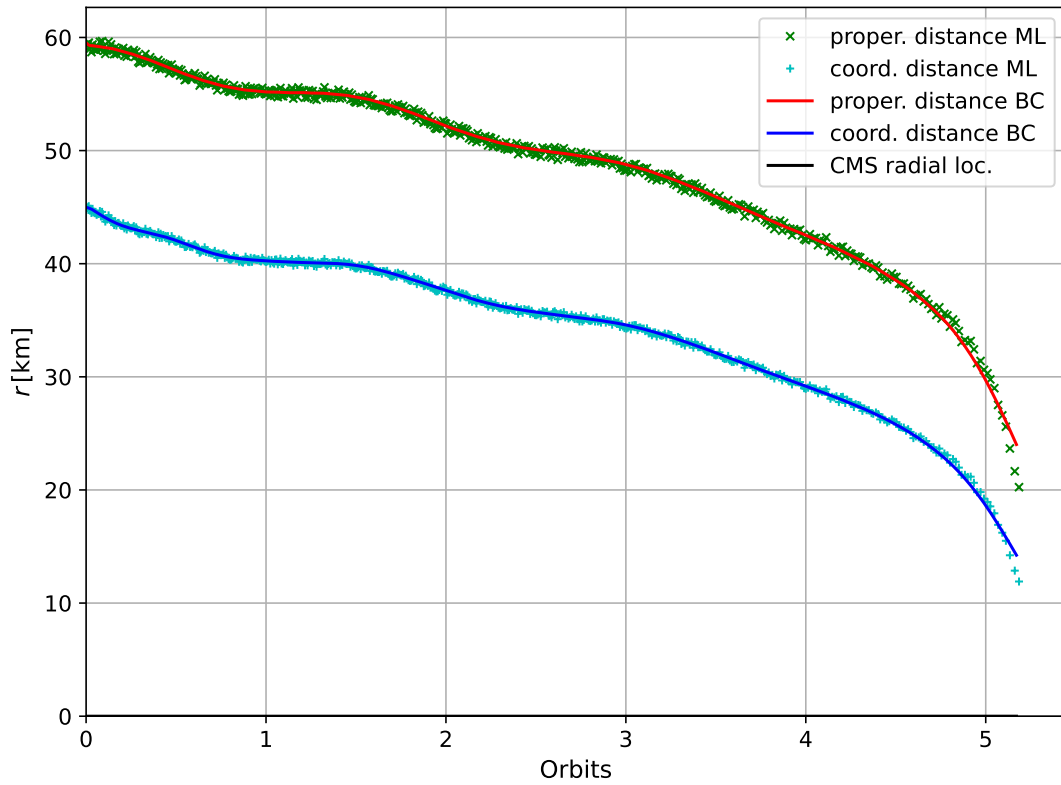


Figure 5.12: Coordinate and proper separation as a function of the number of orbits.

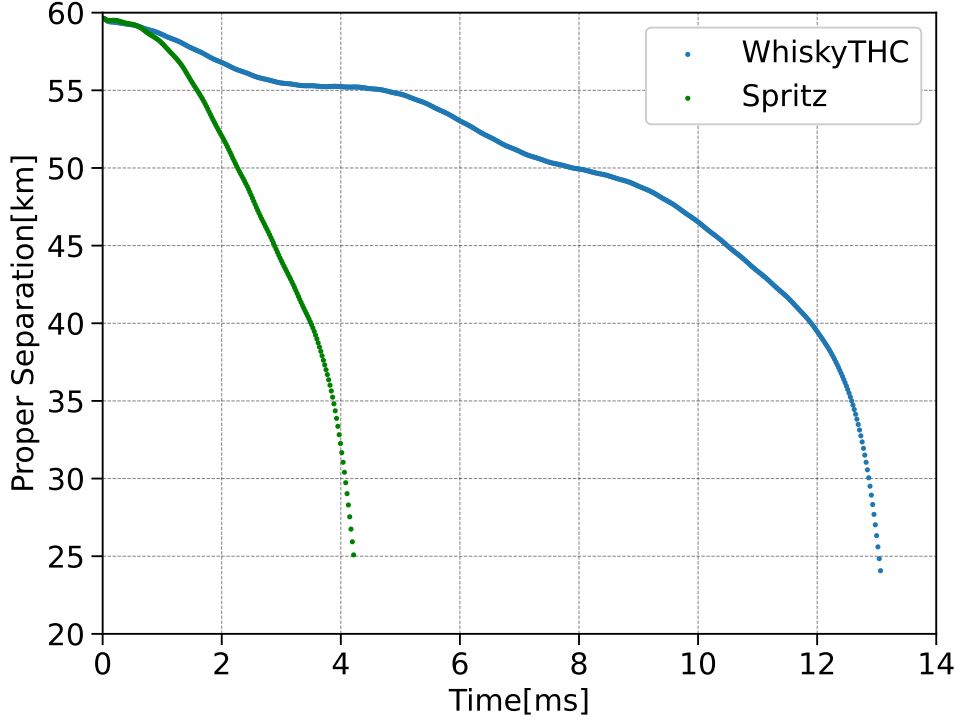


Figure 5.13: Proper separation comparison between **Spritz** and **WhiskyTHC**.

The evolution of the system is similar to the DD2 case in the inspiral phase. In fact, from Figure 5.14 we see that, during the inspiral, the two neutron stars begin to be deformed by tidal forces (see the snapshot at $t = 11.387$ ms) and assume a more elongated shape (see the snapshot at $t = 12.623$ ms). As before, close to the merger, is visible the formation of tidal tails of matter with lower density (snapshot at $t = 13.439$ ms), which then expands outwards in the equatorial plane, forming the disk surrounding the central remnant.

The fate of the central remnant is different from the DD2 case though. Indeed, a few ms after the merger, it collapses to a black hole (see the snapshot at $t = 15.182$ ms). This is coherent with expectations, since the parameters of the system had originally been chosen to produce a short-lived neutron star remnant (Hypermassive Neutron Star) which would have collapsed to a black hole. In particular, the newly formed black hole has a mass $M_{\text{BH}} \sim 2.81M_{\odot}$ and dimensionless spin $a_{\text{BH}} \sim 0.77$. The last snapshot at $t = 17.301$ ms shows that most of this “lower-density” matter forming the accretion disk has fallen into the black hole.

Originally, I had to face some problems in simulating the post-merger phase of the evolution, since after the formation of the apparent horizon I observed a “bad” behaviour of the system with physical quantities abruptly going to zero everywhere in the simulation domain (for instance the rest-mass density and the lapse). Moreover, I encountered infinite values for quantities related to gravitational waves such

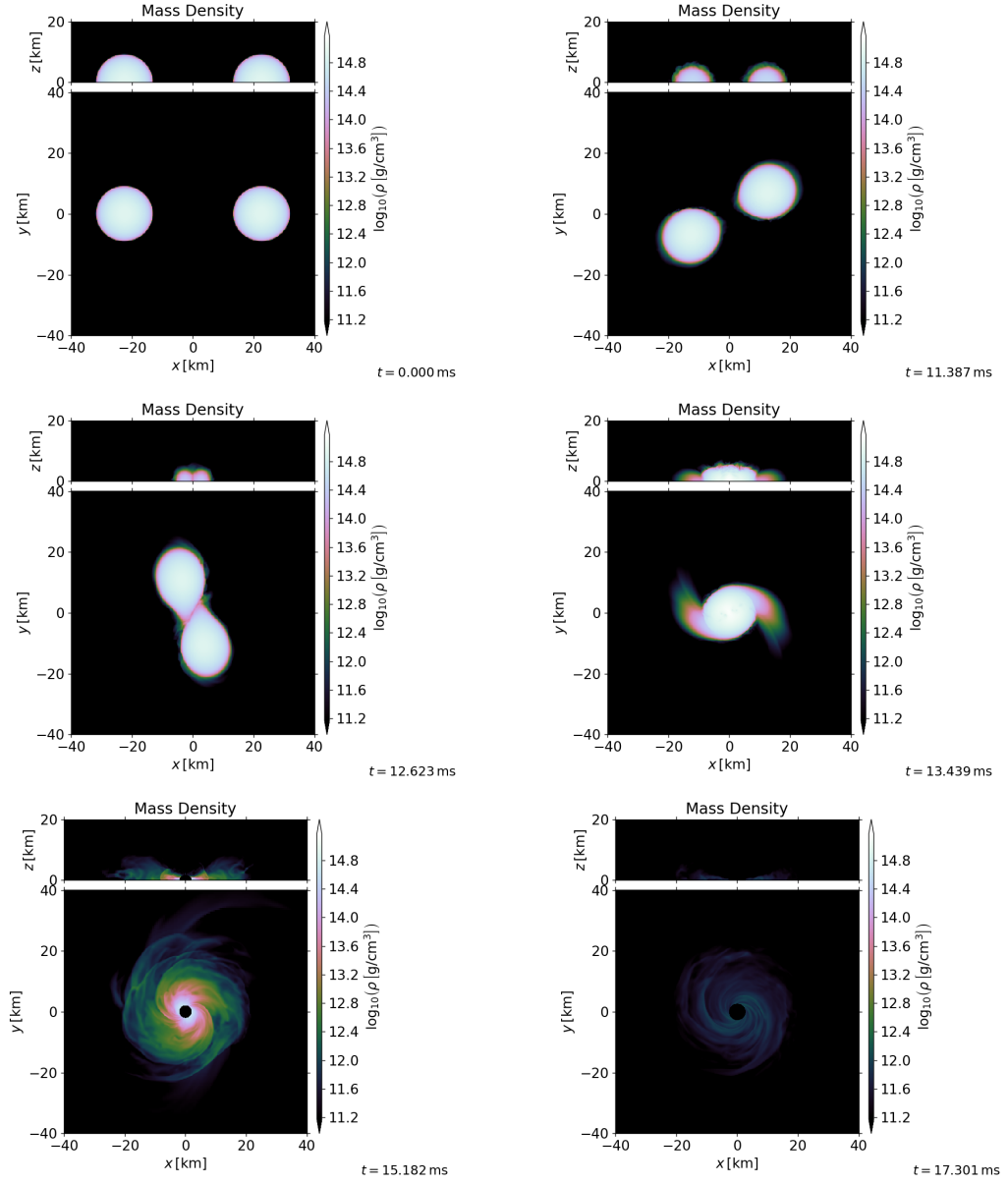


Figure 5.14: Evolution of the rest-mass density (in logarithmic scale) on the xy and xz planes.

as certain multipole components of the Weyl scalar Ψ_4 . Since it there was a possibility that could be a resolution problem - i.e. the finest grid I used in the simulation was not fine enough to resolve the apparent horizon - I tried to overcome the issue by adding two refinement levels for the post-collapse phase with resolutions $dx = 0.1$ and $dx = 0.05$ in **Cactus** units. However, this adjustment did not prove effective and the trouble persisted. Therefore, I performed a *domain excision*. It is not unusual, in Numerical Relativity simulations, that one removes a region of the computational domain where the solution is not physically important (for instance, inside an apparent horizon) and more inclined to numerical errors (e.g. near a singularity). In these cases, *excision boundaries* can be introduced; in this manner, spatial parts of the computational domain can be removed from the time-update algorithms [22]. In my simulation, the excision method has been provided by the WhiskyTHC method `THC_HydroExcision`, which sets the hydrodynamical variables to atmosphere values within the excised region. In particular, I set the excision region to be 0.9 times the radius of the apparent horizon. This solved the issue so that I could examine the evolution of the system after the collapse as well.

For the temperature, considerations analogous to the DD2 case can be made. The temperature evolution is shown in Figure 5.15. During the inspiral, the neutron stars start heating up: again in this case the temperature is higher at the interface between the two merging bodies (see the snapshot at $t = 12.976$ ms). The streams of matter ejected during and after the merger have the highest temperature ($T > 10$ MeV).

Also for the electron fraction the situation is not much different from the DD2 case. Two snapshots of Y_e distribution on xy and xz planes slightly after the collapse are shown in Figure 5.16. We can see again that $Y_e \lesssim 0.1$ (outside the excised region), which is likely to produce nuclei with $A \gtrsim 130$, powering a red kilonova component. The situation would have been different with respect to the DD2 case if we would have taken weak interactions into account. In fact, the lifetime of the central remnant can have a huge impact on the composition of the matter ejected during the merger, because of neutrino irradiation. In this case, since the collapse occurs almost immediately after the merger, there would be little time for raising the electron fraction Y_e of matter ejecta due to neutrino irradiation. Therefore, Y_e would remain relatively low: the red kilonova component would be strongly dominant in this instance.

I end this part with a plot showing the conservation of the total baryonic mass (Figure 5.17). This is an indication of the accuracy of the simulation. From the figure, it is clear that before the merger the total baryonic mass is conserved extremely well: the relative difference between the total baryonic mass and its initial value remains below 10^{-5} . The error grows at the merger since, because of the strong shocks, a significant fraction of the mass is expelled and cross the refinement level boundaries and leaves the finest grid. After the collapse the total baryonic mass is close to zero, meaning that almost all the matter has been swallowed by the BH.

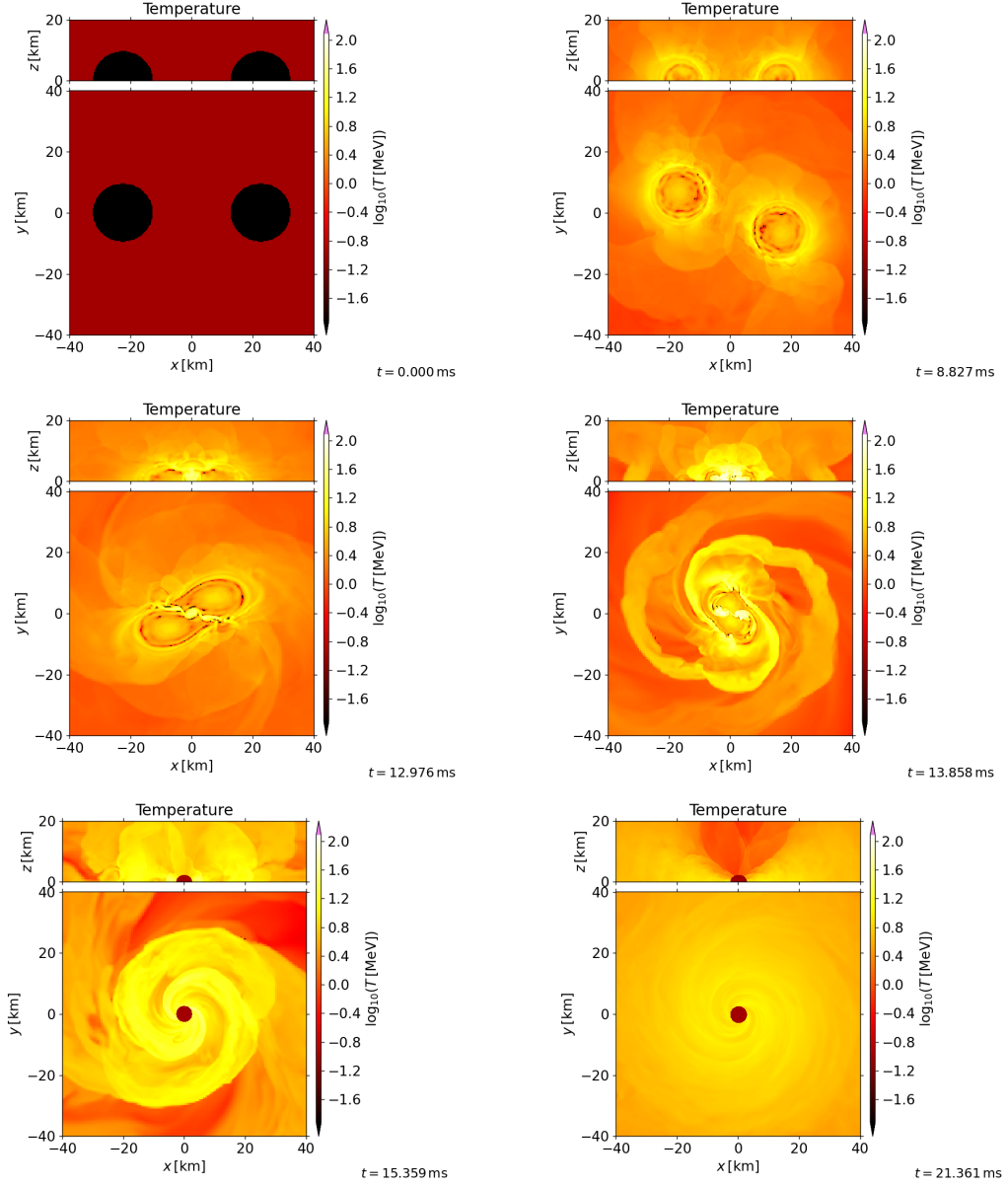


Figure 5.15: Evolution of the temperature (in logarithmic scale) on the xy and xz planes.

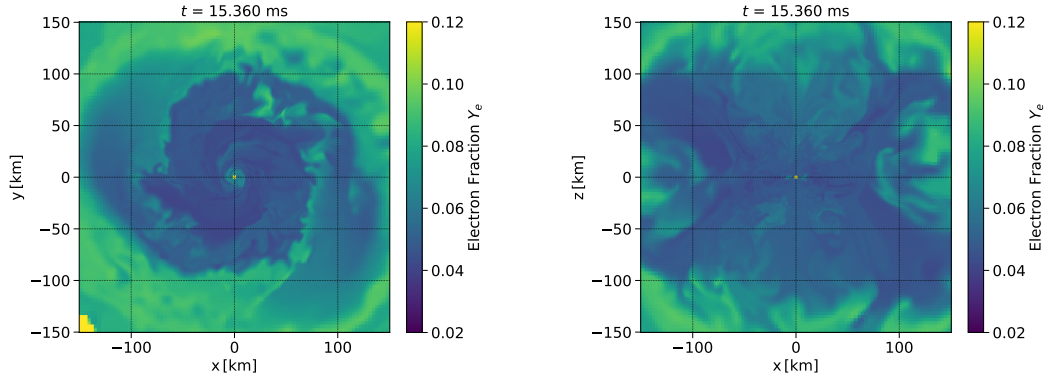
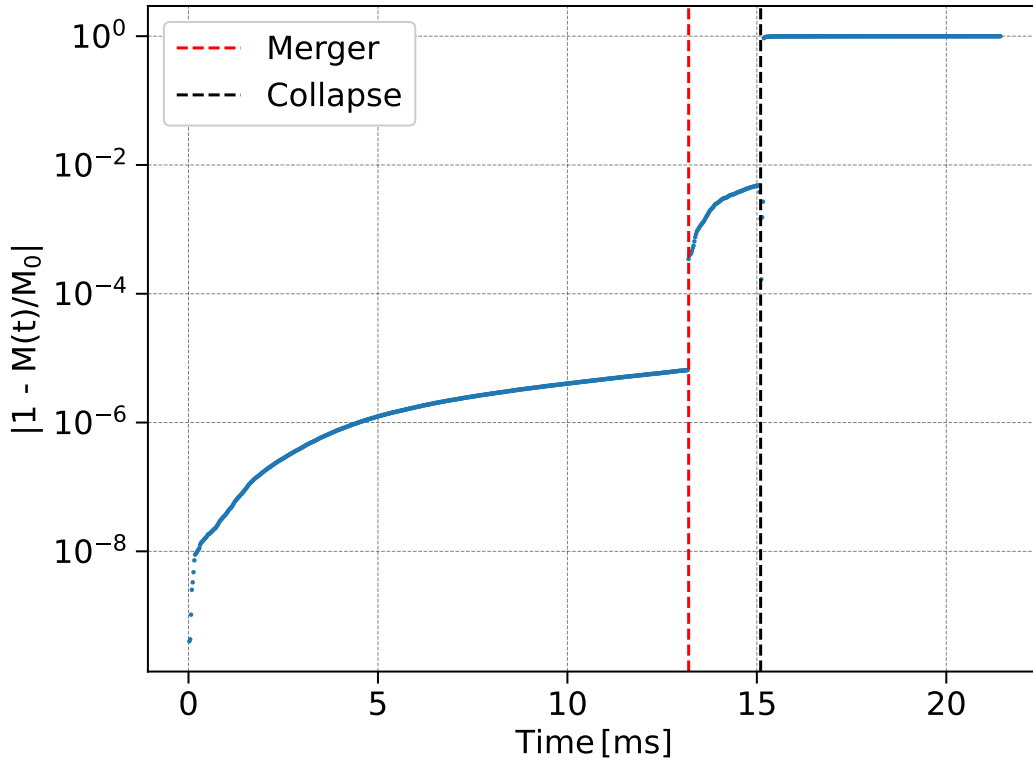
Figure 5.16: Evolution of the electron fraction on the xy and xz planes.

Figure 5.17: Conservation of total baryonic mass.

Gravitational Waves

I am now going to show some significant features of the gravitational wave signals extracted from the simulation. Figure 5.18 shows again that the dominant component is $l = m = 2$, which is the one carrying more energy. In Figure 5.19 we can see the GW strain ($l = m = 2$) and the evolution of the frequency, manifesting the chirp signal characteristic of the inspiral. The only part we can see here is the inspiral because - for these plots - Ψ_4 has been extracted from the outermost detector (distant ~ 2510 km from the source), hence just the “inspiral” portion of the signal has reached the detector.

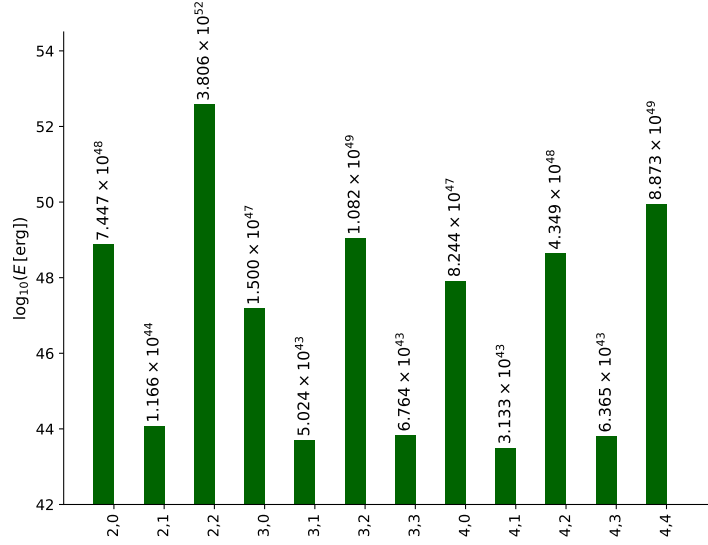


Figure 5.18: Total energy radiated for each multipole component.

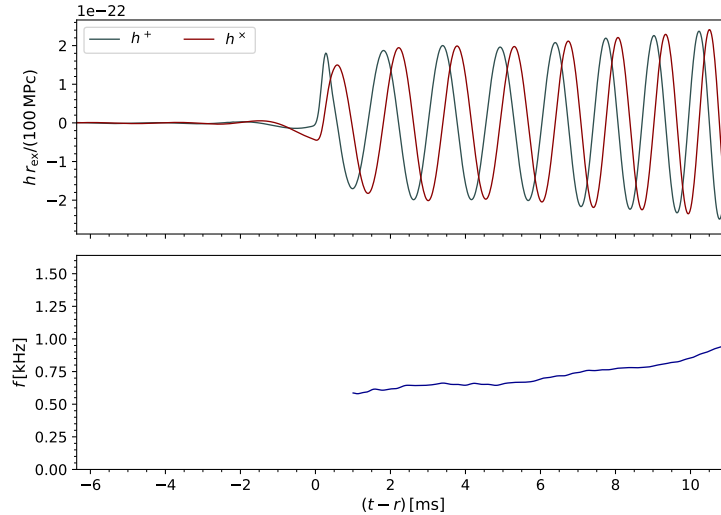


Figure 5.19: Gravitational wave strain for $l = m = 2$ and evolution of instantaneous frequency.

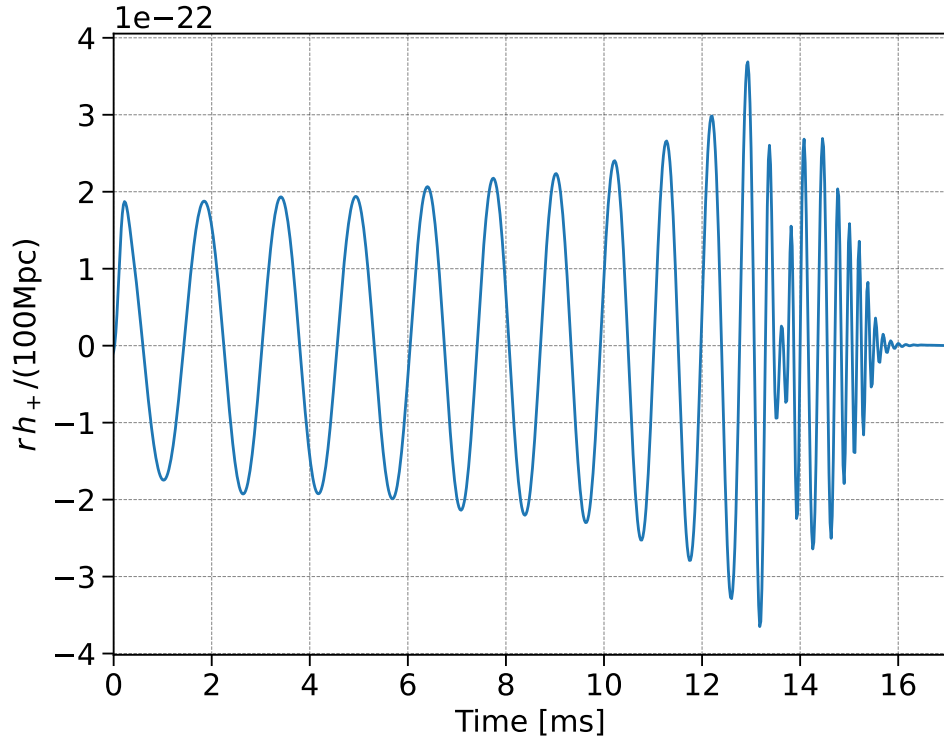
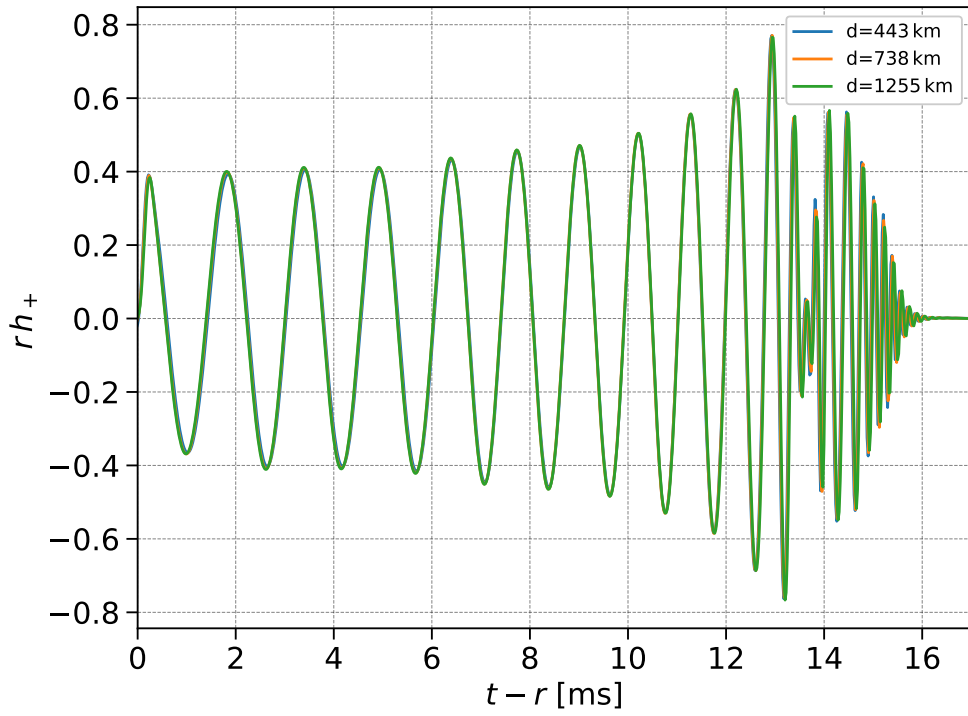
Figure 5.20: Gravitational wave strain ($l = m = 2$).

Figure 5.21: Gravitational wave strain for three different detectors.

Figure 5.20 shows the gravitational wave strain ($l = m = 2$), which in this case has been extracted from a detector closer to the source (443 km). Indeed, the waveform is longer, and we can distinguish the merger (~ 13 ms), reported by the maximum of the amplitude, but also the formation of the black hole (~ 15 ms): in fact, at the time corresponding to the collapse of the central remnant, the signal rapidly goes to zero. This provides a simple illustration of how a gravitational waveform can provide essential information on the evolution of the system.

Again, I have checked whether the signal was actual gravitational radiation, plotting the gravitational wave strain as a function of the retarded time. Also in this case, the signals at three different detectors at 443 km, 738 km and 1255 km overlap extremely well, confirming that we are “measuring” actual gravitational radiation.

Conclusions

In this thesis, I have described a series of simulations of binary neutron star mergers, performed via the general relativistic magnetohydrodynamic code **Spritz** and the general relativistic hydrodynamic code **WhiskyTHC** in the Einstein Toolkit framework. In particular, I carried on the work performed by Alice Gambaro in her master thesis [6], in which she has investigated the origin of the “fast-inspiral problem” encountered evolving binary neutron star systems with tabulated EoS through **Spritz** and initial data generated with **LORENE**.

The first tests aimed to determine whether initial data generated with **Kadath** instead of **LORENE** could lead to a resolution of the fast-inspiral problem. Nevertheless, even in this case, the issue persisted: the two stars reached the merger after ~ 1.5 orbits. Then, I tried to perform other tests, still using **Kadath** for initial data. I supposed that the difficulties encountered in evolving correctly binary neutron stars using tabulated EoS could be related to that specific tabulated equation of state (BLh), so I switched to another tabulated equation of state, known as DD2 EoS, for which I generated new initial data keeping all the other parameters fixed. Also in this case, however, the inspiral was very fast (still less than 2 orbits to reach the merger). After that, in order to explore all the possibilities, I also performed an additional trial by changing the method for spacetime evolution. In the previous simulations, the spacetime was evolved according the BSSN formulation of numerical relativity, so I tried to adopt another method, based on the CCZ4 formulation of numerical relativity. Though, this did not have any effect on the fast inspiral problem, and the only relevant difference compared to the previous simulations was a better suppression of constraint violations. Unfortunately, through the simulations I performed, I did not find a solution for the fast inspiral problem with **Spritz** in evolving binary neutron stars described by tabulated EoS. Solving this issue remains of primary importance for **Spritz** to accurately describe the dynamics of these mergers and working out a solution is left to further investigations.

In the second part of my thesis, I performed simulations using the general relativistic hydrodynamic code **WhiskyTHC** of the same exact binary neutron star system, with same initial data (generated with **Kadath**). In particular, I focused on two distinct simulations, being different only for the equation of state: DD2 and BLh, as before. In this case, I did not have to face the fast inspiral problem, since the system reached the merger after ~ 4.5 orbits in the case of DD2, and slightly more than 5 orbits in the case of BLh. Also, I followed the post-merger evolution of the system: in the simulation with BLh EoS, the central remnant collapsed to a black hole after

a few ms, while the remnant produced in the DD2 case survived for longer times (even if I did not follow the evolution up to the collapse for reasons of time and resources). These two simulations provide an illustration of the impact that the equation of state of matter can have on the evolution of the same system.

Finally, I have presented a series of results, for instance gravitational waves and other physical quantities of interest extracted from the simulations, in order to elaborate considerations about the aftermath of the merger, such as properties of the merger ejecta and their consequences for electromagnetic emissions such as kilonovae. In particular, I performed this task by means of post-processing tools as `PostCactus` [60] and `Kuibit` [61], which allow to analyze and visualize data obtained by simulations performed with the Einstein Toolkit. These kind of analysis are crucial to interpret future multi-messenger observations.

In fact, I am convinced that - in view of near-future multi-messenger observations performed by LIGO, Virgo and other upcoming interferometers as well as electromagnetic telescopes and other major astronomical facilities - it is increasingly important to develop the ability to properly interpret observational data assisted by sophisticated mathematical techniques and computational tools. The theoretical modelling of the binary neutron star mergers through GRMHD simulations performed by means of codes such as `Spritz` or `WhiskyTHC` has already turned out to be an extremely powerful instrument to explore the physics behind these intriguing and fascinating events and will certainly play a critical role in linking the future observational data to the physical processes governing these phenomena.

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